DEVELOPMENT OF STRATEGIC PERFORMANCE MANAGEMENT SYSTEM FOR HUMAN RESOURCE DEVELOPMENT OFFICE OF BENGUET STATE UNIVERSITY

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INTRODUCTION

This chapter provides an overview of the project context, purpose and description, objectives, with emphasis on improvements in operational efficiency. Also, the scope and limitations are established to outline the system's functional boundaries and implementation constraints.

Project Context

Performance management helps organizations improve efficiency by aligning employees' work with organizational goals. According to Fernandes et al. (2025), performance rating systems provide standards for measuring performance and achieving objectives. However, paper-based processes can cause delays, data errors, and inefficient processing (Datamation, 2019; HR Innovators Group, 2024). Digital performance management systems can help improve consistency and support continuous improvement (Capisonda, 2023).

In the Philippines, the Civil Service Commission (CSC) standardized the Strategic Performance Management System in 2012 to align employee performance with organizational goals. The SPMS follows four stages: Performance Planning and Commitment, Performance Monitoring and Coaching,

Performance Review and Evaluation, and Performance Rewarding and Development Planning, as shown in Figure 1.

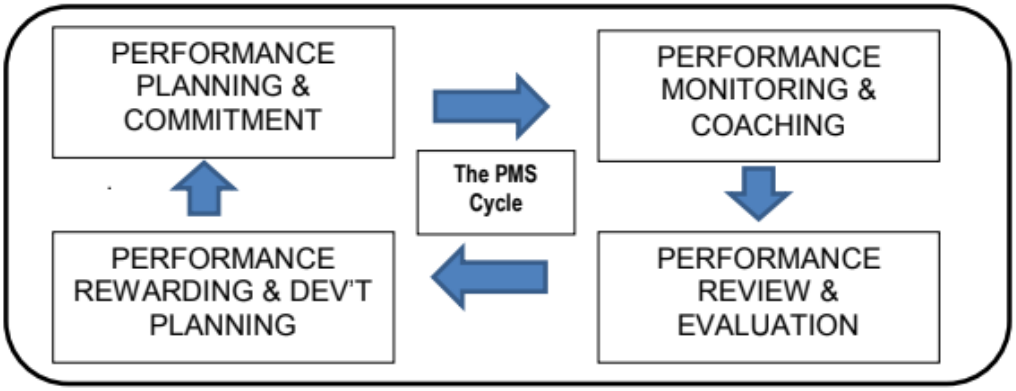


Figure 1. The SPMS Cycle

Source: Benguet State University. (2013). *Strategic Performance Management System (SPMS)*.

SPMS follows a four-stage cycle designed to align individual performance with the goals of the institution. It begins with Performance Planning and Commitment, where supervisors and employees collaboratively set SMART targets that are specific, measurable, achievable, result-oriented, and time-bound, which are then documented through forms such as OPCR, DPCR, IPCR, or IPERF.

This is followed by Performance Monitoring and Coaching, where supervisors continuously monitor employee progress, provide feedback, and offer guidance to address performance gaps and improve work efficiency.

At the end of the performance period, the Performance Review and Evaluation stage is conducted to assess actual accomplishments against the agreed targets through self-ratings and validation by the Performance Management Team (PMT).

Lastly, the Performance Rewarding and Development Planning stage uses the evaluation results as a basis for recognizing high- performing employees and identifying training and development opportunities to further enhance employee competencies and overall organizational performance.

At Benguet State University, the SPMS follows a hierarchical process in which performance targets are cascaded from the University President to the Vice Presidents, Deans, Directors, and Heads of Offices before reaching individual employees. Each level develops its commitments based on the targets established by the level above, as shown in Figure 2.

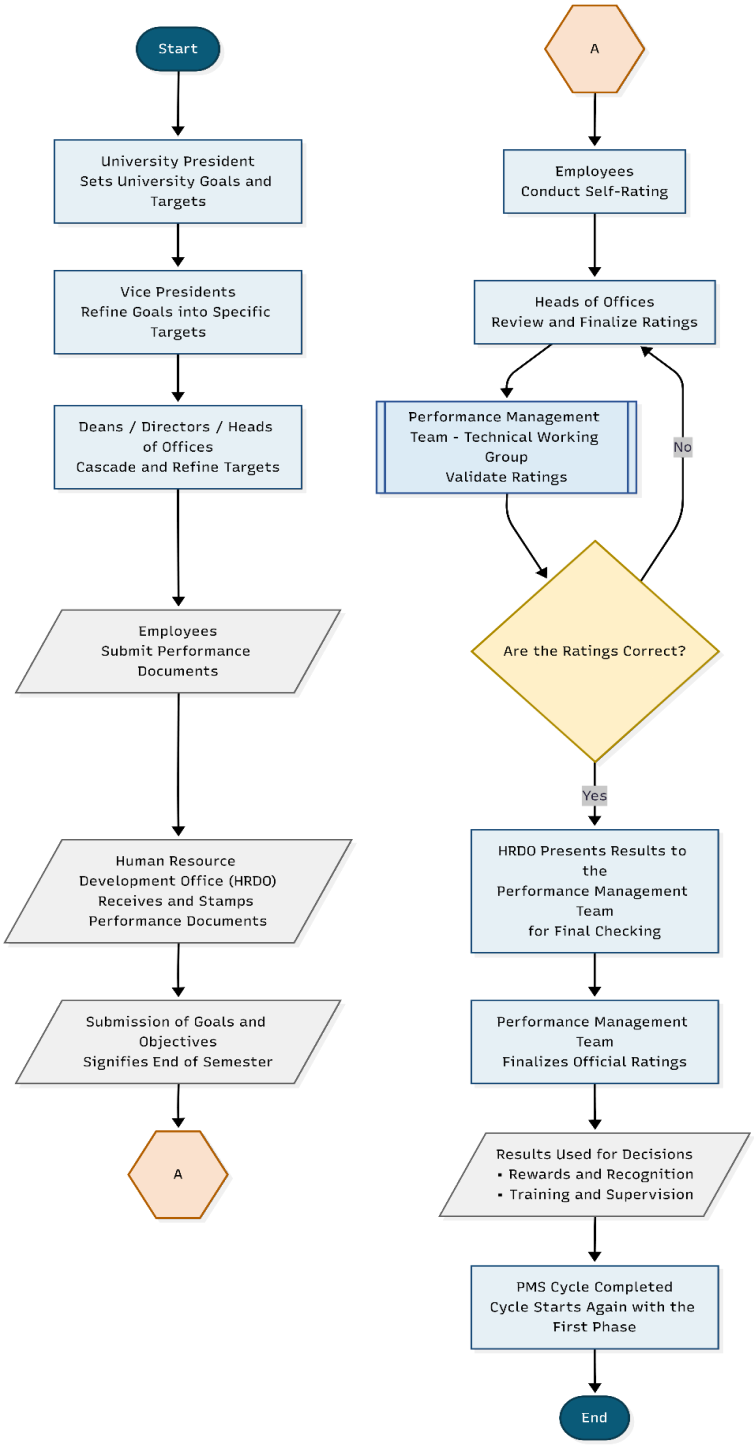


Figure 2. SPMS Process Flow

Source: Symbol conventions based on American National Standards Institute (1970).

The cycle starts with the University President setting the main goals of the university during the Strategic Planning and Operational Planning sessions. These targets serve as the overall direction of the institution for the next four to five years.

Based on initial consultations with HRDO, the cascading of these targets does not always happen on time. When the upper levels release their commitments late, every office below has to wait, and employees who are already prepared cannot proceed with their own paperwork until the cascade reaches them.

After the President sets the targets, they are cascaded down to the Vice Presidents (VPs). The VPs then refine these broad goals into smaller and more specific targets for the deans, directors, and heads of offices under their supervision. These supervisors break the targets down further before passing them on to the employees in their offices.

The process depends on every supervisor being available to review and sign the cascaded targets. In practice, prepared documents often sit on a desk for days or weeks when a signatory is on leave, in a meeting, or attending to other duties, which compresses the time left for the lower offices to complete their own commitments.

When an employee holds two roles that would normally use different forms, such as a faculty member who also handles administrative duties, the

higher-level role is prioritized. These employees are assigned a single OPCR, and the goals from both roles are merged into one set of targets for the performance period.

For HRDO staff, this merging is done by hand for each affected employee, which adds to the workload during the planning stage and creates room for inconsistency when the same employee is classified differently across cycles.

Once the targets are finalized, all employees prepare their performance documents: IPERF for non-regular employees, IPCR for regular employees, OPCR for the University President, Vice Presidents, and Executive Deans, and DPCR for Deans, Directors, and Heads of Offices.

These are submitted to the Human Resource Development Office (HRDO), where each page is stamped to mark the targets as official and to prevent revisions in the middle of the period. The stamping itself is straightforward, but the volume during peak submission periods is heavy. HRDO staff describe being flooded with stacks of forms from all offices within a short window of time.

At the end of the semester, the documents are returned to the offices for evaluation. Employees first conduct a self-rating to assess their performance against their set targets. After that, the respective heads of offices, deans, or directors review and finalize these ratings.

Computing the final scores at this stage is one of the most error-prone parts of the cycle. Each form requires averaging the Quality, Efficiency, and Timeliness scores across several targets and applying weight allocations that change depending on the employee's role. Supervisors compute these by hand, and a single arithmetic mistake on one target can change the final adjectival rating.

When errors are caught later, the affected forms have to be reprinted, re-signed, and re-routed, which adds work for the supervisor, HRDO, and the employee being rated.

The evaluation of performance follows a strict hierarchical structure to ensure accountability:

- Board of Regents: This body is responsible for rating the performance of the University President.
- University President: The President reviews and rates the performance of the Vice Presidents.
- Vice Presidents: These officials rate the Deans, Directors, and Heads of Offices within their respective sectors.
- Deans, Directors, and Heads of Offices: These supervisors rate the individual employees and personnel assigned to their specific units or colleges.

Next, once supervisor ratings are submitted, the Performance Management Team-Technical Working Group (PMT-TWG) validates all ratings against the actual accomplishments and the official rubrics. If the validation is clean, HRDO presents the results to the PMT for final checking. After this, the PMT, composed of Vice Presidents and Heads of Offices, finalizes the ratings as the official performance result of each employee.

Because a single committee validates the ratings of every employee in the university, this stage is a bottleneck. HRDO staff describe rechecking totals multiple times before submission to the PMT, since any computational error caught at this point means returning the document to the originating office for correction, which delays the entire cycle.

The validated ratings then serve as the basis for institutional decisions. Employees who perform well may be recommended for rewards or recognition, while those who need improvement may be recommended for training or further supervision.

The cycle is then completed and begins again with the first phase. Retrieving past performance records to support these decisions is itself a manual task. HRDO staff have to physically locate the relevant files across multiple offices, which becomes particularly difficult during accreditation visits, audits, or urgent report requests from management.

The main causes of the problems in the existing SPMS workflow are summarized in the Ishikawa diagram, as shown in Figure 3.

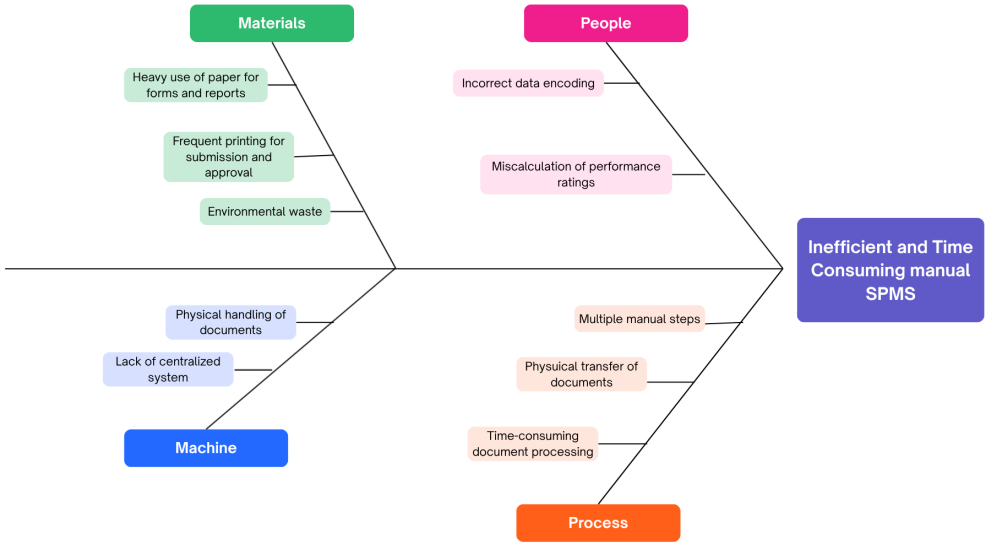


Figure 3. Ishikawa Diagram

Based on the initial consultation with the HRDO and an analysis of the existing SPMS workflow, five problems were identified that affect performance evaluation and record management.

1. Miscalculation of Performance Ratings. During the SPMS evaluation process, performance ratings in the OPCR, DPCR, IPCR, and IPERF forms are passed through different offices and manually encoded and computed at each level. Because the scores are repeatedly written,

transferred, and re-entered by hand, there is a higher chance of mistakes in the calculations. These errors are usually noticed only when all reports are being checked and combined. When this happens, HR staff and evaluators need to go back to the forms, compare records, and recompute the ratings to make sure the results are correct. This makes the process slower and more difficult, especially when many employees are being evaluated at the same time. Since the system depends on manual encoding and repeated computation, even small mistakes can affect the final results. This shows the need for an automated system that can compute ratings correctly, reduce errors, and make the evaluation process faster and more reliable.

2. Slow document processing: This refers to the slow handling of documents caused by the one-directional flow of the SPMS process. Targets start with the President, then move to the Vice Presidents, then to Deans and Heads of Offices, and finally to the individual employees. Targets start with the President, then go to the Vice Presidents, then to Deans and Heads of Offices, and finally reach the individual employees. Each level cannot finalize until the level above it has done so. When any office in the upper levels is late, every office below is also late, even if their own work is already ready. During peak evaluation periods, this turns into a long queue of forms waiting for a signature.

3. Excessive use of paper and printing materials. Every cycle uses thousands of pages between OPCR, DPCR, IPCR, and IPERF forms, accomplishment reports, and supporting documents. Reprints due to errors add even more. The university then has to keep these records on file for years, which takes up cabinet space across HRDO and the individual offices. This also conflicts with BSU's institutional commitment to establishing a smart and green campus environment.
4. Laborious Reporting and Tracking: The current workflow centers on the manual preparation of reports, where information must be encoded and subsequently verified and rechecked by other staff members before finalization. This manual handling of documents makes data management difficult and prone to duplication or confusion, especially when files are misplaced with other documents. Such misplacement directly impacts the accuracy of performance evaluations and institutional reporting. Furthermore, urgent requests for accreditation or management meetings still require the manual generation of reports, creating significant bottlenecks and delays. The absence of a centralized system also makes it challenging to monitor and track employee performance trends over time.
5. Limited Transparency: The lack of a centralized digital platform makes it difficult for employees to track the status of their submissions or for management to maintain a clear audit trail of the evaluation process.

Without real-time access to performance data, the system struggles to provide the transparency required for effective professional development and institutional accountability.

These problems affect employees, supervisors, and HRDO personnel throughout the SPMS cycle. Manual processing, delays, and fragmented records make it difficult to maintain accurate information and provide timely reports for institutional decision-making. They also limit the timely retrieval of records needed for employee development, recognition, and institutional planning.

The development of a proposed web-based SPMS is therefore intended to provide a centralized platform for managing performance targets, evaluations, ratings, and records. It aims to reduce manual work, improve access to performance information, and support more efficient monitoring and reporting. The proposal also aligns with BSU's Goal 5 of promoting integrity-based governance and efficient resource management, as well as its commitment to establishing a smart and green campus environment.

Purpose and Descriptions

This study aims to design and develop a web-based Strategic Performance Management System for the Human Resource Development Office of Benguet State University to address issues in the existing performance management

process. The system supports BSU's Goal 5 of promoting integrity-based governance and efficient resource management, consistent with studies by Armas et al. (2025) and Volante and Nicart (2025), which highlight the benefits of digital performance management in improving efficiency, data reliability, and institutional responsiveness.

The system integrates the major stages of the Strategic Performance Management System into a single platform, from the submission of performance targets to the completion of employee evaluations. It automates rating computation, organizes records in a centralized database, and provides monitoring tools that support timely tracking of performance activities.

The system is designed to provide specific benefits for its primary users:

- HRDO and the Performance Management Team (PMT): The system provides tools for managing employee performance records, monitoring the progress of submissions and evaluations, validating records, and generating reports needed for administrative and institutional requirements.
- Supervisors (Deans, Directors, and Heads of Offices): The system enables supervisors to review employee submissions, conduct performance evaluations, approve ratings, monitor pending activities, and access performance information through a unified interface.

- Employees: The system allows employees to submit performance commitments, view the status of their evaluations, access performance records, and receive evaluation results through the platform.

The system is intended to support a more organized and reliable performance management process at BSU.

Objectives of the Study

This study aims to design and develop a functional, digitally secure, and user-friendly SPMS that will digitize the HRDO performance evaluation process and supports the implementation of the SPMS at BSU.

Specifically, the study aims to attain the following:

1. To determine the functional and non-functional requirements of the SPMS.
2. To design and develop the system architecture, database structure, and user interface of the SPMS.
3. To evaluate the developed SPMS in terms of system usefulness, information quality, and interface quality based on the PSSUQ standards.

Scope and Limitations of the Study

This study focuses on the design and development of a SPMS for the HRDO of BSU. The system is built as a cloud-based web application that can be

accessed through laptops and desktop computers with an internet connection. It provides three separate dashboards for HRDO personnel, supervisors, and employees, so that each user can access the functions and information related to their role.

The system supports three parts of the SPMS cycle. The first is Performance Planning and Commitment, which is handled through the digital submission and cascading of performance targets using the IPCR, OPCR, DPCR, and IPERF forms. The second is Performance Review and Evaluation, which includes the digital encoding of accomplishments, automatic computation of weighted ratings, supervisor review, and validation by the Performance Management Team–Technical Working Group. The third is Performance Rewarding and Development Planning, which is supported through summary reports that can be generated from the stored evaluation data. These reports can be used for PRAISE Committee deliberations, training recommendations, and accreditation submissions.

To protect the data managed by the system, several security measures are included. The system uses Role-Based Access Control (RBAC) to limit access to modules based on the user's role, Cross-Site Request Forgery (CSRF) protection on form submissions, and Two-Factor Authentication (2FA) through email OTP during login. Administrative controls also allow HRDO personnel to disable

accounts of separated or compromised users through session controls. Standard password reset and password change procedures are also provided.

The digital workflow ends at the supervisor's final approval. Once the form is approved, the system generates a printable copy of the form, which can then be printed and signed manually based on the applicable BSU and CSC requirements. The signed printed copy serves as the official record, while the digital copy is kept in the system for analytics, reporting, and audit purposes.

The system also has some limitations and boundaries. The SPMS forms used in the system have predefined fields and structures that cannot be freely modified by users. The fields are connected to the system's validation rules and automated calculations, especially the computation of performance ratings. Because of this, users cannot freely add, remove, or change fields or modify the structure of the forms, since doing so may affect the data and calculations used by the system. The forms are therefore designed to follow a fixed structure to maintain consistency in the processing and computation of performance information.

The Performance Monitoring and Coaching phase of the SPMS cycle is only partially supported. The system provides a journal logging feature that allows supervisors to document coaching sessions, feedback, and other monitoring activities. However, it does not include automatic coaching workflows, continuous performance monitoring tools, or in-cycle target

amendments. These activities still require discussions, observations, and decisions between supervisors and employees and cannot be fully handled by the system.

The office planning module is limited to the digital submission and management of office-level targets. It does not include higher-level institutional planning activities such as strategic plan formulation, policy development, or top-level administrative decision-making. These activities involve broader processes that are outside the purpose of the developed system.

The system does not integrate with other existing BSU systems or databases. Integration with other university systems would require coordination with the BSU ICT department, access to existing databases, and additional development and testing time beyond what is available for this capstone project.

Electronic signatures are not included in the system. The system instead generates printable copies of the forms that can be manually signed. The signed printed copy remains the official record, while the digital version is kept in the system for reporting, analytics, and other related purposes.

A formal Privacy Impact Assessment (PIA) is also not included in the study. Data privacy is considered during the design and development of the system, but conducting a formal PIA would require additional legal, technical, and organizational assessment with the appropriate privacy authorities. These activities are beyond the technical development focus and timeframe of the study.

The system is designed mainly for use through laptops and desktop computers. A separate mobile interface is not included because the SPMS forms contain many fields and columns for targets, accomplishments, and ratings that require enough screen space for encoding and reviewing information. The current system therefore focuses on providing the required SPMS functions through a desktop-based web interface.

Even with these limitations, the system addresses the main operational needs identified in the current SPMS implementation at BSU. It provides a digital way to manage performance targets, accomplishments, evaluations, and reports while also helping organize the records used in the SPMS process. The system can also serve as a basis for future improvements, such as integration with other BSU systems, additional features, or further assessment for formal implementation.

REVIEW OF RELATED LITERATURE/SYSTEMS

This chapter reviews studies and existing systems related to performance management, the shift toward digital human resource processes, and web-based information systems used in higher education institutions. The review is organized to first establish the theoretical basis of performance management as a process, then to examine how digital tools have changed this process, and finally to look at how performance management systems have been built and used at the international, national, and local levels. Each grouping ends with a short synthesis that draws out the common patterns and the gaps that remain. The chapter closes with a clear statement of the research gap and an explanation of how the present study responds to it.

The review covers three main areas. The first is the concept of performance management as a strategic and continuous process, not a once-a-year activity. The second is the shift from paper-based to digital approaches in managing employee performance. The third is the design and use of web-based systems for performance evaluation, with particular attention to Philippine state universities and colleges. These three areas together form the foundation for the design and development of the SPMS for BSU.

Performance management is more than an annual rating activity. Fernandes et al. (2025) examined how performance management systems function as strategic tools rather than as plain appraisal procedures. Using the Balanced

Scorecard framework and performance indicator matrices across several organizational case studies, they found that organizations which treat performance management as a continuous strategic process achieve better long-term outcomes than those which treat it as a once-a-year administrative requirement. The study, however, did not test these ideas inside a specific information system, leaving the question of implementation open.

Walhekar and Khatke (2020) focused on educational institutions and analyzed how performance management systems support the alignment of individual goals with broader institutional targets. Their work pointed to goal-setting, monitoring, and structured feedback as the three core elements that make a performance management system effective in education. They concluded that when these three elements are aligned, employee motivation and overall institutional performance both improve. The study is useful as a theoretical foundation, but it does not provide a working model of how these elements are operationalized in a government-mandated framework.

Himes et al. (2024) developed and validated a human resource performance management model that tested five dimensions: goal-setting, performance development, feedback, reward systems, and skills development. The model was tested through interviews with HR experts and a survey of 384 employees in knowledge-based organizations using structural equation modeling. The findings confirmed strong alignment between all five dimensions and positive

outcomes such as improved employee motivation and data-driven HR decisions. While the study produced a validated model, it was carried out without a dedicated information system and did not address how the model would behave inside an operational digital platform.

Taken together, these studies show that a performance management system is structured, continuous, and built around clearly defined components. It is not just an evaluation activity but a planning, monitoring, evaluation, and feedback cycle that drives both individual and institutional improvement. What these studies do not address is how this cycle is actually operationalized through a working information system in a government institution such as a Philippine state university, which is the focus of the present study.

The shift from paper-based to digital performance management has been documented in several recent studies, particularly in the Philippine higher education setting.

Capisonda (2023) examined the implementation of the SPMS within the Philippine State College of Aeronautics and identified the main administrative challenges faced by institutions that still rely on paper-based processes. The study found that delays in the evaluation cycle, inconsistent data management across departments, and difficulty in tracking long-term performance trends were directly traceable to the manual handling of records. The main contribution of Capisonda's work is the careful documentation of these problems in an actual

SUC setting, although the study itself did not produce a working digital solution to address them.

Galorio et al. (2024) conducted a systematic review of human resource management practices in education and identified the absence of integrated digital tools as a primary obstacle to effective HR performance management. The review consolidates findings from multiple studies and concludes that paper-based and fragmented systems consistently produce slower decision cycles, weaker accountability, and limited transparency. The study does not present a new system, but it strengthens the broader argument that digital transformation in HR is no longer optional for institutions that want to operate efficiently.

Volante and Nicart (2025) studied the digital transformation of performance management in Philippine higher education through the development of an electronic SPMS. Their work demonstrated that institutions which adopted a digitalized framework achieved greater transparency, better data integrity, and clearer alignment between individual contributions and institutional targets. The study provides a strong precedent for the present work, but it focused on the broader concept of digitalization and did not address the operational design of a system tailored to the workflow of a specific SUC.

These three studies, taken together, point to the same conclusion. Manual SPMS implementation in Philippine higher education leads to systemic delays, weak transparency, and lost institutional value, and digital systems clearly

improve these areas. However, the implementations described so far either stop at a conceptual or evaluative level, or are not designed around the specific cycle prescribed by CSC Memorandum Circular No. 06 (2012) for state universities. The present study takes the next step by designing and developing a working web-based SPMS that is built around this specific cycle.

Beyond the Philippine setting, several international studies offer useful references for the design of web-based performance and HR information systems.

Ranjetha et al. (2022) developed an online employee performance management system supporting three user roles, each with its own interface for Employees, Project Managers, and Administrators. The study addressed the inefficiency of paper-based and email-dependent performance tracking, which caused delays in reporting and difficulty in maintaining organized employee records. The system improved operational efficiency through role-based access control and digitized reporting, providing a useful reference for multi-role web-based system design. Its limitation, with respect to the present work, is that it was built for a private-sector context and not for a government-mandated framework like the SPMS.

Andrian and Massaty (2025) developed a web-based internship management system for a university and integrated User-Centered Design with their development process. The system addressed the inefficiency and limited accessibility of traditional internship management, where manual handling was

error-prone and lacked the transparency that students and administrators needed. The study confirmed that a digital system significantly improves efficiency and transparency in institutional management when each functional area is properly identified and built. While the application area is internship management rather than employee performance, the study is relevant for showing how an institutional digital system can be developed through clearly defined functional modules.

These international studies collectively show that role-based web systems significantly improve performance tracking, transparency, and reporting in institutional settings. However, none of them address integration with a government-mandated framework such as the SPMS used in Philippine state universities, which has specific compliance requirements and a fixed top-down cascade structure. This is one of the gaps that the present study seeks to address.

At the national level, several studies have examined SPMS implementation and HR information systems in Philippine institutions, and most point to similar operational challenges.

Armas et al. (2025) developed an integrated SPMS model for a Philippine state university that aligns key performance indicators with strategic institutional goals. The framework used individual performance modules and reporting dashboards, and improved accountability and transparency in the institution where it was applied. The model is locally validated, but it is presented as a conceptual

framework rather than as a complete working web-based system tailored to the SPMS cycle of a specific SUC.

Arcigal (2021) conducted a benchmarking study across SUCs in the CALABARZON region and identified inconsistencies in evaluation processes and limited use of data as common gaps in SPMS implementation. The study highlights the need for standardized and digitally supported frameworks to support consistency and accessibility across SUCs. While useful as a benchmark, the study did not produce a system; it ended at the level of recommendations for enhancement.

Barayuga et al. (2024) explored a web-based Human Resource Information System designed to manage employee performance records at Ilocos Sur Polytechnic State College. The system improved data accuracy and the speed of report generation for HR administrators. The study is useful as a national precedent for HR digitalization in a state college setting, but it focused on general HRIS functions and not on the full four-stage cycle of the SPMS.

Torneo and Mojica (2020) conducted an assessment of SPMS implementation across various national government agencies and identified poorly structured performance indicators and a lack of adequate digital monitoring mechanisms as systemic problems. Their recommendations called for digital integration as a long-term policy direction, but the work itself is evaluative and did not produce a system.

The national policy direction that supports these studies is anchored in Republic Act No. 12254 (2025), which recognizes the vital role of information and communication technology in nation-building and aims to establish a digitally empowered and integrated government that delivers responsive and transparent online services. This policy reinforces the need for digital transformation in government performance management, where transparency, efficiency, and data accessibility have become essential rather than optional. The present study is consistent with this national direction and supports the policy goal at the level of an actual institutional system.

Taken together, these national studies show that the challenges of SPMS implementation in the Philippines are well documented, and that digital transformation has been recognized at the policy level. What remains missing is a fully integrated SPMS that is tailored to the specific workflow, compliance requirements, and form structure used in Philippine state universities.

Studies conducted in the Cordillera region and in nearby academic institutions provide the closest reference points for the operational context of BSU.

Ralutin (2024) developed a web-based Human Resource Management System for Kalinga State University to address operational inefficiencies in HR processes. The system automated several repetitive administrative tasks, which improved overall HR workflow efficiency. The study is the closest geographic

precedent for the present work and confirms that web-based HR systems are feasible and useful in Cordillera SUCs. Its limitation, with respect to the present study, is that it focused on broader HR functions rather than on the SPMS cycle in particular.

Abrojena and Ormilla (2023) evaluated the implementation of the Results-Based Performance Management System (RPMS) within the Department of Education in Ifugao. The study found gaps such as infrequent feedback and a clear disconnect between teachers and administrators in the review cycle. The work did not produce a system; instead, it documented user-level operational issues that any future digital RPMS or SPMS in the region would have to address.

Kasan and Buenavides (2022) developed a web-mobile evaluation system that addressed the inefficiency of traditional paper-based Quality Course Evaluation forms in an academic setting. While the application area is course evaluation rather than employee performance, the study confirms the technical feasibility of digitalizing evaluation forms in a Philippine higher education environment.

These local studies show that there is strong evidence of manual process inefficiencies in Cordillera and nearby institutions, and that web-based digitalization is feasible at the SUC scale. Their collective implication is that the need for a localized, SPMS-specific system at BSU is consistent with the documented experience of neighboring institutions.

Across the studies reviewed, several patterns emerge consistently. Performance management is most effective when it is treated as a continuous strategic process and not as a once-a-year activity. Manual implementations of this process produce delays, inconsistencies, and weak transparency, regardless of the institution. Digital transformation has been recognized at the international, national, and local levels as the necessary direction. Web-based systems with role-based access have been shown to improve performance tracking, reporting, and transparency in both private-sector and institutional settings.

At the same time, the differences across the studies highlight a clear gap. International systems are often built for private-sector or general HR contexts and are not aligned with the SPMS framework used in Philippine state universities. National studies have documented the problems with manual SPMS implementation, but most stop at the level of frameworks, benchmarks, or policy recommendations rather than working systems. Local studies confirm the operational pain points and the feasibility of digital systems in the region, but none has produced an SPMS-specific platform built around the full four-stage cycle of the SPMS.

Despite the availability of digital performance management and HR information systems, the existing studies reveal several limitations that the present study aims to address. Many of the available systems are designed for general HR functions and are not tailored to the structured requirements of the SPMS used in

Philippine state universities. Most current implementations also lack centralized platforms that integrate planning, evaluation, and reporting into a single system. There is also limited focus on real-time submission tracking, automated computation of weighted ratings, and audit-friendly transparency, which are exactly the operational challenges identified in local SPMS contexts. These gaps point to the need for a customized SPMS that directly responds to the institutional workflow and administrative requirements of a Philippine state university.

Unlike previous studies, which focus on general HR functions, benchmarking exercises, or partial digitalization efforts, the present study develops a SPMS designed specifically for the operational structure of BSU. The system integrates performance planning, review and evaluation, and reporting into a single platform that follows the SPMS cycle prescribed by CSC Memorandum Circular No. 06 (2012). It also responds directly to the operational problems documented in the local and national studies reviewed above, including computational errors in weighted ratings, slow document routing, paper waste, manual report preparation, and limited submission visibility. By doing so, this study extends the body of work on SPMS digitalization in the Philippines by producing a working, deployable system that is grounded in both the policy framework and the documented realities of a Cordillera-based SUC.

A brief note on the technology choices is included here, with the full justification placed in the Technical Background chapter that follows. The

frontend uses JavaScript, Tailwind CSS, and Axios because these tools support efficient data exchange and responsive interfaces, which are essential for real-time performance monitoring and evaluation. The backend uses CodeIgniter 4 with SQLite 3 because this combination meets the BSU ICT department's compatibility requirements and provides a secure, organized structure for handling sensitive performance data.

TECHNICAL BACKGROUND

The development of the SPMS requires a specific technical architecture, software stack, and infrastructure designed to address the operational limitations within the HRDO of BSU. The selected technologies are grounded in an analysis of the existing workflow to ensure accessibility, data integrity, and rapid deployment.

Currently, the SPMS operates entirely offline, relying on physical paper for data entry, manual calculation for data processing, and physical filing cabinets for storage. Technically, this system possesses no network capability, no automated validation, and no centralized data retrieval mechanism. When institutional reports are required, data must be manually extracted from physical pages and re-encoded into standalone spreadsheet files, introducing a high risk of data duplication and human error.

The primary technical gap is the absence of a unified, accessible data pipeline. To resolve this, the system transitions the institution from a fragmented, offline physical state to a centralized, digital database architecture.

To bridge the identified gaps, the system's technological stack is optimized for the specific administrative constraints of BSU:

- Web-Based Architecture: A web-based platform was selected over a standalone desktop application to eliminate the need for individual software installations across multiple colleges and offices. Because the

evaluation process involves users ranging from the University President down to individual personnel, a web-based approach ensures that any authorized user can instantly access their performance dashboard through standard web browsers on existing office computers.

- Database Management (SQLite): SQLite serves as the primary database engine for this implementation. Unlike heavier database servers that require complex network configuration and continuous maintenance, SQLite provides a lightweight, serverless, and self-contained database. This supports the rapid prototyping and local deployment phases of the system, ensuring that performance data (such as user credentials, targets, and evaluation scores) is securely stored and easily transportable without burdening the university's IT infrastructure.
- Development and Project Management (Trello): To efficiently manage the system's creation, Trello is utilized as the primary project management tool. In alignment with the Feature-Driven Development (FDD) methodology, Trello organizes the comprehensive feature list, assigns specific development responsibilities, prioritizes tasks, and tracks the progress of the SPMS modules throughout all iterations. This ensures that development remains strictly aligned with the operational needs of the HRDO.

Because performance evaluations contain sensitive institutional and personnel data, security is a primary technical requirement. The system implements Role-Based Access Control (RBAC) to ensure that users can only access data relevant to their specific clearance level (e.g., an employee cannot view another employee's ratings, while the HRDO can view institutional summaries). Furthermore, the architecture includes backend security measures such as Cross-Site Request Forgery (CSRF) protection and secure password hashing to prevent unauthorized access and data manipulation across the university's local network.

Ultimately, the selected technologies serve as calculated solutions to the university's administrative bottlenecks. By combining a highly accessible web architecture with a secure, self-contained database and strict access controls, the technical infrastructure directly addresses the inefficiencies of the manual workflow, transforming the SPMS into a reliable and secure institutional tool.

Hardware Specifications

The hardware specifications define the minimum and recommended computing resources required to ensure the SPMS operates efficiently. These components are allocated to support both the development process and the actual deployment of the system for HRDO personnel, supervisors, and employees.

Table 1. Hardware Specifications for Development

Hardware/Specification	Description
Device	Laptop or Desktop Computer
Memory (RAM)	At least 8 GB
Storage	At least 256 GB SSD
Processor	Intel Core i3 / AMD Ryzen 3 or higher
Internet Connection	Stable internet connection for updates, testing, and collaboration

Table 2. Hardware Specifications for Deployment or Actual Use

Hardware/Specification	Description
User Devices	Desktop computers or laptops for BSU personnel, supervisors, and employees
Backup Device	External storage or backup device for data backup and recovery
Server/Hosting Device	Server computer or hosting device
Memory (RAM)	At least 4 GB
Storage	At least 10 GB of free disk space
Processor	Intel Core i5 / AMD Ryzen 5 or higher
Internet Connection	Stable internet connection for system access, updates, and data synchronization

Software Specifications

The selected frontend technologies, JavaScript, Tailwind CSS, and Axios, are highly compatible with one another, forming a connected and efficient development stack for the SPMS. JavaScript serves as the core runtime language in the browser, allowing both Tailwind CSS and Axios to integrate seamlessly within the same environment without requiring additional compilation layers. Tailwind CSS is purely a utility-first styling framework that works directly within HTML and JavaScript-driven components, making it fully compatible with JavaScript-based frontend structures such as dynamic forms and dashboards. It does not interfere with application logic, which ensures clean separation between design and functionality.

Table 3. Frontend Technologies Used

Technology/Software		Description	Justification
JavaScript ECMAScript (ES2025)	2025	This is the main programming language used for creating interactive and dynamic elements of the system such as buttons, forms, dashboards, and real-time user actions.	JavaScript will be used as it is the language the development team is most accustomed to, ensuring smoother development without external dependency issues. It remains the industry standard for immediate browser compatibility and rapid iteration, maintaining modern usability standards without the overhead of additional

		compilation steps (Hrebenyk et al., 2025; Eternitech, 2025).
Tailwind CSS v4.2.4	Used for styling and responsive design, this helps make the system look clean and visually appealing	Tailwind CSS will be selected to provide a lightweight UI architecture and granular design control, avoiding the generic appearance of traditional frameworks (Parul, 2024). It will also provide a fast, consistent design language using utility classes that offer superior flexibility for customized design iterations compared to Bootstrap (Zaveri, 2024).
Axios v1.15.1	This tool helps the frontend and backend communicate with each other to send and receive data smoothly.	Axios will be implemented to reduce boilerplate code and handle asynchronous requests more efficiently than the native Fetch API (Khatri, 2023). It is further chosen for its simplified, reliable data exchange capabilities, including automatic JSON transformation and robust error-handling mechanisms (Mishchenko, 2024).

Table 4 presents the backend technologies selected for the server-side development and data management components of the system. CodeIgniter 4 was chosen as the primary backend framework to comply with the BSU ICT

department's server compatibility requirements. For data storage, SQLite 3 managed through Laragon was adopted as the relational database engine. It offers a self-contained, file-based structure that supports high-speed data processing through Write-Ahead Logging configurations and allows straightforward backup procedures without requiring a separate server installation (SQLite, 2024; Sam700007, 2023).

The combination of CodeIgniter 4 and SQLite 3 provides a reliable, efficient, and locally deployable backend infrastructure appropriate for the scale and institutional context of the SPMS. Furthermore, the built-in security features of CodeIgniter 4, such as CSRF protection and data encryption, provide a robust defense for sensitive employee performance records stored within the database. By utilizing this specific technology stack, the system achieves a balance between high-speed data retrieval and low server overhead, which is essential for maintaining stability during peak academic evaluation periods.

Table 4. Backend Technologies Used

Technology/Software	Description	Justification
CodeIgniter4 v4.7.2	This handles the system's operations behind the scenes, such as processing data and managing user accounts.	CodeIgniter 4 will be used primarily to meet the ICT department's requirements for system compatibility and easier integration with existing servers. It offers high speed and reduced development time while providing the necessary security and organization for database-driven institutional management. Furthermore, it is specifically effective for small-scale projects requiring seamless RESTful API integration. (Ahmed et al., 2024; Putra et al., 2020).
SQLite3 v3.49.0 (used with Laragon)	This is where all the system's information is stored in an organized and secure way. Laragon is a tool that helps manage the database and other development components.	SQLite 3 will be optimized for high-speed relational data processing through specialized configurations like Write-Ahead Logging (Sam700007, 2023). It will also be used because it is a self-contained engine that allows for easy backups through simple file copying while maintaining strict relational integrity (SQLite, 2024).

METHODOLOGY

This chapter presents the methodology that will be used in the development of the SPMS. It describes the systematic procedures followed to design, develop, and evaluate the system in an organized manner. The methodology is structured to address the specific objectives of the study, which include determining system requirements, designing the system architecture and database, developing the application, and evaluating the system in terms of functionality, usability, and performance.

Software Development Life Cycle Paradigm

The development of the SPMS utilizes the Feature-Driven Development methodology. FDD is an agile framework that focuses on building software through small, client-valued, and manageable features. This methodology was specifically chosen because the SPMS is a highly modular system, containing distinct but interconnected evaluation tools (e.g., IPCR, DPCR, OPCR, IPERF) that require constant, iterative validation from different stakeholders, including HRDO personnel, supervisors, and individual employees.

Unlike traditional, linear development models where testing only occurs at the very end, FDD allows the development team to build, present, and validate the system incrementally. This ensures that the specific workflow rules for each

university role are accurately captured and refined before being integrated into the final, complete system.

The FDD methodology operates through five sequential phases: Develop an Overall Model, Build a Feature List, Plan by Feature, Design by Feature, and Build by Feature (O'Connor, 2025). This structured approach directly dictates how the development team addresses the operational bottlenecks identified in Chapter 1. By breaking down the complex, paper-based workflow into specific digital features, the team systematically resolves the core issues of the manual system, such as delayed document routing, mathematical miscalculations, and the lack of real-time transparency.

The five processes of the Feature-Driven Development methodology that will guide the development of the proposed web-based SPMS are presented in Figure 4.

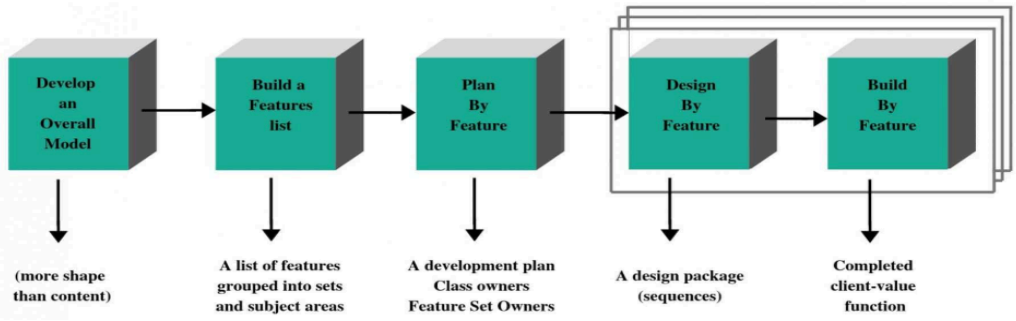


Figure 4. Feature-Driven Development Process Model

Source: ToolsQA. (2021). *Feature Driven Development (FDD) process model diagram*. In Agile Methodology: Feature Driven Development. https://www.toolsqa.com/agile/feature-driven-development/

Phase 1. Develop Overall Model

Objective: To map the existing manual workflow and define the system's structural blueprint.

This phase translates the operational problems identified in Chapter 1 into high-level digital requirements. Three core problem areas guided the modeling process: (1) the absence of automated rating computation, which causes recurring arithmetic errors during evaluation; (2) the sequential document routing bottleneck, which delays the entire SPMS cycle when any upper-level office is late; and (3) the lack of a centralized platform, which prevents employees and HRDO from tracking submission status in real time.

Drawing from these identified gaps, the researchers will establish an initial system model involving three core user roles: HRDO personnel, Supervisors, and Employees. The needs and responsibilities of each role will serve as the basis for identifying the requirements of the SPMS. The outputs of this phase will be the FDD Requirements Overview Template and the System Requirements, which will include the functional and non-functional requirements of the system. These outputs will serve as the basis for identifying, organizing, and prioritizing the system features in the succeeding phase.

Phase 2. Build Feature List

Objective: To decompose the high-level requirements into actionable and manageable software features directly related to the identified operational needs.

In this phase, the researchers will use the FDD Requirements Overview Template and the System Requirements from Phase 1 as the basis for identifying and organizing the software features. The high-level requirements will guide the identification of the main features, while the functional and non-functional requirements will help determine the specific functions and system constraints that need to be addressed. Each identified feature will be organized according to its purpose and relevance to the needs of HRDO personnel, supervisors, and employees. The researchers will also assign priority levels to the features based on their importance to the system's core functions, operational needs, and performance evaluation process. The output of this phase will be the Feature List, which will serve as the basis for planning the development sequence and assigning tasks in Phase 3.

Phase 3. Plan by Feature

Objective: To establish the development schedule and assign iterations based on feature dependencies.

In this phase, the feature list developed in Phase 2 will be used to organize the planned development tasks of the SPMS. The researchers will use Trello as a project management tool to arrange tasks, group related features into development groups, and organize work according to their functions and dependencies. According to Trello (2026), its boards, lists, and cards help users organize tasks and manage project activities, making it suitable for planning software development work.

The researchers will use Trello to organize the work for the succeeding development phases. Related features will be grouped according to their primary functions to establish a systematic structure for managing the planned development activities. The Trello board will contain lists representing the task workflow, while cards will represent the main development tasks. Checklists within each card will identify the corresponding system features to be addressed. This organization will help the researchers monitor task progress and identify activities that are pending, ongoing, awaiting testing, under review, or completed.

The input for this phase will be the feature list developed in Phase 2, while the output will be the organized development plan in Trello, illustrated in Figure 5. This presents the format of the Trello board, including the lists and cards used to organize and manage the planned development tasks of the SPMS.

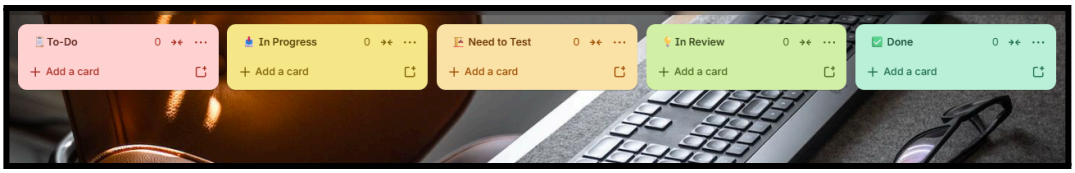


Figure 5. Trello Board for SPMS Feature Development Planning

Phase 4. Design by Feature

Objective: To design the user interfaces, process logic and data structures for each scheduled feature before coding begins.

Process & Output: This phase focuses on the structural and visual design of the system. To ensure a cohesive and organized structure, the system is designed following a three-layer Model-View-Controller (MVC) architecture:

- Data Layer (Model): Structuring the underlying database. An Entity-Relationship Diagram (ERD) will be prepared to illustrate the relationships among SPMS entities, such as users, user roles, roles, units, positions, plantillas, documents, document folders, document attachments, evaluation routings, activity logs and notifications. A database schema will also be developed to specify the planned database tables, columns, data types, constraints, default values and descriptions.
- Presentation Layer (View): Designing the frontend interfaces, such as the digital forms and role-specific dashboards where HRDO, supervisors, and employees will interact.

- Application Layer (Controller): This layer focuses on defining the backend logic required to process user inputs, manage system workflows, enforce access permissions, and compute performance ratings. The process flow of important system functions will be planned to show how information moves between the user interface, application logic, and database.

  In addition, a use case diagram will be prepared to illustrate the functions available to HRDO personnel, supervisors, and employees. This diagram will provide an overview of how each user role interacts with the SPMS.

Phase 5. Build by Feature

Objective: To develop, integrate, and test the designed features of the SPMS and evaluate based on user feedback.

This phase focuses on implementing the system features based on the requirements and designs established in the preceding phases. The researchers will develop the SPMS through three major iterations: Foundation, Core Modules, and Refinement. Each iteration will involve coding, feature integration, and testing to identify and correct errors before proceeding to the next set of features.

- Iteration 1 (Foundation): Focuses on implementing the core infrastructure of the SPMS. This includes setting up the SQLite database, implementing secure user authentication, and configuring Role-Based Access Control

middleware. These features will establish the system's data storage and access control mechanisms.

- Iteration 2 (Core Modules): Focuses on implementing the primary SPMS modules such as OPCR, DPCR, IPCR and IPERF. The researchers will implement the functions for entering performance targets and accomplishments, submitting forms, reviewing performance records and processing evaluations according to each user's assigned role. Automated rating computation will also be implemented based on the applicable SPMS rating rules and prescribed criteria for Quality, Efficiency, and Timeliness.
- Iteration 3 (Refinement): Focuses on implementing reporting and monitoring dashboards, automated notification functions, and improvements to the user interface and overall usability of the working system.

Testing will be conducted throughout the development process. Each feature will be tested after implementation to identify errors and verify that it meets its intended requirements. The researchers will perform unit testing, integration testing, and system testing, followed by User Acceptance Testing (UAT).

- Unit Testing: It will be used to verify individual SPMS components, focusing on user authentication, input validation, automated rating computation, database operations, and role-based access control.

Table 5. Unit Testing Plan Template

Feature	Test Procedure	Expected Result	Actual Result	Status PASS/FAIL
User Authentication	Enter valid and invalid login credentials.	The system accepts valid credentials and rejects invalid credentials.		
Input Validation	Submit forms with missing or invalid required fields.	The system identifies invalid inputs and provides appropriate validation messages.		
Automated Rating Computation	Enter test values for the applicable rating criteria and compare the computed result with the expected calculation.	The system calculates ratings according to the implemented SPMS rules.		

Database Operations	Save, retrieve, and update test records	The system processes the records correctly and preserves the stored information.
Role-Based Access Control	Attempt to access functions using accounts with different assigned roles.	The system permits or restricts access according to the assigned permissions.

- Integration Testing: Integration testing will be conducted to verify the interaction between SPMS modules, focusing on authentication and role-based access control, performance forms and database operations, form submission and supervisor review, rating computation, monitoring and notifications, and report generation.

Table 6. Integration Testing Plan Template

Feature	Test Procedure	Expected Result	Actual Result	Status
Authentication and Role-Based Access Control	Log in using accounts assigned to different roles and navigate to their	Each user is directed to the appropriate pages and can access		

	permitted pages.	only authorized functions.
Performance Forms and Database	Enter information in an applicable performance form and save the record.	The entered information is stored and retrieved correctly.
Form Submission and Supervisor Review	Submit a performance form and check whether it becomes available for the appropriate review process.	The submitted form is recorded and made available to the authorized reviewer according to the workflow.
Rating Computation and Evaluation Records	Enter or update rating information and examine the resulting evaluation record.	The computed rating and related information are reflected correctly in the record.
Monitoring and Notifications	Perform an action that should update evaluation status or	The applicable status and notification are updated according to

trigger a notification.	the implemented rules.

- System Testing: System testing will be conducted to evaluate the overall functionality of the SPMS, focusing on access control and security, organizational unit management, evaluation cycles and deadlines, target cascading, target validation and approval, MOV evidence handling, automated rating computation, PMT calibration and audit, performance monitoring, and report generation with Excel export.

Table 7. System Testing Plan Template

Feature	Test Scenario	Expected Result	Actual Result	Status
Access Control, Authentication, and Security	Log in using authorized and unauthorized credentials and check access permissions.	The system authenticates users and enforces the applicable access restrictions.		
Organizational Unit and Department Transfer	Access organizational records and examine applicable office or department assignments.	Organizational information is displayed and processed according to the user's permissions		

		and implemented rules.
Semester Evaluation Cycle and Deadline	Check the evaluation period, deadline information, and evaluation status.	The system displays the configured cycle, deadlines, and applicable status information.
Institutional Target Cascading and Performance Formulation	Enter or review applicable targets and performance forms.	The system supports the implemented target-setting and performance -formulation workflow.
Target Weight Validation and Approval	Test the applicable target validation and approval functions.	The system applies the implemented validation and approval rules.
MOV Evidence and Encryption	Attach or access supporting evidence using the applicable functions.	Evidence is handled and protected according to the implemented security mechanisms.

Automated CSC Scoring and Rating	Enter test ratings and compare the resulting calculations with the expected values.	The system calculates and displays ratings according to the implemented scoring rules.
PMT Calibration and Audit	Examine the applicable calibration functions and evaluation records.	Authorized users can access the implemented calibration and audit functions.
Executive Compliance and Analytics	Access the available monitoring dashboard and examine its displayed information.	The system displays the implemented compliance and performance monitoring information.
Official Performance Report and Excel Export	Generate a performance report and export eligible records.	The system generates the appropriate report and exports the selected information correctly.

- User Acceptance Testing (UAT): User Acceptance Testing will be conducted to determine whether the SPMS meets the identified functional requirements. The test scenarios will cover the ten feature groups listed in the table, including access control, authentication and security, organizational unit management, evaluation cycle and deadline management, institutional target cascading, target validation and approval, MOV evidence handling, automated rating computation, PMT calibration and audit, executive compliance and analytics, and performance report generation with Excel export. Each scenario will be checked against its expected result to assess whether the system functions according to the specified requirements.

Table 8. User Acceptance Testing Template

Functional Requirement ID	Feature Group	UAT Test Scenario	Expected Result	Actual Result	Status
FR 1-11	Access Control, Authentication, and Security Governance	Log in using authorized and unauthorized credentials and check the functions available to each role.	The system authenticates users and applies the appropriate access permissions and security controls.		

FR 12-18	Organizational Unit and Department Transfer Management	Examine organizational records and test the applicable department assignment or transfer functions.	The system processes organizational records and department transfers according to the implemented rules.
FR 19-26	Semester Evaluation Cycle and Deadline Management	View the evaluation cycle, submission deadlines, evaluation deadlines, and completed evaluation records.	The system displays the applicable cycle and deadlines and handles completed evaluation records according to the implemented rules.
FR 27-37	Institutional Target Cascading and Performance Formulation	Access the applicable OPCR, DPCR, IPCR, or IPERF form and test target entry, cascading, and submission.	The system supports the applicable target-setting and performance-formulation workflow.
FR 38-45	Target Weight Validation and Approval Management	Test the applicable target validation and approval functions.	The system applies the configured validation and approval rules.

FR 46-53	MOV Evidence and Encryption	Attach or access supporting evidence using the available functions.	The system processes supporting evidence according to the implemented functions and security controls.
FR 54-60	Automated CSC Scoring and Rating	Enter test rating values and compare the computed results with the expected calculations.	The system calculates and displays ratings according to the implemented scoring rules.
FR 61-68	PMT Calibration and Audit	Examine the applicable calibration functions, evaluation records, and audit information.	Authorized users can access the applicable calibration and audit functions.
FR 69-74	Executive Compliance and Analytics	Access the available monitoring dashboard and examine the displayed compliance and performance information.	The system displays the implemented monitoring and analytics information correctly.

System Evaluation

After ensuring that the developed Strategic Performance Management System (SPMS) functions correctly through system testing, the researchers will evaluate its usability using the Post-Study System Usability Questionnaire (PSSUQ). The PSSUQ is a standardized 16-item questionnaire developed to measure users' perceived satisfaction after interacting with a system. It evaluates usability based on three dimensions: system usefulness, information quality, and interface quality. The instrument is widely used for assessing the overall usability of information systems because it measures how effectively users can accomplish tasks, the quality of information provided by the system, and the ease of using the system interface.

The evaluation will involve thirty (30) respondents from Benguet State University (BSU) who are directly involved in the implementation and use of the Strategic Performance Management System. The respondents will consist of six (6) HRDO personnel who serve as OPCR and DPCR consolidators, eight (8) supervisors composed of Deans, Directors, or Heads of Offices, twelve (12) regular employees who accomplish the Individual Performance Commitment and Review (IPCR), and four (4) Contract of Service (COS) personnel who accomplish the Individual Performance Evaluation and Review Form (IPERF). These respondents represent the primary users of the system and are expected to provide reliable feedback based on their actual responsibilities and experiences.

The study will employ purposive sampling in selecting the respondents. This sampling technique is appropriate because the evaluation requires participants who possess firsthand knowledge and experience in the Strategic Performance Management System process. Only individuals who regularly perform SPMS-related activities are capable of evaluating whether the developed system effectively supports their assigned functions. Through purposive sampling, the researchers can ensure that all major user groups of the SPMS are adequately represented during the evaluation.

To qualify as respondents, participants must meet the following inclusion criteria: (1) they must be currently employed at Benguet State University; (2) they must belong to one of the identified SPMS user groups, namely HRDO personnel, supervisors, regular employees, or Contract of Service personnel; (3) they must have actual responsibilities involving the preparation, submission, review, consolidation, monitoring, or evaluation of SPMS documents; and (4) they must be willing to participate in the usability evaluation of the developed system.

During the evaluation, respondents will use the SPMS to perform realistic tasks such as submitting IPCR forms, reviewing OPCR targets, and generating performance reports. This procedure ensures that the assessment is based on actual system interaction and practical user experience.

After completing the tasks, respondents will answer the PSSUQ questionnaire using a 7-point Likert scale, where 1 represents “Strongly Agree”

and 7 represents “Strongly Disagree.” Lower scores indicate better usability and higher user satisfaction. The completed questionnaires will then be collected and analyzed by the researchers.

The gathered data will be analyzed using the weighted mean to determine the usability and acceptability of the system in terms of system usefulness, information quality, and interface quality. The computed mean scores will be interpreted using the descriptive scale shown in Table 19, where lower mean scores indicate higher usability and greater user satisfaction.

To evaluate the system’s effectiveness and user satisfaction, the researchers will follow these procedures:

1. Selected HRDO personnel, supervisors, regular employees, and COS personnel will perform actual SPMS-related tasks using the system.
2. After the testing session, respondents will answer the 16-item PSSUQ questionnaire.
3. Responses will be rated using a 7-point Likert scale ranging from “Strongly Agree” to “Strongly Disagree.”
4. The weighted mean will be computed for each evaluation dimension and for the overall usability of the system.
5. The computed results will be interpreted using the descriptive equivalence scale to determine the level of usability and user satisfaction.

The formula for the weighted mean is shown below:

$$X = \frac{\Sigma(w_i x_i)}{\Sigma w_i}$$

Where:

- x_i = each value (e.g., ratings or scores)
- w_i = weight of each value
- $\sum(w_i x_i)$ = sum of each value multiplied by its weight
- $\sum w_i$ = total of all weights

For the purpose of interpretation, the computed mean scores are categorized into usability levels as follows as presented in Table 9:

Table 9. PSSUQ Mean Score Interpretation Guide for System Usability

Point	**Range**	**Descriptive Equivalence**
1	1.00 - 1.86	Strongly Agree
2	1.87 - 2.72	Agree
3	2.73 - 3.58	Somewhat Agree
4	3.59 - 4.44	Neutral
5	4.45 - 5.30	Somewhat Disagree
6	5.31 - 6.16	Disagree
7	6.17 - 7.00	Strongly Disagree

Table 9 presents the descriptive equivalence scale used to interpret the computed mean scores from the PSSUQ. Since lower scores in the PSSUQ indicate better usability, mean scores falling within the “Strongly Agree” and “Agree” ranges signify high user satisfaction and excellent system usability, while higher scores indicate areas that may require improvement. The interpretation ranges were adapted to provide clearer descriptive analysis while maintaining the original principle of the PSSUQ that lower scores correspond to better usability.

The first dimension, system usefulness, measures how effectively the SPMS helps users accomplish performance evaluation tasks such as submitting IPCR forms, reviewing OPCR targets, and generating reports. Lower scores in this dimension indicate that the system improves efficiency and reduces the time and effort required to complete tasks.

The second dimension, information quality, evaluates the clarity, accuracy, and organization of information presented in the system, including performance ratings, supervisor feedback, and generated reports. Good information quality supports reliable decision-making and accurate evaluation processes.

The third dimension, interface quality, measures the ease of navigation and overall user-friendliness of the system. This includes the consistency of layouts, organization of forms and dashboards, responsiveness of the interface, and clarity of system prompts and error messages. Lower scores in this dimension indicate that the system is easy to use and requires minimal user training.

RESULTS AND DISCUSSION

This chapter presents the results in this study. It discusses the requirements analysis, the design and development of the system and the evaluation of the system's usefulness, information quality, and interface quality.

Requirements Analysis

The requirements analysis began during Develop an Overall Model, when the researchers examined the existing SPMS process at Benguet State University. This involved identifying the problems encountered in the current process and determining the requirements needed to address them through the SPMS. The researchers analyzed the Strategic Performance Management System process at Benguet State University to identify the operational problems that needed to be addressed by the SPMS. The analysis identified five main problems.

- First, miscalculation of performance ratings may occur because the ratings in the OPCR, DPCR, IPCR and IPERF forms are manually encoded and computed at different levels, increasing the possibility of errors and repeated computations.
- Second, slow document processing results from the sequential flow of performance targets and forms across different offices where delays at one level can affect the succeeding offices.

- Third, excessive use of paper and printing materials increases printing costs and requires considerable storage space for physical records.
- Fourth, laborious reporting and tracking makes it difficult to prepare, verify and consolidate performance reports efficiently, especially when urgent reports are needed.
- Lastly, limited transparency makes it difficult for employees, supervisors and the HRDO personnel to monitor submission statuses, access previous performance records and track the progress of the evaluation process. These problems highlighted the need for a centralized web-based system that could automate rating computations, improve document processing, reduce paper usage, simplify reporting, and provide better monitoring of performance records.

Based on these findings, the researchers identified HRDO personnel, supervisors and employees as the primary users of the SPMS. Their responsibilities in preparing, submitting, reviewing, evaluating and managing performance records served as the basis for identifying the major system modules and their high-level requirements. These requirements were organized using the FDD Requirements Overview Template, as presented in Table 10.

Table 10. FDD Requirements Overview Template

Module Name	Target User	High-Level Requirement
Performance Planning	HRDO Personnel	The system must allow administrators to set university-wide targets and cascade them downwards
Evaluation and Rating	Supervisors	The system must provide a digital interface to review employee targets and compute final ratings
Self-Assessment	Employees	The system must allow personnel to digitally input their IPCR/IPERF accomplishments

Table 10 presents the FDD Requirements Overview Template, which summarizes the high-level requirements of the SPMS according to its main modules and target users. The Performance Planning module is intended for HRDO personnel to establish university-wide targets and cascade them to the different offices and personnel. The Evaluation and Rating module supports supervisors in reviewing employee targets and computing performance ratings. Meanwhile, the Self-Assessment module allows employees to enter their IPCR and IPERF accomplishments digitally. These requirements define the primary responsibilities of each module and serve as the basis for identifying the system's functional and non-functional requirements. The functional requirements specify

the system's expected functions and are organized by subsystem, as presented in Table 11.

Table 11. Functional Requirements by Subsystem

Subsystem 1: Identify, Authentication and Security Governance

Requirement ID	Functional Requirement
FR-1.1	The system shall authenticate personnel using their official @bsu.edu.ph institutional email and encrypted password credentials.
FR-1.2	The system shall validate automated Cloudflare Turnstile CAPTCHA tokens on every login and registration attempt to prevent scripted attacks.
FR-1.3	The system shall enforce Two-Factor Authentication (2FA) by generating and emailing a time-delimited 6-digit One-Time Password (OTP) with a strict 5-minute validity window.
FR-1.4	The system shall enforce brute-force throttling: 5 consecutive invalid credentials from an IP address or device fingerprint shall trigger a 15-minute authentication lockout.
FR-1.5	The system shall support a 30-day "Remember Me" session issued via a cryptographically random token whose SHA-256 hash is validated at the database layer.
FR-1.6	The system shall provide a 3-step secure email update workflow requiring password re-verification, new address verification code dispatch, and confirmation.
FR-1.7	The system shall facilitate password recovery via single-use, 60-minute cryptographic tokens delivered to verified institutional inboxes.

Subsystem 2: Organizational Hierarchy and Academic Department Transfers

Requirement ID	Functional Requirement
FR-2.1	The system shall model the authentic hierarchical structure of Benguet State University, including the 15 constituent colleges, executive divisions (OP, OVPAA, OVPAF, HRDO), and academic departments.
FR-2.2	The system shall provide an administrative workflow for reassigning academic departments between colleges with mandatory Board of Regents (BOR) justification logging.
FR-2.3	Department transfers shall update current and future reporting trees while maintaining historical evaluation records under their original parent colleges.
FR-2.4	The system shall maintain Plantilla positions, distinguishing teaching positions (eligible for IPCR) from non-teaching positions (eligible for IPERF).

Subsystem 3: Semester Evaluation Cycle and Deadlines Governance

Requirement ID	Functional Requirement
FR-3.1	Administrators shall create semestral evaluation cycles specifying Academic Year, Semester term, and distinct deadlines for target formulation and accomplishment evaluation.
FR-3.2	The system shall automatically compute time-based folder states and enforce target and accomplishment locking when institutional deadlines expire.

FR-3.3	The system shall synchronize all operational cutoffs and audit timestamps with Philippine Standard Time (PST / UTC+8) in compliance with Republic Act No. 10535.
FR-3.4	Completed cycles shall support administrative archiving, permanently freezing underlying forms against further alteration.

Subsystem 4: Tier Target Cascading and Superior Basis

Requirement ID	Functional Requirement
FR-4.1	The system shall enforce a top-down 4-tier target cascade: VPAA (OPCR) -> College Deans (DPCR) -> Department Chairs (DPCR) -> Faculty / Staff (IPCR / IPERF).
FR-4.2	Ratees shall formulate targets categorized into Core Functions, Strategic Priorities, and Support Functions.
FR-4.3	The system shall provide a side-by-side Superior Basis split pane displaying the immediate supervisor's approved commitments as a live reference during drafting.
FR-4.4	The system shall save matrix modifications automatically in real time using an 800ms debounced background autosave service.
FR-4.5	The system shall provide a Revoke Submission function allowing ratees to pull back submitted targets into draft mode prior to supervisor approval.

Subsystem 5: Statutory Weight Balancing and Approval

Requirement ID	Functional Requirement
FR-5.1	For IPCR and DPCR forms, the system shall validate that the sum of percentage weights across Core, Strategic, and Support functions equals exactly 100.0%.
FR-5.2	If the total category weight does not equal 100%, the system shall display an alert and disable the Submit Targets button.
FR-5.3	For IPERF forms (Job Order / COS personnel), deliverables shall automatically lock to 100% Committed Outputs.
FR-5.4	Supervisors shall approve targets with a single action (Approve Target), which locks commitments and authorizes accomplishments reporting.
FR-5.5	Supervisors shall return targets for revision (Return Target for Revision) with inline remarks preserved directly in the Remarks column.

Subsystem 6: Evidence Management (Means of Verification / MOV)

Requirement ID	Functional Requirement
FR-6.1	Ratees shall attach verifiable evidence per deliverable row in the slide-out MOV Evidence Drawer.
FR-6.2	The system shall support local file uploads (PDF, DOCX, XLSX, PNG, JPG) up to 20 MB per file.
FR-6.3	Uploaded evidence files shall be encrypted at rest using the AES-256 cipher in the secure server storage directory.

FR-6.4	The system shall support external cloud links from Google Drive, Microsoft OneDrive, SharePoint, and Dropbox.
FR-6.5	The system shall provide an in-app document viewer allowing supervisors to inspect evidence without downloading files locally.
FR-6.6	The system shall allow exporting pre-filled official rubric templates and uploading completed rubric matrices.

Subsystem 7: Automated CSC Scoring Engine (Q-E-T Dimensions)

Requirement ID	Functional Requirement
FR-7.1	Deliverables shall be scored across three dimensions: Quality (Q), Efficiency (E), and Timeliness (T) on an official 1.00 to 5.00 scale.
FR-7.2	The system shall automatically compute row averages: Row Average = (Q + E + T) / 3.
FR-7.3	Dimension Gating (N/A): If a dimension is toggled N/A, the average recalculates across active dimensions without penalizing the ratee.
FR-7.4	The system shall calculate composite scores using category percentage weights: Final Rating = (Core Ave * W_core) + (Strategic Ave * W_strat) + (Support Ave * W_supp).
FR-7.5	The system shall map numerical scores to official CSC adjectival ratings: Outstanding (4.500 to 5.000), Very Satisfactory (3.500 to 4.499), Satisfactory (2.500 to 3.499), Unsatisfactory (1.500 to 2.499), Poor (< 1.500).

Subsystem 8: PMT Institutional Calibration and Forensic Audit

Requirement ID	Functional Requirement
FR-8.1	The PMT and TWG shall review university-wide score distributions to prevent grade inflation or compression.
FR-8.2	Calibrated score adjustments shall require mandatory justification notes and append an immutable Audit Dot displaying original vs. adjusted scores.
FR-8.3	Every mutating action (logins, submissions, evaluations, score overrides, invitations) shall be recorded in an immutable forensic audit log with user ID, IP address, user agent, and PST timestamp.

Subsystem 9: Executive Analytics and Institutional Compliance

Requirement ID	Functional Requirement
FR-9.1	The system shall provide institutional dashboards tracking completion progress across all 15 colleges in real time.
FR-9.2	The system shall calculate Plantilla compliance percentages against active campus employee rosters.
FR-9.3	Administrators shall export university-wide compliance masterlists in Excel (.xlsx) and CSV formats.

Subsystem 10: Statutory Report Generation and Excel Exporter

Requirement ID	Functional Requirement

FR-10.1	The system shall generate official Microsoft Excel workbooks (.xlsx) compliant with CSC MC No. 6, s. 2012 Annexes (IPCR Annex A, DPCR Annex B, OPCR Annex C).
FR-10.2	Exported Excel workbooks shall embed active Excel formulas (AVERAGE, SUM, weighted multipliers) and official BSU letterhead headers.
FR-10.3	Exported forms shall include standard CSC signature blocks for Ratee, Immediate Supervisor, and Head of Agency.

The functional requirements are organized into ten subsystems, each addressing specific operational needs of the SPMS. Together, these subsystems define the system's required functions and serve as the basis for its design and development.

Meanwhile, Table 12 presents the non-functional requirements guided by the ISO/IEC 25010 standards, specifying the expected quality characteristics of the SPMS.

Table 12. Non-functional Requirements under ISO/IEC 25010 Standards

NFR ID	Quality Characteristic	Requirement Specification	Metric/Benchmark
NFR-01	Performance Efficiency	The system shall render matrix rows and process user requests within standard campus latency limits under peak loads.	Page load

NFR ID	Quality Characteristic	Requirement Specification	Metric/Benchmark
NFR-02	Autosave Responsiveness	The background autosave service shall debounce user keystrokes to prevent database saturation while eliminating loss of work.	Debounce 800ms; payload
NFR-03	Data-at-Rest Security	Uploaded evidentiary documents (MOVs) containing sensitive personnel or academic records must be encrypted on disk.	AES-256-CTR / GCM cipher via Encrypter
NFR-04	Credential Protection	User authentication credentials must never be stored in plaintext. Session tokens must be cryptographically hashed.	Bcrypt cost 10; SHA-256 for remember tokens
NFR-05	Attack Mitigation	The application shall defend against common web application vulnerabilities outlined in OWASP Top 10 guidelines.	Parameterized queries, CSRF tokens, esc() escaping
NFR-06	Reliability and Recovery	Multi-row evaluation operations, cascade folder generation, and score recalculations must maintain atomic consistency.	ACID SQLite transactions (BEGIN/COMMIT)

NFR ID	Quality Characteristic	Requirement Specification	Metric/Benchmark
NFR-07	Usability and Operability	Faculty members with varying technical proficiency shall complete target formulation without external technical assistance.	Standard table layouts, contextual prompts, human errors
NFR-08	Time Synchronization	All audit trails, semestral deadline cutoffs, and system alerts must synchronize with the official national time standard.	PST / UTC+8 synchronization under RA 10535
NFR-09	Responsive Viewport	The user interface shall adapt and render clearly across common office screen resolutions and tablet computers.	Responsive layout supporting 1280x720 to 1920x1080
NFR-10	Modularity and Maintainability	The software architecture shall adhere to MVC design principles, separating routing, business logic, and presentation.	Modular services for Auth, Cascading, Scoring, Excel.

Table 12 presents the ten non-functional requirements of the SPMS under the ISO/IEC 25010 standards. These requirements define the technical

specifications and quality characteristics that guide the system's operation, data protection, and software structure.

Performance Efficiency requires the system to render matrix rows and process user requests within standard campus latency limits, including during peak loads. Autosave Responsiveness specifies an 800-millisecond debounce for user keystrokes to help prevent database saturation while avoiding the loss of user input. For data protection, Data-at-Rest Security requires uploaded Means of Verification (MOVs) containing sensitive personnel or academic records to be encrypted on disk using AES-256-CTR or GCM through the Encrypter. Credential Protection prohibits storing authentication credentials in plaintext and specifies Bcrypt with a cost factor of 10 and SHA-256 hashing for remember tokens. Attack Mitigation requires safeguards against common web application vulnerabilities identified in the OWASP Top 10 guidelines, using parameterized queries, CSRF tokens, and esc() escaping.

Reliability and Recovery requires multi-row evaluation operations, cascade folder generation, and score recalculations to maintain atomic consistency through ACID SQLite transactions using BEGIN and COMMIT. Usability and Operability aims to help faculty members with varying levels of technical proficiency complete target formulation without external technical assistance, using standard table layouts and contextual prompts. Time Synchronization requires audit trails, semestral deadline cutoffs, and system alerts to follow the

official national time standard, specified as Philippine Standard Time (PST) or UTC+8 under Republic Act No. 10535. Responsive Viewport requires the interface to adapt to common office screen resolutions ranging from 1280×720 to 1920×1080, as well as tablet computers. Lastly, Modularity and Maintainability requires the software to follow the Model-View-Controller design principles, separating routing, business logic, and presentation, with modular services for authentication, cascading, scoring, and Excel operations.

Requirements Documentation

This section discusses Phase 2 of the Feature-Driven Development methodology, which focuses on identifying and organizing the core features of the SPMS. Based on the overall model developed in Phase 1 and the established system requirements, the researchers identified the necessary features and grouped them into categories and subcategories. This organization provides a structured overview of the system's functions and serves as a reference for the succeeding phases of development. The identified features are presented in the following list.

I. Access Control, Authentication and Security Governance

A. Managing User Authentication and Verification

1. Restricting Registration and Invitations to BSU Institutional Email Addresses

2. Resetting Forgotten Passwords using Time-Limited Reset Tokens
3. Verifying Users through Cloudflare Turnstile CAPTCHA
4. Sending Six-Digit Email One-Time Passwords for Two-Factor Authentication

B. Securing User Accounts

5. Temporarily Locking Accounts after Five Failed Login Attempts
6. Keeping users signed in through secure 30-day Remember-Me Tokens
7. Verifying user identity before changing email addresses
8. Requiring current password verification before changing passwords

C. Managing User Roles and Permissions

9. Assigning Permissions Based on Role-Based Access Control
10. Controlling System Access Based on Plantilla Positions and Organizational Levels
11. Supporting Admin, Supervisor, Employee, Technical Working Group and Performance Management Team Roles

II. Organizational Unit and Department Transfer Management

A. Managing Organizational Records

12. Registering the Fifteen Official BSU Colleges and Executive Divisions
13. Organizing Colleges and Departments Using Parent-Child Relationships
14. Viewing the University Organizational Hierarchy

B. Processing Department Transfers

15. Reassigning Departments from One College to Another
16. Recording Board of Regents Justifications for Department Transfers
17. Updating Organizational Relationships for Future Target Cascading
18. Preserving Historical Organizational Records for Completed Semesters

III. Semester Evaluation Cycle and Deadline Management

A. Managing Evaluation Cycles

19. Creating Academic Year and Semester Evaluation Cycles
20. Setting Target Submission Deadlines
21. Setting Accomplishment Rating Deadlines

B. Monitoring Deadlines and Evaluation Status

22. Synchronizing System Time with Philippine Standard Time

23. Automatically Updating Target Submission Status Based on the Evaluation Schedule
24. Automatically Updating Evaluation Period Status Based on the Evaluation Schedule

C. Archiving Completed Evaluation Cycles

25. Freezing Completed Evaluation Folders to Prevent Further Changes
26. Preserving Read-Only Archives of Completed Evaluation Records for Civil Service Compliance Audits

IV. Institutional Target Cascading and Performance Formulation

A. Managing Institutional Target Cascading

27. Generating Subordinate Performance Folders Based on the University Organizational Hierarchy
28. Cascading OPCR Targets to DPCR and IPCR/IPERF Forms
29. Viewing Approved Performance Commitments of Immediate Supervisors

B. Managing Standardized Performance Forms

30. Preparing IPCR Forms
31. Preparing IPERF Forms
32. Preparing DPCR Forms

33. Preparing OPCR Forms

C. Formulating and Submitting Performance Targets

34. Entering Performance Targets and Success Indicators

35. Defining Quality, Efficiency and Timeliness Indicators for Performance Targets

36. Automatically Saving Changes Made to Performance Forms

37. Withdrawing Submitted Performance Targets Before Supervisor Review

V. Target Weight Validation and Approval Management

A. Managing Target Weight Distribution

38. Calculating Weights for Core, Strategic and Support Functions

39. Validating Target Weights Against Civil Service Commission Category Requirements

40. Ensuring that the Total Weight of All Deliverables Equals 100%

41. Restricting IPERF Deliverables for Non-Teaching Staff to 100% of Committed Outputs

B. Reviewing and Approving Performance Targets

42. Approving Submitted Performance Targets

43. Locking Approved Performance Commitments Against Further Changes
44. Returning Performance Targets to Employees for Revision
45. Providing Specific Revision Remarks for Individual Performance Target Entries

VI. Means of Verification Evidence Management and Encryption

A. Managing Performance Evidence

46. Opening Evidence Management Panels for Individual Performance Deliverables
47. Uploading Supporting Evidence in PDF, Word, Excel and Image Formats
48. Enforcing a Maximum File Upload Size of 20 MB
49. Encrypting Stored File Contents Using AES-256 Encryption
50. Linking Supporting Evidence from Google Drive, OneDrive and Dropbox
51. Previewing Uploaded Supporting Documents Within the System

B. Managing Evaluation Rubrics

52. Downloading Pre-Filled Excel Evaluation Rubric Templates

53. Uploading Completed Excel Evaluation Rubric Files

VII. Automated CSC Scoring and Rating Computation

A. Managing Performance Ratings

54. Encoding Actual Semester Accomplishments and Deliverables Before Evaluation

55. Entering Performance Scores Using the Official Five-Point Rating Scale

56. Automatically Calculating Performance Indicator Averages for Each Deliverable Row

57. Excluding Non-Applicable Criteria from Rating Calculations

58. Applying Weighted Category Multipliers and Assigning Adjectival Performance Ratings

B. Managing Evaluation Remarks

59. Identifying Supervisors in Evaluation Comments Using Identification Tags

60. Categorizing Evaluator Feedback According to Administrative Role

VIII. PMT Calibration and Audit Management

A. Managing Institutional Rating Calibration

61. Monitoring Performance Rating Distributions Across the Fifteen Colleges
62. Reviewing and Calibrating Performance Scores
63. Recording Original Scores, Calibrated Scores and Reasons for Rating Adjustments
64. Displaying Audit Indicators for Adjusted Performance Ratings

B. Managing Evaluation Audit Records

65. Identifying PMT and TWG Contributors in Audit Notes Using Identification Tags
66. Searching System Activity Logs by keyword, user and timestamp
67. Filtering System Activity Logs by Authentication, Folder and Evaluation Activities
68. Exporting System Audit Records in CSV Format

IX. Executive Compliance Monitoring and Analytics

A. Monitoring College Employee Compliance

69. Displaying Performance Progress Across the Fifteen Colleges in Real Time
70. Monitoring the Status of Submitted, Evaluated and Certified Performance Records

71. Tracking Plantilla Roster Submission Compliance
72. Calculating Institutional Performance Submission Percentages

B. Generating Institutional Compliance Records

73. Generating Semester Performance Compliance Masterlists
74. Exporting Institutional Compliance Records in Excel Format

X. Official Performance Report Generation and Excel Report

A. Generating Official CSC Performance Reports

75. Generating IPCR Reports Based on CSC Annex A
76. Generating DPCR Reports Based on CSC Annex B
77. Generating OPCR Reports Based on CSC Annex C
78. Formatting Exported Reports for Short Bond Paper with the Official BSU Letterhead

B. Managing Exported Spreadsheet Formulas

79. Embedding Average and Sum Formulas in Exported Excel Workbooks
80. Protecting Formula Cells in Exported Spreadsheets Against Unauthorized Editing

C. Formatting Official Signature Blocks

81. Synchronizing Signatory Blocks for the Ratee, Immediate Supervisor and Head of Agency

Design of the System

This section discusses Phases 3 and 4 of the Feature-Driven Development methodology. Phase 3 focused on planning and organizing the development activities for the 81 identified features. The researchers grouped the features into five development groups based on their functions and dependencies and used Trello to organize the tasks and monitor their progress. The Trello board was organized into five lists: To-Do, In Progress, Need to Test, In Review, and Done, which were used to track the status of the development tasks. The corresponding task cards and checklists under these lists are presented in the Appendix H. This provided a structured plan for the succeeding design activities.

Following the planning activities, Phase 4 focused on designing the structure, interface, and database of the SPMS before its actual implementation. The researchers prepared the system architecture, user interface design, database schema, entity-relationship diagram, process flow diagrams, and use-case diagram. These design outputs served as the blueprint for developing the system and defining the functions and processes according to each user role.

The system design followed the Model-View-Controller architecture to organize the presentation, application logic, and data management components. This approach helped define how users would interact with the system, how requests would be processed, and how information would be stored and retrieved. Together, the planning and design activities provided a structured basis for the succeeding development and implementation phase.

System Architecture

The system architecture was designed to illustrate the overall structure of the SPMS and the interaction among its major components. It identifies the technologies planned for the presentation, application and data layers as well as the user roles that would access the system as shown in Figure 6.

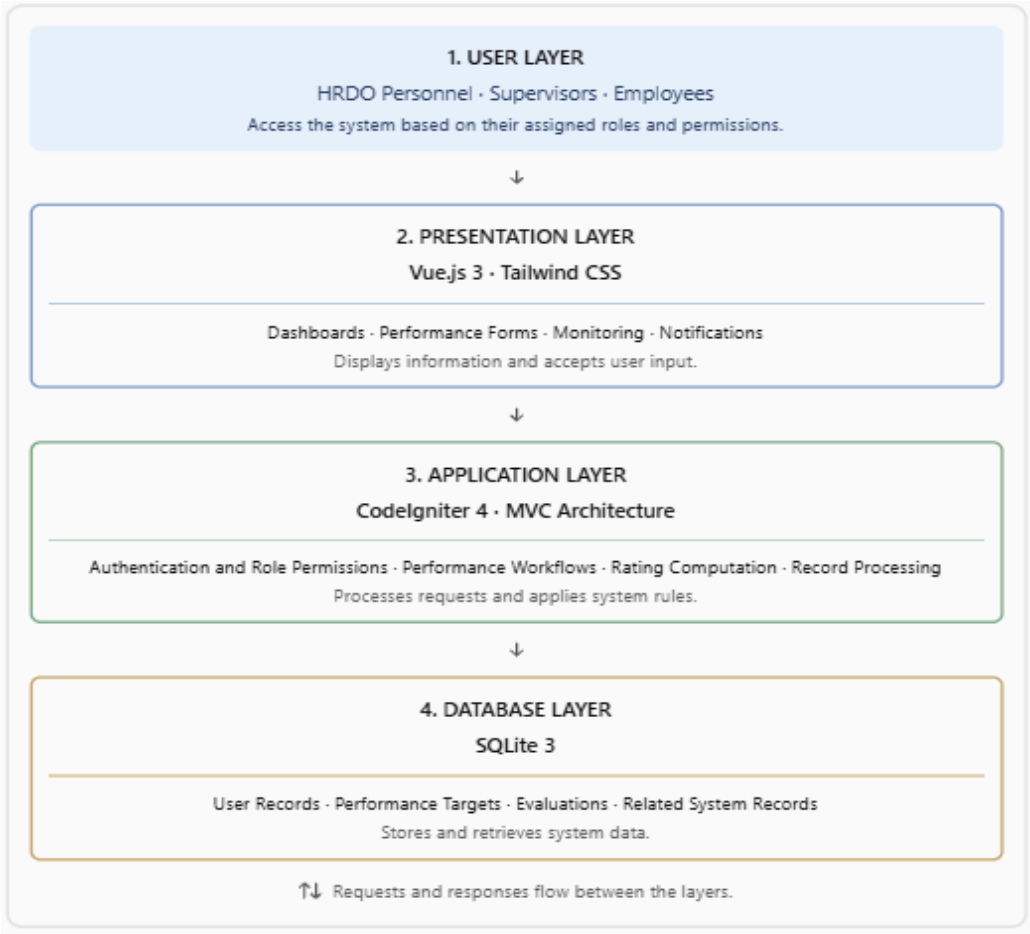


Figure 6. System Architecture of the SPMS

Figure 6 illustrates the structure of the SPMS and how its components work together. It consists of four layers: User, Presentation, Application, and Database. Each layer has a specific function in managing user interactions, processing requests, and storing system records.

The User Layer consists of HRDO personnel, supervisors, and employees. Each user accesses the system based on their assigned role and permissions.

Employees manage their performance targets and accomplishments, supervisors review and evaluate performance submissions, and HRDO personnel manage and monitor performance records.

The Presentation Layer provides the interface through which users interact with the system. It uses Vue.js 3 for interactive components and Tailwind CSS for the layout and design. It includes role-specific pages, performance forms, monitoring views, and notifications. Axios handles communication between the frontend and backend by sending requests and receiving responses.

The Application Layer uses CodeIgniter 4 and follows the Model-View-Controller architectural pattern. It processes user requests, manages authentication and role permissions, handles performance workflows, and performs rating computations. Controllers handle requests and coordinate system operations, while models manage database access and related data operations.

The Database Layer uses SQLite 3 to store and retrieve user records, performance targets, evaluation records, ratings, and other related system data. The application accesses the database when information needs to be saved, updated, or retrieved for system operations. Overall, these four layers work together to support the SPMS's performance planning, submission, evaluation, rating, and monitoring processes. User requests are processed by the application layer, which accesses the database when necessary. The results are then returned

to the presentation layer for display. This structure separates the system's main responsibilities and supports organized data management and system maintenance.

User-Interface Design

The user interface was designed to provide an organized and accessible platform for HRDO personnel, supervisors, and employees through role-specific dashboards, navigation menus, clear labels, status indicators, and accessible controls.

The implemented interface included a login page for authorized BSU email accounts, digital performance forms, notifications, audit trails, user management, and team management. The forms covered OPCR, DPCR, IPCR, and IPERF, organizing performance targets, success indicators, accomplishments, ratings, remarks, and signatory details.

Employees could access performance folders, forms, submission status, and approval progress. Supervisors could manage assigned personnel, review submissions, and evaluate accomplishments, while HRDO personnel could manage accounts, monitor records, and oversee evaluations. Notifications and audit trails supported activity monitoring. Screenshots of the implemented features and their functions are presented in the Build by Feature phase.

Database Design

The database design presents the structure of the SPMS through its entity-relationship diagram and database schema, as shown in Figure 7.

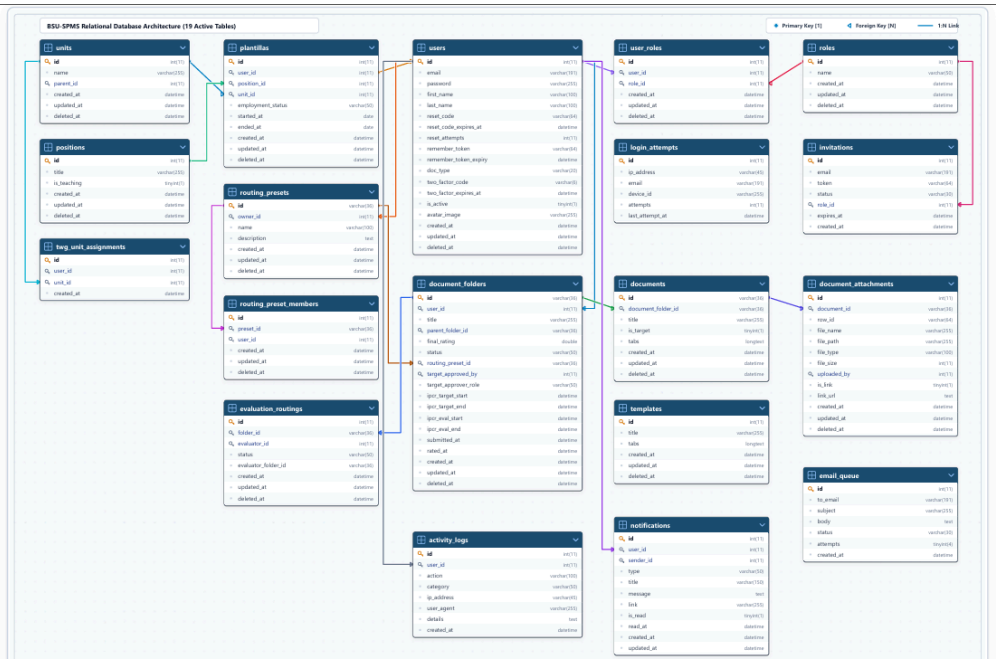


Figure 7. Entity-Relationship Diagram of the SPMS

Figure 7 illustrates the database structure of the SPMS. It presents the database tables, their attributes, primary keys, foreign keys and relationships that connect the different records. The diagram shows how the database supports user account management, organizational assignments, performance evaluation routing, document management, notifications and system security.

The database consists of 19 active tables, each serving a specific purpose in the system. The users, roles and user_roles tables manage user accounts, available roles and role assignments. The positions, plantillas, units and twg_unit_assignments tables support personnel information, organizational units, plantilla records and unit assignments. These tables help organize user information according to the university's organizational structure.

For performance evaluation routing, the routing_preset and routing_preset_members tables store predefined routing configurations and their assigned members. The evaluation_routings table records the routing of evaluation documents, including the evaluator, ratee, status, and related evaluation information. These tables support the movement and monitoring of performance evaluation documents throughout the required process.

The document_folders table organizes users' folders and stores references to related routing presets and evaluation records. The documents table stores document information, such as its title, type, target, status, and associated folder. Meanwhile, the document_attachments table stores attachment details, including filenames, file paths, file types, file sizes and upload information. The templates table stores reusable document templates that can support standardized forms and document preparation.

The remaining tables provide supporting functions for system operation and security. The activity_logs table records user activities, while the notifications table stores messages and updates intended for users. The login_attempts table records login attempts to support account security, and the invitations table manages invitation-related information. The email_queue table stores email messages awaiting processing or delivery. Together, these tables support monitoring, communication, and authentication-related activities.

The relationships between the tables are established through primary and foreign keys. Primary keys uniquely identify records while foreign keys connect related records across tables. These relationships help maintain data consistency and allow the system to retrieve related information when processing user accounts, organizational assignments, evaluation routing and document records.

In addition to the ERD, the database schema provides detailed information about the columns, data types, constraints, default values and descriptions of the fields in each database table. The complete database schema and data dictionary of the SPMS are presented in Appendix I for reference.

Process Flow Diagram

The process flow diagrams illustrate the main processes of the proposed web-based SPMS using the Model-View-Controller (MVC) architecture. They

show how the system processes user requests, verifies authentication and access permissions, manages performance planning and review, computes and validates ratings, and handles monitoring, evaluation status updates, and notifications, as shown in Figures 8–12.

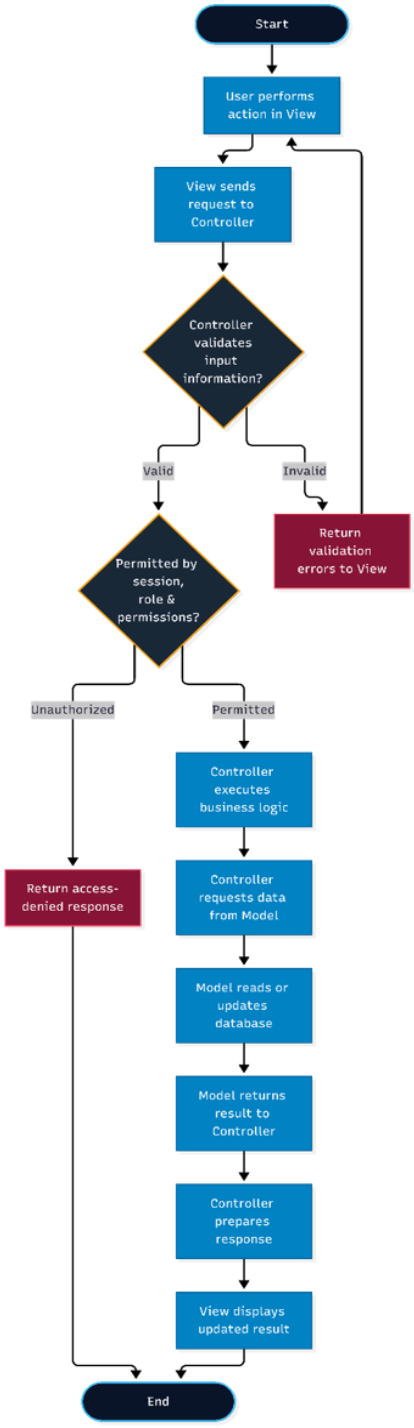


Figure 8. General Request Processing Flow of the SPMS .

Figure 8 presents the general request-processing flow of the SPMS using the MVC architecture. The process begins when a user performs an action on the View, which sends the request to the Controller for input validation. If the input is invalid, the system returns validation errors to the View. If the input is valid, the Controller checks whether the request is permitted based on the user's session, role, and permissions. Unauthorized requests receive an access-denied response, while permitted requests proceed to the business logic. The Controller then requests the necessary data from the Model, which reads or updates the database and returns the result to the Controller. Finally, the Controller prepares the response and the View displays the updated result to the user. This flow demonstrates how the SPMS organizes request processing, validates user inputs, enforces access permissions, and manages database interactions to support the system's functions.

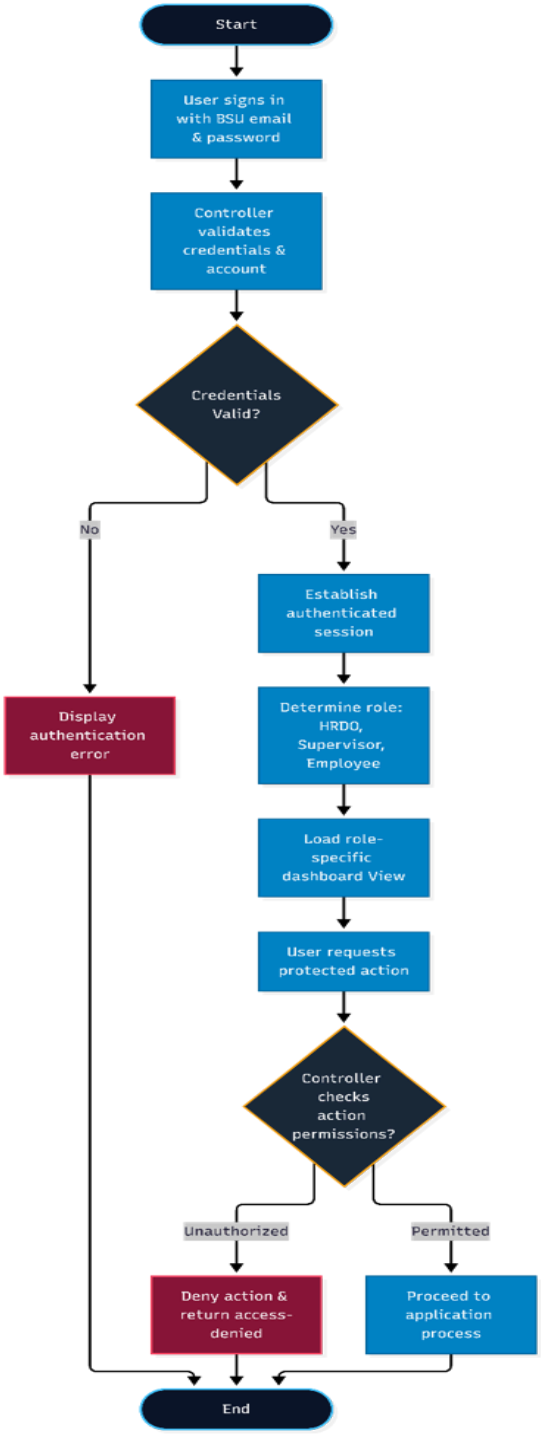


Figure 9. Authentication and Role-Based Access Control Flow

Figure 9 illustrates how the Controller handles authentication and role-based access control. The process begins when a user signs in using their BSU email address and password. The Controller validates the submitted credentials and account information. If the credentials are invalid, the system displays an authentication error. Otherwise, an authenticated session is established, and the system determines the user's role as HRDO, Supervisor, or Employee. The corresponding role-specific dashboard view is then loaded based on the user's assigned permissions. When the user requests a protected action, the Controller checks whether the action is authorized. Unauthorized requests are denied, and an access-denied response is returned, while permitted requests proceed to the appropriate application process. This flow demonstrates how the SPMS verifies user identities and enforces role-based permissions to restrict access to authorized functions.

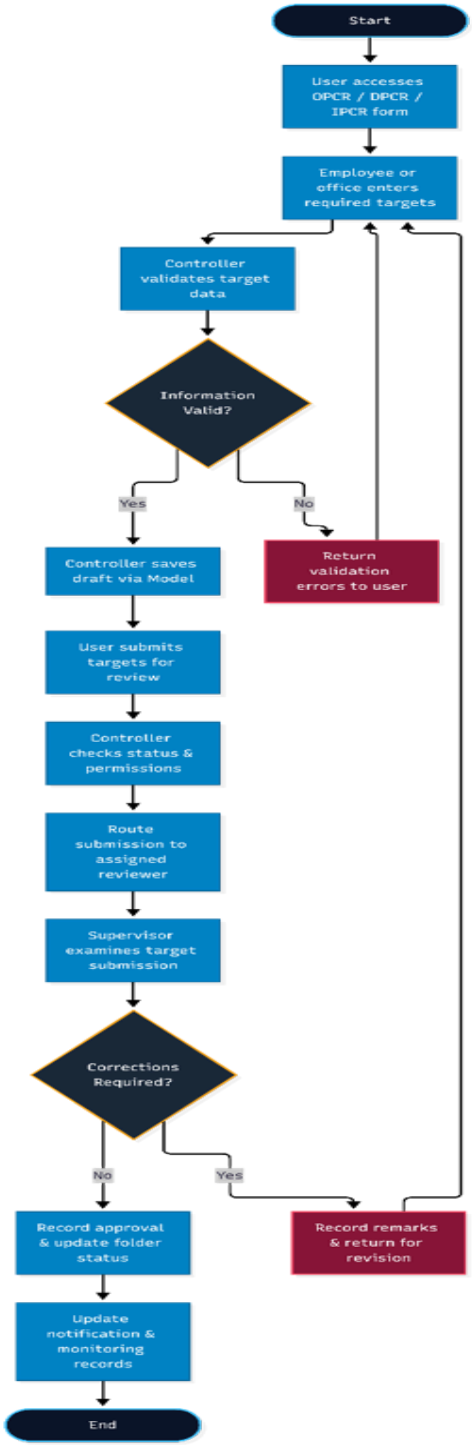


Figure 10. Performance Planning, Submission and Review Workflow

Figure 10 shows the application logic involved in performance planning, submission, and review. The process begins when a user accesses the OPCR, DPCR, or IPCR form and enters the required performance targets. The Controller validates the submitted target information. If the information is invalid, the system returns validation errors to the user. Otherwise, the Controller saves the draft through the Model, allowing the user to submit the targets for review. The Controller then checks the submission status and permissions before routing the targets to the assigned reviewer. The supervisor examines the submitted targets and determines whether corrections are required. If revisions are necessary, the supervisor records remarks and returns the submission to the user. If no corrections are required, the system records the approval and updates the folder status. Finally, the system updates the notification and monitoring records to reflect the submission's progress. This workflow demonstrates how the SPMS facilitates target preparation, submission, review, and approval while supporting status monitoring and communication between users and reviewers.

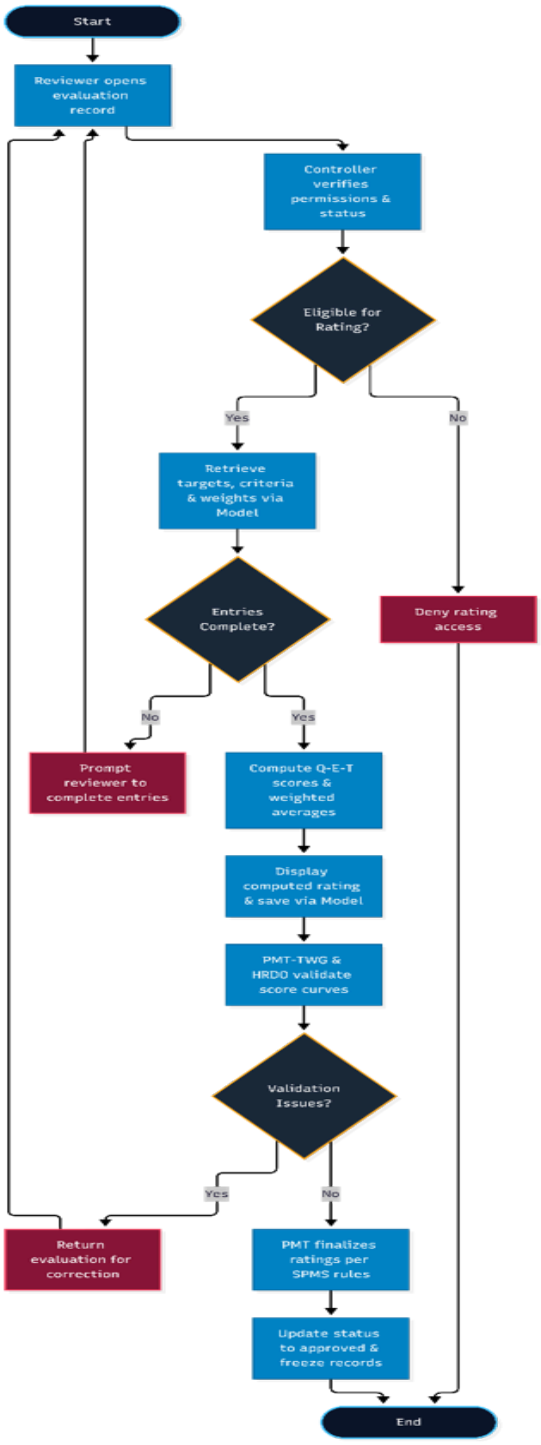


Figure 11. Performance Rating Computation and Validation Flow

Figure 11 presents the process for computing and validating performance ratings. The process begins when the reviewer opens an evaluation record, and the Controller verifies the user's permissions and the record's status to determine whether the evaluation is eligible for rating. If the record is not eligible, the system denies access to the rating function. Otherwise, the Model retrieves the performance targets, rating criteria, and applicable weights. The system then checks whether all required entries are complete. If some entries are missing, the reviewer is prompted to complete them before proceeding. Once the entries are complete, the system computes the Quality, Efficiency, and Timeliness (Q-E-T) scores and weighted averages, displays the computed rating, and saves the result through the Model. The PMT-TWG and HRDO then validate the score curves to ensure that the ratings comply with the applicable SPMS rules. If validation issues are identified, the evaluation is returned for correction. Otherwise, the PMT finalizes the ratings in accordance with SPMS rules, and the system updates the status to approved and freezes the records. This workflow demonstrates how the SPMS supports automated rating computation, evaluation validation, and the finalization of performance ratings.

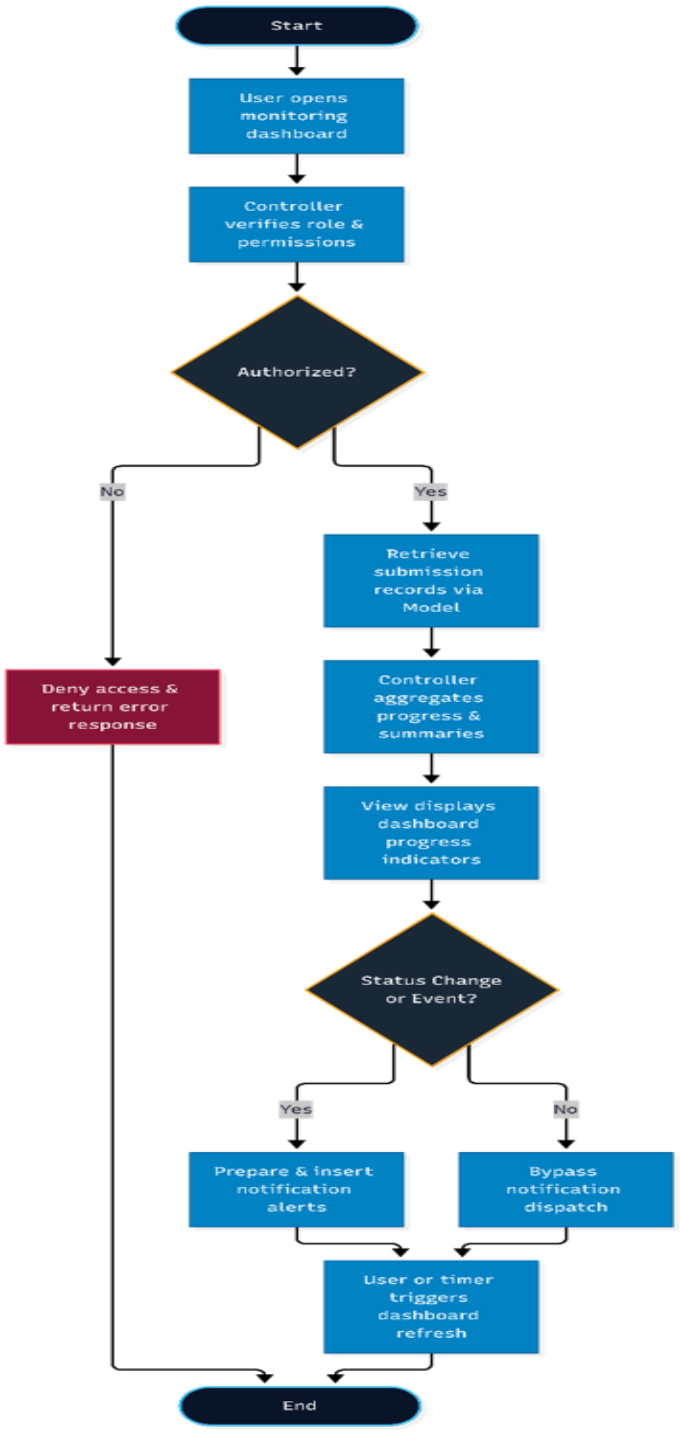


Figure 12. Monitoring, Notifications & Evaluation Status Management Flow

Figure 12 illustrates how the Controller manages monitoring information and notifications. The process begins when a user opens the monitoring dashboard, and the Controller verifies the user's role and permissions. If the user is unauthorized, the system denies access and returns an error response. Otherwise, the Model retrieves the submission records, and the Controller aggregates the progress information and summaries for display on the dashboard. The View then presents the progress indicators to help users monitor the status of performance submissions. The system checks whether a status change or event has occurred. If a change or event is detected, the system prepares and inserts the corresponding notification alerts; otherwise, notification dispatch is bypassed. Finally, the dashboard is refreshed through a user action or timer to display the latest available information. This workflow demonstrates how the SPMS supports performance submission monitoring, status tracking, and timely notifications for relevant system events.

Use-Case Diagram

The use case diagram presents the interactions between the users and the SPMS. It identifies the system's main actors and the functions they can access based on their roles and responsibilities.

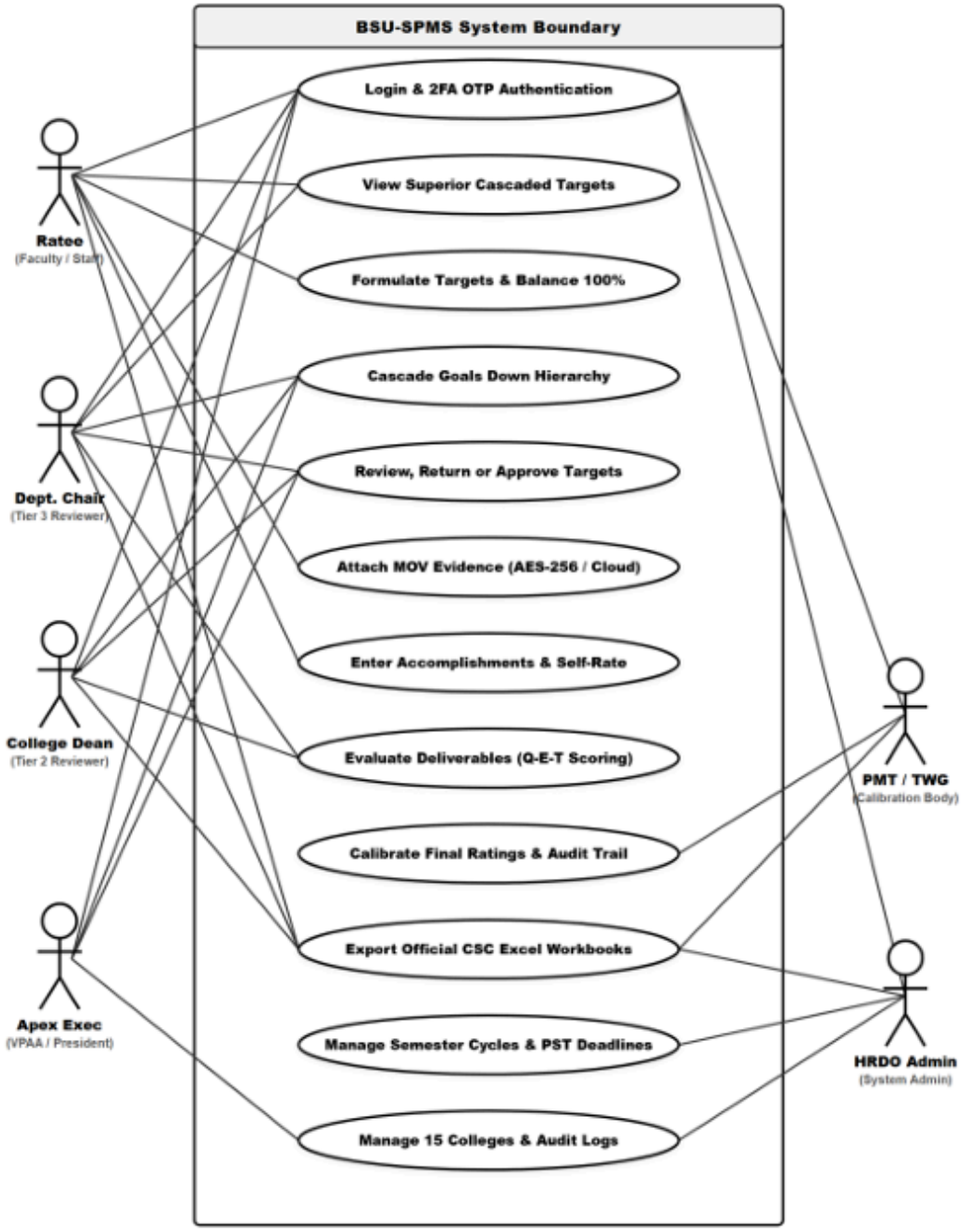


Figure 13. BSU-SPMS Use Case Diagram

The use case diagram illustrates the interactions of the SPMS with its primary actors: Ratee, Department Chair, College Dean, Apex Executive, PMT/TWG, and HRDO Admin. The Ratee is responsible for viewing superior-cascaded targets, formulating targets, attaching Means of Verification evidence, and entering accomplishments and self-ratings. Department Chairs and College Deans perform supervisory functions, including cascading goals, reviewing submissions, returning them for revision, approving targets, and evaluating deliverables based on Quality, Efficiency, and Timeliness (Q-E-T) criteria. The Apex Executive represents the higher-level executive responsible for overseeing performance planning and goal alignment.

The PMT/TWG supports the validation and calibration of performance ratings, while the HRDO Admin manages system administration, semester cycles, deadlines, organizational records, audit logs, and official CSC Excel workbook exports. Login and two-factor authentication (2FA) provide an authentication mechanism for authorized users, while the audit trail supports accountability and traceability of system activities.

Development and Testing

Development

This section discusses the Build by Feature phase of the Feature-Driven Development methodology. It presents the development and implementation of the SPMS. The researchers implemented the features based on the requirements and designs established in the previous phases. The development process was divided into three iterations: Foundation, Core Modules, and Refinement. Each iteration focused on specific system components to support an organized development process.

The system was developed using Vue.js 3 for the user interface, Tailwind CSS for styling, Axios for communication between the front end and back end, CodeIgniter 4 for server-side processing, and SQLite 3 for database management. The Model-View-Controller architecture was followed to separate the presentation, application processing, and data management components. The following sections discussed the implementation of each iteration.

Iteration 1: Foundation

The first iteration established the technical foundation of the SPMS. The researchers configured the SQLite database and implemented user authentication and role-based access control. The system also used role-specific dashboard

routing to direct authorized users to the appropriate interface after successful authentication. Input validation and applicable security checks were incorporated to help prevent invalid submissions and unauthorized access. These components provided the foundation for the implementation of the performance management modules in the next iteration.

Table 13. Implemented Components of the Foundation Iteration

Component	Implementation Result
Database	The SQLite database was configured to store the systems operational records
User authentication	The login process was implemented to verify user credentials
Role-based access control	Access permissions were configured according to the assigned user roles
Dashboard Routing	After successful login, the admin was directed to the admin dashboard while supervisors and employees were directed to their respective folders pages.
Input Validation	Validation rules were applied to supported inputs to help prevent invalid submission,
Security checks	Applicable security mechanisms were implemented and subsequently examined during testing.

Table 13 presented the components implemented during the Foundation iteration of the SPMS. The SQLite database was configured to store the system's operational records, while user authentication was implemented to verify user credentials during login. Role-based access control was established to manage access permissions according to assigned user roles. After successful login, the

admin was directed to the admin dashboard, whereas supervisors and employees were directed to their respective folders pages. Input validation was also applied to supported fields to help prevent invalid submissions. Lastly, applicable security mechanisms were implemented and examined during testing to assess their behavior and compliance with the system's security requirements. These components established the foundation for the implementation of the core performance management modules in the succeeding iteration.

Iteration 2: Core Modules

The second iteration focused on developing the core performance management modules of the SPMS: OPCR, DPCR, IPCR, and IPERF. The researchers followed the existing templates for each form to ensure that the digital forms reflected the required performance information and structure. The modules supported the entry of targets and accomplishments, submission and review of performance records, and rating computation based on the implemented SPMS rules and applicable Quality, Efficiency, and Timeliness criteria.

Table 14. Implemented Components of the Core Modules Iteration

Module or Function	Implementation Result
OPCR	The module followed the existing OPCR template for office-level performance commitments and accomplishments.
DPCR	The module followed the existing DPCR template

	for division-level performance commitments and accomplishments.
IPCR	The module followed the existing IPCR template for individual targets, accomplishments, and performance evaluation.
IPERF	The module followed the existing IPERF template for the applicable individual performance evaluation process.
Form Submission and Review	Functions were implemented to support form submission and the review of assigned records by authorized supervisors.
Automated computation rating	The system computed performance ratings based on the implemented SPMS rules and applicable Quality, Efficiency, and Timeliness criteria.

Table 14 presented the core modules implemented during the second iteration. Each module followed its corresponding existing performance management template to maintain the required structure and information. Form submission and review functions were also implemented to support the processing of performance records, while automated rating computation calculated performance ratings based on the implemented SPMS rules and applicable Quality, Efficiency, and Timeliness criteria.

Iteration 3: Refinement

The third iteration focused on refining the monitoring, reporting, notification, and interface features of the SPMS. The researchers improved the dashboards to present performance-related information, including monitoring and

coaching activities, target calibration, and personnel performance summaries. Reporting functions were also refined to support the generation and export of performance records. Notifications were implemented to inform users about relevant system activities, while interface improvements enhanced the presentation and accessibility of system functions.

Table 15. Implemented Components of the Refinement Iteration

Component	Implementation Result
Monitoring dashboards	The dashboard was refined to display Stage 2 monitoring and coaching, target calibration information by office as well as a leaderboard of top-performing personnel.
Reporting functions	Report generation and Excel export functions were implemented to support the preparation of performance records and reports.
Notifications	Notification functions were implemented to inform users about relevant system activities.
User interface refinements	Interface layouts and elements were improved to provide a more organized presentation of information and support easier navigation.
Workflow and permission consistency	The refined features were aligned with the existing user roles, access permissions, and performance management workflows.

Table 15 presented the improvements implemented during the Refinement iteration. The monitoring dashboard displayed Stage 2 monitoring and coaching information, target calibration by office, and a leaderboard of top-performing personnel. Report generation and Excel export functions supported the

preparation of performance records, while notifications informed users about relevant system activities. The researchers also refined the interface for clearer information presentation and easier navigation while maintaining the established user roles, permissions, and workflows.

Testing

Testing was conducted to check whether the developed SPMS functioned according to its identified requirements. Since the researchers used the Feature-Driven Development methodology, the testing activities were organized around the features identified during the development process. The 81 features, grouped into ten feature groups, served as references for identifying the functions and processes to be tested.

The testing activities covered unit testing, integration testing, system testing, and user acceptance testing. Each type of testing had a different purpose. Unit testing focused on individual functions or components, integration testing checked the interaction between related components, system testing examined the overall operation of the developed system, and UAT assessed whether the implemented features met the needs of the intended users. The results were documented by comparing the expected results with the actual results observed during testing.

Unit Testing

Unit testing focused on checking individual functions and components of the SPMS. The researchers examined functions such as user authentication, input validation, rating computation, database operations, and role-based access control. These tests helped determine whether individual components produced the expected results before or as they were integrated with other parts of the system.

Table 16. Unit Testing Results

Feature/Component	Test Procedure	Expected Result	Actual Result	Status
User Authentication	Enter valid and invalid login credentials.	The system accepts valid credentials and rejects invalid credentials.	The system accepted valid credentials and rejected invalid credentials.	PASSED
Input Validation	Submit forms with missing or invalid required fields.	The system identifies invalid inputs and provides appropriate validation messages.	The system detected invalid inputs and displayed the appropriate validation messages.	PASSED
Automated Rating Computation	Enter test values for the	The system calculates ratings	The system calculated the ratings	PASSED

	applicable rating criteria and compare the computed result with the expected calculation.	according to the implemented SPMS rules.	according to the implemented scoring rules.	
Database Operations	Save, retrieve, and update test records.	The system processes the records correctly and preserves the stored information.	The system successfully saved, retrieved, and updated the tested records.	PASSED
Role-Based Access Control	Attempt to access functions using accounts with different assigned roles.	The system permits or restricts access according to the assigned permissions.	The system provided access to authorized functions and restricted access to functions outside the assigned permissions.	PASSED

Table 16 presents the results of unit testing, which focused on checking individual functions and components of the SPMS. The tests covered user authentication, input validation, automated rating computation, database operations, and role-based access control. Based on the recorded results, the tested

components performed according to their expected functions. The system accepted valid credentials, rejected invalid inputs, processed records correctly, computed ratings according to the implemented rules, and applied the appropriate access permissions. These results indicate that the tested components were functioning as intended.

Integration Testing

Integration testing was conducted to verify whether the related features and components of the SPMS worked together correctly. The researchers tested the connections between authentication and role-based access control, performance forms and database operations, form submission and supervisor review, rating computation and evaluation records, monitoring and notifications, and reporting and database functions. This testing aimed to identify issues in the interaction between components and ensure that information was processed correctly across the connected features.

Table 17. Integration Testing Results

Features.Components	Test Procedure	Expected Result	Actual Result	Status

Authentication and Role-Based Access Control	Log in using accounts assigned to different roles and navigate to their permitted pages.	Each user is directed to the appropriate pages and can access only authorized functions.	The system directed users to the appropriate pages and applied the access permissions associated with their roles.	PASSED
Performance Forms and Database	Enter information in an applicable performanc e form and save the record.	The entered information is stored and retrieved correctly.	The system successfully saved the tested form information and retrieved the stored records.	PASSED
Form Submission and Supervisor Review	Submit a performanc e form and check whether it becomes available for the appropriate review process.	The submitted form is recorded and made available to the authorized reviewer according to the workflow.	The submitted form was recorded and made available for the applicable review process.	PASSED
Rating Computation and Evaluation Records	Enter or update rating information and	The computed rating and related information	The computed rating and related evaluation	PASSED

	examine the resulting evaluation record.	are reflected correctly in the record.	information were reflected correctly in the tested record.	
Monitoring and Notifications	Perform an action that should update evaluation status or trigger a notification.	The applicable status and notification are updated according to the implemented rules.	The system updated the applicable evaluation status and displayed the corresponding notification.	PASSED
Reporting and Database	Generate a report or export eligible records to Excel.	The generated report or exported file contains the appropriate stored information.	The generated report and exported file contained the expected information from the tested records.	PASSED

Table 17 presents the results of integration testing, which examined how related features and components worked together. The tests covered authentication and access control, performance forms and database operations, form submission and supervisor review, rating computation and evaluation records, monitoring and notifications, and reporting and database functions. The

results indicate that the tested components exchanged and processed information as expected. Submitted forms were made available for the applicable review process, rating information was reflected in evaluation records, and the tested reporting functions generated the expected information. These results suggest that the tested components were properly connected.

System Testing

System testing was conducted to evaluate the overall functionality of the SPMS and verify whether its implemented features worked according to the identified requirements. The researchers tested the system based on the ten feature groups identified during the feature-list development phase, covering authentication and security, organizational management, evaluation cycles, target formulation, approval, evidence management, rating computation, calibration, monitoring, and reporting. This testing aimed to determine whether the system performed its intended functions correctly when the different components operated together.

Table 18. System Testing Result

Features	Test Scenario	Expected Result	Actual Result	Status

Access Control, Authentication, and Security	Log in using authorized and unauthorized credentials and check access permissions.	The system authenticates users and enforces the applicable access restrictions.	The system accepted authorized credentials, rejected invalid credentials, and applied the configured access restrictions.	PASSED
Organizational Unit and Department Transfer	Access organizational records and examine applicable office or department assignments.	Organizational information is displayed and processed according to the user's permissions and implemented rules.	The system displayed the tested organizational information according to the assigned permissions.	PASSED
Semester Evaluation Cycle and Deadline	Check the evaluation period, deadline information, and evaluation status.	The system displays the configured cycle, deadlines, and applicable status information.	The system displayed the configured evaluation period, deadlines, and applicable status information.	PASSED
Institutional Target	Enter or review	The system supports the	The system supported the	PASSED

Cascading and Performance Formulation	applicable targets and performance forms.	implemented target-setting and performance-formulation workflow.	tested target entry and performance-formulation process.	
Target Weight Validation and Approval	Test the applicable target validation and approval functions.	The system applies the implemented validation and approval rules.	The system applied the configured validation and approval rules to the tested records.	PASSED
MOV Evidence and Encryption	Attach or access supporting evidence using the applicable functions.	Evidence is handled and protected according to the implemented security mechanisms.	The system processed the tested supporting evidence according to the implemented functions and security controls.	PASSED
Automated CSC Scoring and Rating	Enter test ratings and compare the resulting calculations with the expected values.	The system calculates and displays ratings according to the implemented scoring rules.	The system displayed rating results consistent with the expected calculations for the tested values.	PASSED

PMT Calibration and Audit	Examine the applicable calibration functions and evaluation records.	Authorized users can access the implemented calibration and audit functions.	The tested calibration and evaluation record functions were accessible to authorized users.	PASSED
Executive Compliance and Analytics	Access the available monitoring dashboard and examine its displayed information.	The system displays the implemented compliance and performance monitoring information.	The system displayed the available monitoring and performance information for the tested records.	PASSED
Official Performance Report and Excel Export	Generate a performance report and export eligible records.	The system generates the appropriate report and exports the selected information correctly.	The system generated the tested report and exported the selected records to Excel.	PASSED

Table 18 presents the results of system testing, which examined the overall operation of the SPMS based on its ten feature groups. The tests covered security, organizational management, evaluation cycles and deadlines, target formulation, approval, evidence management, rating computation, calibration, monitoring, and

reporting. Based on the recorded results, the tested functions operated according to their expected behavior. The system supported the applicable performance management processes, applied the configured access restrictions, displayed monitoring information, and generated the tested reports and Excel exports. These results indicate that the tested features worked as intended within the developed system.

User Acceptance Testing

User Acceptance Testing was conducted to determine whether the SPMS met the needs and requirements of its intended users. The researchers organized the test scenarios according to the ten feature groups identified during the feature-list development phase. The testing focused on checking whether the implemented features supported the users' tasks, including managing performance forms, reviewing and rating performance, monitoring evaluation status, and generating reports. The actual results were compared with the expected results to determine whether each tested feature performed as intended.

Table 19. User Acceptance Testing Result

Functional Requirement ID	Feature Group	UAT Test Scenario	Expected Result	Actual Result	Status
FR 1-11	Access Control, Authenticati	Log in using authorized	The system authenticates users and	Users were able to log in using	PASSED

	on, and Security Governance	and unauthorized credentials and check the functions available to each role.	applies the appropriate access permissions and security controls.	valid credentials, while invalid credentials were rejected and role-based access restrictions were applied.	
FR 12-18	Organizational Unit and Department Transfer Management	Examine organizational records and test the applicable department assignment or transfer functions.	The system processes organizational records and department transfers according to the implemented rules.	The tested organizational records and department assignment functions operated according to the configured rules.	PASSED
FR 19-26	Semester Evaluation Cycle and Deadline Management	View the evaluation cycle, submission deadlines, evaluation deadlines, and completed evaluation records.	The system displays the applicable cycle and deadlines and handles completed evaluation records according to the implemented rules.	The tested evaluation cycle, deadline, and record management functions operated as expected.	PASSED

FR 27-37	Institutional Target Cascading and Performance Formulation	Access the applicable OPCR, DPCR, IPCR, or IPERF form and test target entry, cascading, and submission.	The system supports the applicable target-setting and performance-formulation workflow.	The tested performance forms supported the applicable target entry and submission activities.	PASSED
FR 38-45	Target Weight Validation and Approval Management	Test the applicable target validation and approval functions.	The system applies the configured validation and approval rules.	The tested validation and approval functions operated according to the implemented rules.	PASSED
FR 46-53	MOV Evidence and Encryption	Attach or access supporting evidence using the available functions.	The system processes supporting evidence according to the implemented functions and security controls.	The tested evidence-related functions processed the submitted or accessed files as expected.	PASSED
FR 54-60	Automated CSC Scoring and Rating	Enter test rating values and compare the computed	The system calculates and displays ratings	The computed results matched the expected	PASSED

		results with the expected calculations.	according to the implemented scoring rules.	calculations for the tested rating values.	
FR 61-68	PMT Calibration and Audit	Examine the applicable calibration functions, evaluation records, and audit information.	Authorized users can access the applicable calibration and audit functions.	The tested calibration and audit functions were accessible according to the assigned permissions.	PASSED
FR 69-74	Executive Compliance and Analytics	Access the available monitoring dashboard and examine the displayed compliance and performance information	The system displays the implemented monitoring and analytics information correctly.	The tested monitoring functions displayed the applicable compliance and performance information.	PASSED
FR 75-81	Official Performance Report and Excel Export	Generate a performance report and export the applicable records to Excel.	The system generates the appropriate report and exports the selected	The tested reporting and export functions generated the expected report	PASSED

	information correctly.	information and Excel file.

Table 19 presents the results of User Acceptance Testing based on the ten feature groups identified in the feature list. The Functional Requirement IDs column links each test scenario to the corresponding features, allowing the researchers to trace the tested functions back to the system requirements. The test scenarios covered access control and security, organizational management, evaluation cycles, target formulation, approval, evidence management, rating computation, calibration, monitoring, and reporting. Based on the recorded results, the tested features performed according to their expected functions. The results indicate that the SPMS supported the intended activities, including managing performance forms, processing ratings, monitoring evaluation status, and generating reports. The UAT findings complement the other testing activities by examining whether the implemented features meet the intended users' requirements.

System Evaluation

The usability of the developed SPMS was evaluated using the Post-Study System Usability Questionnaire (PSSUQ), which consisted of 16 items. The evaluation involved 30 respondents who assessed the system in terms of system

usefulness, information quality, and interface quality. These dimensions were used to determine how the respondents perceived the system's ability to support performance management activities, present relevant information, and provide a usable interface.

Responses were rated using a seven-point Likert scale, where 1 represented Strongly Agree and 7 represented Strongly Disagree. Based on the interpretation guide presented in the methodology, lower mean scores indicated more favorable usability perceptions. The results were analyzed using the mean scores for each questionnaire item, each evaluation dimension, and the overall evaluation. Responses marked N.A. were excluded from the computation of the corresponding item and dimension means.

Table 20. Overall PSSUQ Evaluation Results of the SPMS

Evaluation Dimension	Mean Score	Descriptive Equivalence
System Usefulness (SYSUSE)	2.52	Agree
Information Quality (INFOQUAL)	2.67	Agree
Interface Quality (INTERQUAL)	2.66	Agree
Overall Mean	**2.61**	**Agree**

Table 20 presents the overall results of the PSSUQ evaluation conducted with 30 respondents. The developed SPMS obtained an overall mean score of 2.61, which falls within the Agree range of 1.87–2.72 based on the descriptive

equivalence scale presented in the methodology. This result indicates that the respondents generally provided favorable assessments of the system's usability.

Among the three evaluation dimensions, system usefulness obtained the lowest mean score of 2.52, followed by interface quality at 2.66 and information quality at 2.67. Since lower scores indicate more favorable responses, system usefulness received the most favorable assessment among the three dimensions. This suggests that the respondents generally viewed the system positively in terms of the usefulness of its functions for supporting their assigned SPMS-related activities. Interface quality also received a favorable assessment, indicating that the respondents generally agreed with the positive aspects of the interface measured by the questionnaire. Information quality obtained the highest mean score among the three dimensions, although it remained within the Agree range. This suggests that the information-related aspects of the system may benefit from further review and refinement.

System Usefulness: It measures the respondents' perceptions of how well the developed SPMS supports their work and helps them perform their assigned tasks. This dimension covers the first six items of the PSSUQ and provides an assessment of the system's usefulness in supporting performance management activities.

Table 21. PSSUQ Results for System Usefulness

Questionnaire Item	Mean Score	Descriptive Equivalence
Question 1	2.43	Agree
Question 2	2.63	Agree
Question 3	2.50	Agree
Question 4	2.53	Agree
Question 5	2.43	Agree
Question 6	2.57	Agree
Overall Mean	**2.52**	**Agree**

Table 21 presents the PSSUQ results for system usefulness. The dimension obtained an overall mean score of 2.52, which falls within the Agree range. This indicates that the respondents generally agreed with the positive usability statements associated with this dimension. Questions 1 and 5 both obtained the lowest mean score of 2.43, followed by Question 3 at 2.50, Question 4 at 2.53, Question 6 at 2.57, and Question 2 at 2.63. All six items fell within the Agree range of the adopted interpretation scale. The results indicate generally favorable responses across the system usefulness items, with Questions 1 and 5 receiving the most favorable ratings. The overall mean of 2.52 suggests that the respondents generally perceived the system positively in terms of the usefulness measured by the PSSUQ. The consistently favorable results across the six items indicate that this dimension was positively assessed by the respondents. These findings support the intended role of the system in assisting users with their performance management responsibilities.

Information Quality: It evaluates the respondents' perceptions of the information presented by the developed SPMS. This dimension covers Questions 7 to 12 of the PSSUQ and assesses the information-related aspects of the system that support users in performing their assigned tasks.

Table 22. PSSUQ Results for Information Quality

Questionnaire Item	Mean Score	Descriptive Equivalence
Question 7	3.24	Somewhat Agree
Question 8	2.77	Somewhat Agree
Question 9	2.46	Agree
Question 10	2.50	Agree
Question 11	2.43	Agree
Question 12	2.63	Agree
Overall Mean	**2.67**	**Agree**

Table 22 presents the PSSUQ results for information quality. The dimension obtained an overall mean score of 2.67, which falls within the Agree range. This indicates that the respondents generally provided favorable assessments of the information-related aspects measured by the questionnaire.

Among the six items, Question 11 obtained the lowest mean score of 2.43, followed by Question 9 at 2.46, Question 10 at 2.50, Question 12 at 2.63, Question 8 at 2.77, and Question 7 at 3.24. Questions 7 and 8 fell within the Somewhat Agree range, while Questions 9 to 12 fell within the Agree range.

Since lower scores represent more favorable responses, Question 11 received the most favorable rating, whereas Question 7 received the least favorable rating within this dimension. The overall mean of 2.67 indicates that information quality was generally assessed favorably by the respondents. However, the higher mean scores for Questions 7 and 8 suggest that the corresponding aspects of the system may require further examination.

Interface Quality: It evaluates the respondents' perceptions of the interface-related aspects of the developed SPMS. This dimension covers Questions 13 to 16 of the PSSUQ and provides an assessment of the interface as experienced by users during their interaction with the system.

Table 23. PSSUQ Results for Interface Quality

Questionnaire Item	Mean Score	Descriptive Equivalence
Question 13	2.87	Somewhat Agree
Question 14	2.67	Agree
Question 15	2.73	Somewhat Agree
Question 16	2.37	Agree
Overall Mean	**2.66**	**Agree**

Table 23 presents the PSSUQ results for interface quality. The dimension obtained an overall mean score of 2.66, which falls within the Agree range. This indicates that the respondents generally assessed the interface-related aspects of the SPMS favorably.

Question 16 obtained the lowest mean score of 2.37, followed by Question 14 at 2.67, Question 15 at 2.73, and Question 13 at 2.87. Questions 14 and 16 fell within the Agree range, while Questions 13 and 15 fell within the Somewhat Agree range. Based on the adopted scale, Question 16 received the most favorable assessment, while Question 13 received the least favorable assessment among the four items. The overall mean of 2.66 suggests that the respondents generally agreed with the positive interface quality statements measured by the PSSUQ. However, the comparatively higher mean scores for Questions 13 and 15 indicate that the corresponding interface-related aspects may benefit from further review.

Description of the Prototype

The developed SPMS prototype includes interfaces for user authentication, role-based dashboards, performance evaluation forms, notifications, audit trail monitoring, user management, and team management. Each interface supports specific system functions based on the user's role and responsibilities, as shown in Figures 14–24.

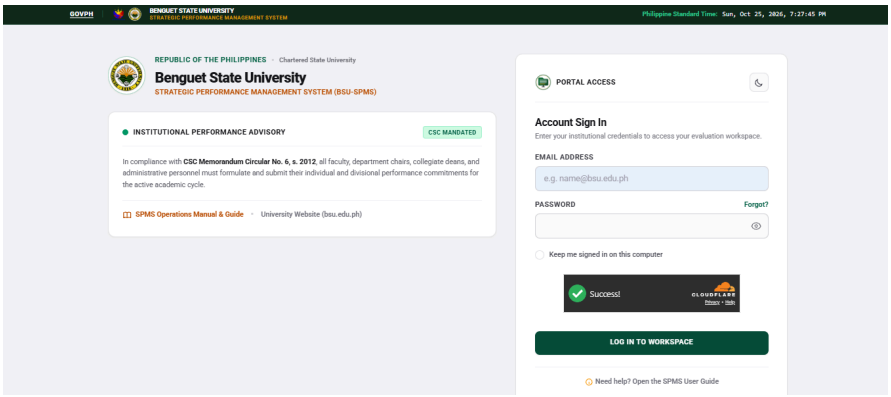


Figure 14. Login Interface of the SPMS

Figure 14 presents the login interface of the SPMS of Benguet State University. The interface was designed to provide authorized users with a centralized and accessible entry point to their respective performance evaluation workspaces. It featured the university's official branding, institutional information, and a login panel with a clean layout that separated the institutional advisory section from the account authentication section.

The upper portion of the interface displayed the university name, seal, and SPMS identification to establish the system's institutional identity. On the left side, the Institutional Performance Advisory section presented information regarding compliance with CSC Memorandum Circular No. 6, s. 2012. It also provided links to the SPMS Operations Manual and Guide and the university website, allowing users to access relevant references.

On the right side, the Portal Access panel contained the account sign-in form, which included fields for the institutional email address and password. A password visibility control was provided to allow users to view or hide their entered passwords. The Forgot? option provided access to the password recovery process, while the Keep me signed in on this computer checkbox allowed users to select their preferred sign-in experience. The Cloudflare verification component was also integrated into the interface to help protect the login process against automated access. The Log In to Workspace button allowed users to submit their credentials to access the system. Additionally, the interface included a dark mode toggle, which provided an alternative display appearance based on user preference.

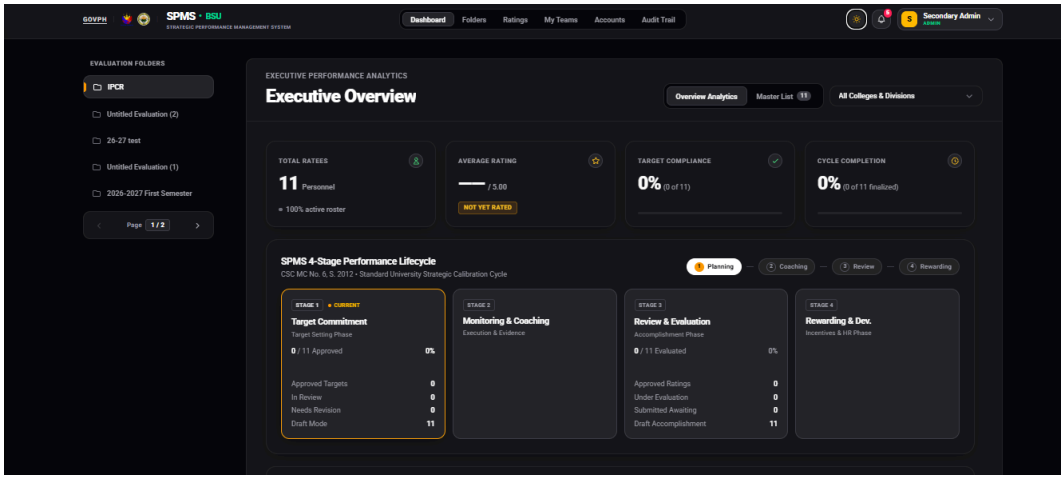


Figure 15.1. Admin Dashboard Interface of the SPMS

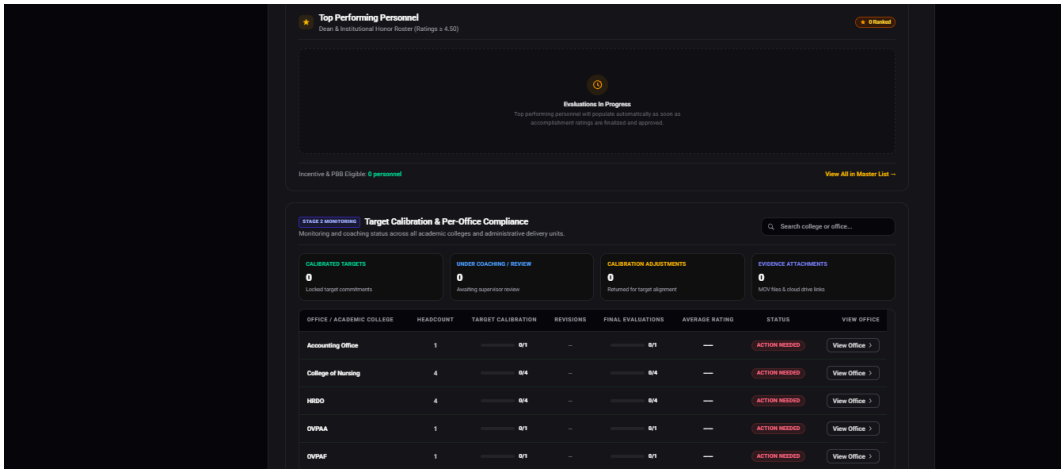


Figure 15.2. Admin Dashboard Interface of the SPMS

Figure 15.1 and 15.2 presents the admin dashboard of the SPMS. The dashboard was designed to provide administrators with an overview of personnel performance, target compliance, evaluation progress, and monitoring activities across the university. It also provided navigation options for accessing evaluation folders, ratings, teams, accounts, and audit trail records.

The upper portion of the dashboard displayed the main navigation menu and the account of the currently logged-in Secondary Admin. On the left side, the Evaluation Folders panel displayed the available performance evaluation folders, including IPCR and other evaluation records. This section allowed the administrator to navigate between different evaluation folders and access the corresponding performance records.

The main section presented the Executive Overview, which contained four summary indicators: Total Ratees, Average Rating, Target Compliance, and Cycle

Completion. These indicators provided information about the number of personnel included in the evaluation, the average performance rating, the percentage of approved targets, and the completion status of the evaluation cycle. The dashboard also included options for viewing overview analytics, accessing the master list, and filtering records by college or division.

Below the summary indicators, the SPMS 4-Stage Performance Lifecycle section presented the four stages of the performance management process: Planning, Coaching, Review, and Rewarding. Each stage displayed relevant information about its corresponding activities. For example, the Target Commitment stage showed the number of approved targets, targets under review, targets requiring revision, and draft targets. Meanwhile, the Review and Evaluation stage displayed the progress of accomplishment evaluations and the status of submitted ratings. This section helped the administrator monitor the progress of performance management activities throughout the evaluation cycle.

The lower portion of the dashboard contained the Top Performing Personnel section, which was intended to display personnel who achieved ratings of 4.50 or higher. It also presented information about incentive and Performance-Based Bonus (PBB) eligibility. Since evaluations were still in progress in the displayed interface, the section indicated that rankings would become available once accomplishment ratings had been finalized and approved.

Another section, Target Calibration and Per-Office Compliance, presented monitoring information across academic colleges and administrative offices. It included summary indicators for calibrated targets, targets under coaching or review, calibration adjustments, and evidence attachments. A table provided office-level information, including headcount, target calibration progress, revisions, final evaluations, average ratings, and status. The View Office option allowed the administrator to access more detailed information for a specific office. The Action Needed indicators highlighted offices that required further attention based on their displayed evaluation progress.

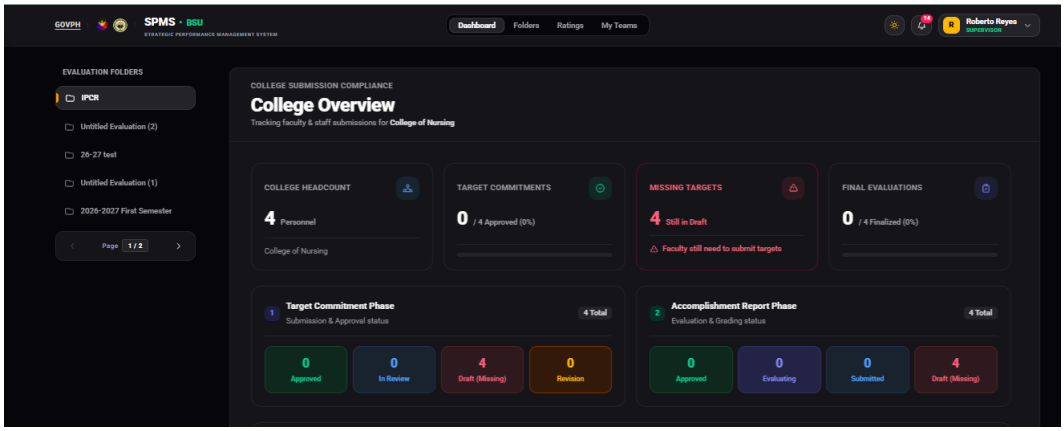


Figure 16.1. Supervisor Dashboard Interface of the SPMS

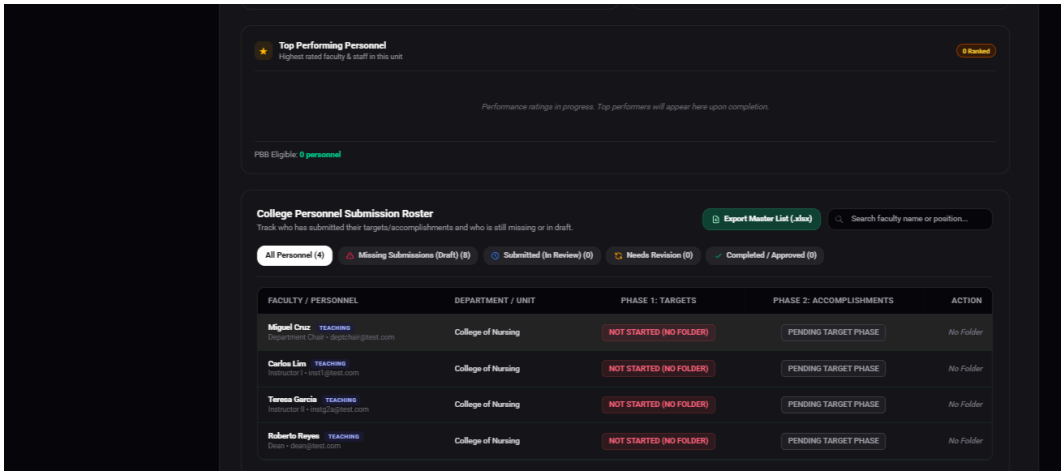


Figure 16.2. Supervisor Dashboard Interface of the SPMS

Figure 16.1 and 16.2 presents the supervisor dashboard of the SPMS. The dashboard was designed to help supervisors monitor personnel performance, track target submissions, review accomplishment reports, and oversee the performance evaluation progress within their assigned college or unit. It provided a centralized view of submission compliance and personnel evaluation records, allowing supervisors to identify pending submissions and determine which activities required attention.

The upper portion of the interface displayed the system navigation menu, evaluation folders, and the profile of the currently logged-in supervisor. The navigation menu provided access to the Dashboard, Folders, Ratings, and My Teams sections. On the left side, the Evaluation Folders panel displayed available evaluation folders, including IPCR and other evaluation records. These folders

allowed the supervisor to navigate through the performance evaluation records available in the system.

The main section displayed the College Overview, which presented the submission status of faculty members and staff in the College of Nursing. It contained four summary indicators: College Headcount, Target Commitments, Missing Targets, and Final Evaluations. The College Headcount indicated the number of personnel within the college, while Target Commitments showed the number and percentage of approved targets. Missing Targets identified personnel whose target submissions remained in draft status, and Final Evaluations displayed the number of finalized evaluations. These indicators provided the supervisor with an overview of the college's compliance with the performance planning and evaluation process.

Below the summary indicators, the dashboard presented the Target Commitment Phase and Accomplishment Report Phase. The Target Commitment Phase displayed the number of personnel whose targets were approved, under review, in draft or missing, or requiring revision. Meanwhile, the Accomplishment Report Phase presented the status of performance accomplishment reports, including approved, under evaluation, submitted, and draft or missing records. These sections helped the supervisor monitor the progress of both target submission and accomplishment evaluation.

The dashboard also included a Top Performing Personnel section, which was intended to display the highest-rated faculty members and staff within the assigned college or unit. The section indicated that performance ratings were still in progress and that top performers would appear after the ratings had been completed. It also displayed the number of personnel eligible for Performance-Based Bonus (PBB), allowing the supervisor to view the relevant performance-related information.

The lower portion of the dashboard contained the College Personnel Submission Roster, which provided a detailed list of faculty members and staff and their respective submission statuses. The roster included the personnel's names, positions, email addresses, and department or unit, together with their target and accomplishment statuses. The available status filters included All Personnel, Missing Submissions (Draft), Submitted (In Review), Needs Revision, and Completed or Approved. These filters allowed the supervisor to narrow the displayed records according to their current status. A search field was also provided to locate personnel by name or position, while the Export Master List (.xlsx) option allowed the supervisor to export the displayed personnel roster for reporting, monitoring or printing purposes. The Action column indicated whether an evaluation folder was available for a particular personnel record.

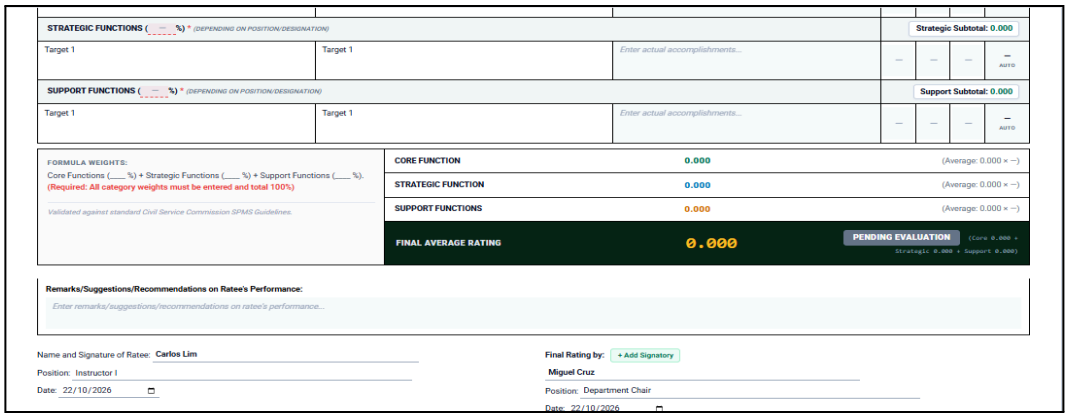


Figure 17.1. Individual Performance Commitment and Review Form

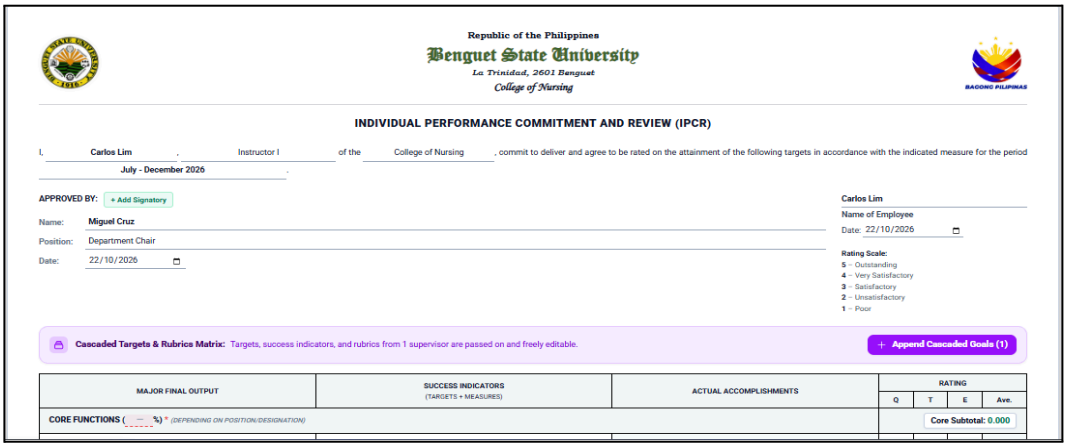
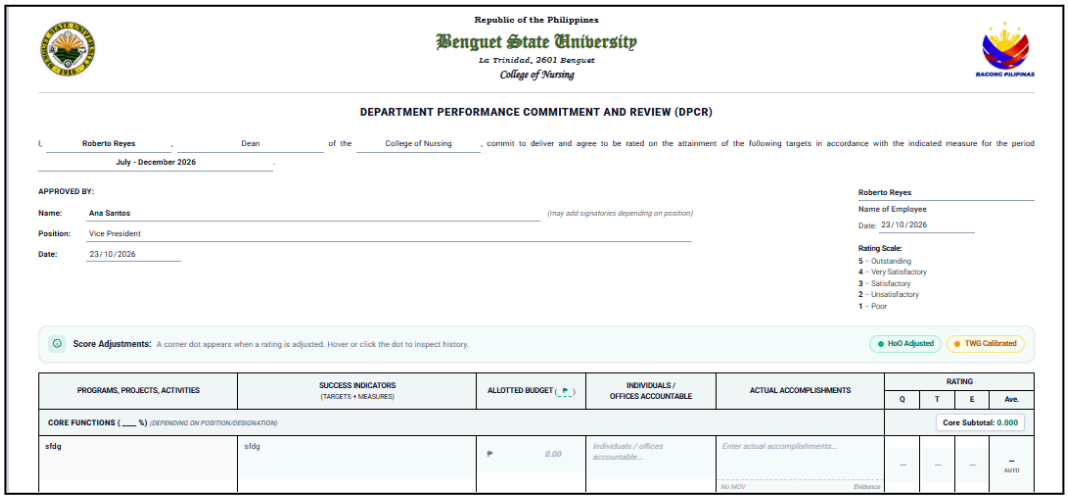


Figure 17.2. Individual Performance Commitment and Review Form

The Individual Performance Commitment and Review (IPCR) Form Interface displayed the employee's information, performance targets, actual accomplishments, and evaluation ratings. The upper section presented the employee's name, position, office, performance period, approving authority, and rating scale from 1 to 5. It also included the Cascaded Targets & Rubrics Matrix and the Append Cascaded Goals button for adding targets cascaded from

supervisors. The performance table organized targets into Core Functions, Strategic Functions, and Support Functions, with fields for Major Final Output, Success Indicators, Actual Accomplishments, and ratings for Quality, Efficiency, and Timeliness. The Formula Weights section displayed the weight allocation and calculated results for each function category, together with the Final Average Rating and evaluation status. A remarks section was provided for performance feedback, while the bottom portion displayed the ratee's and evaluator's details, including their names, positions, dates, and signature fields.



Republic of the Philippines

Benguet State University

La Trinidad, 2601 Benguet

College of Nursing

BAGONG PILIPINAS

DEPARTMENT PERFORMANCE COMMITMENT AND REVIEW (DPCR)

I, Roberto Reyes , Dean of the College of Nursing , commit to deliver and agree to be rated on the attainment of the following targets in accordance with the indicated measure for the period July - December 2026 .

APPROVED BY:

Name: Ana Santos (may add signatories depending on position)

Position: Vice President

Date: 23/10/2026

Roberto Reyes

Name of Employee

Date: 23/10/2026

Rating Scale:

5 – Outstanding
4 – Very Satisfactory
3 – Satisfactory
2 – Unsatisfactory
1 – Poor

Score Adjustments: A corner dot appears when a rating is adjusted. Hover or click the dot to inspect history.

HoO Adjusted | TWG Calibrated

PROGRAMS, PROJECTS, ACTIVITIES	SUCCESS INDICATORS (TARGETS + MEASURES)	ALLOTTED BUDGET (₱)	INDIVIDUALS / OFFICES ACCOUNTABLE	ACTUAL ACCOMPLISHMENTS	RATING			
					Q	T	E	Ave.
CORE FUNCTIONS (___ %) (DEPENDING ON POSITION/DESIGNATION)								Core Subtotal: 0.000
sfdg	sfdg	₱ 0.00	Individuals / offices accountable...	Enter actual accomplishments... No MOV Evidence	–	–	–	– AUTO

Figure 18.1. Department Performance Commitment and Review Form

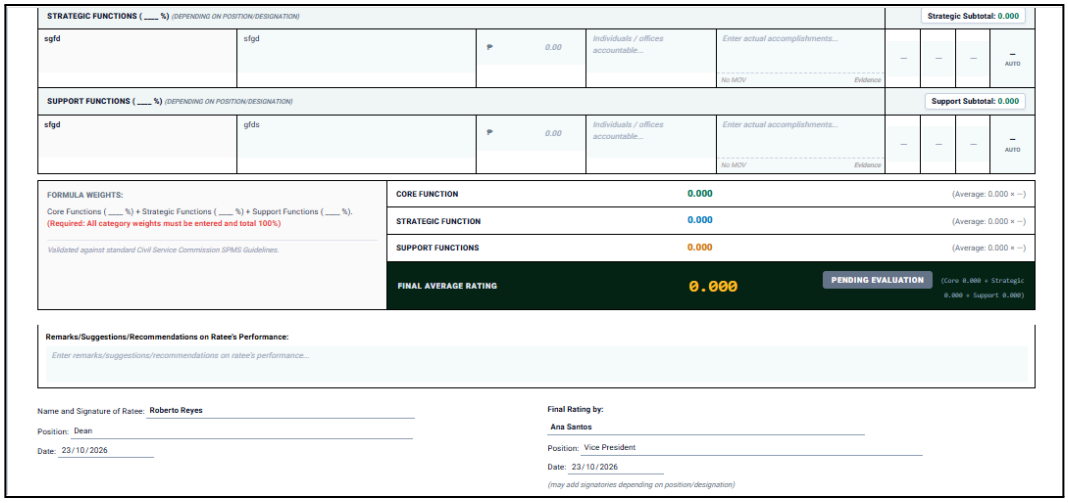
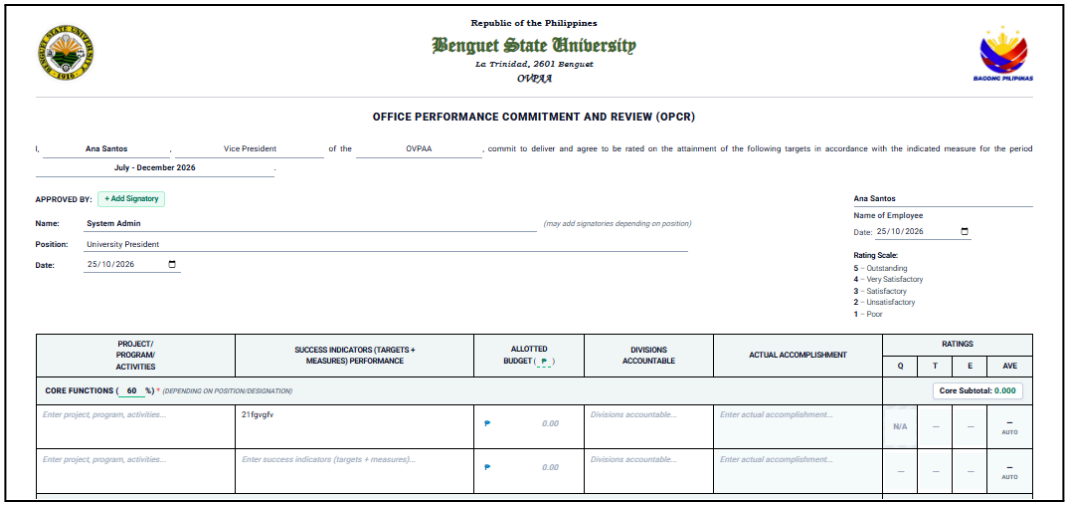


Figure 18.2. Department Performance Commitment and Review Form

The Department Performance Commitment and Review (DPCR) Form Interface displays the department's performance commitments, success indicators, budget allocations, accountable individuals or offices, actual accomplishments, and rating results. The upper section presented the official Benguet State University header, the department head's name and position, the college or office, and the covered performance period. It also included the Approved By section, which displayed the supervising authority's name, position, and date, as well as the employee's details and the rating scale from 1 to 5. The interface included a Score Adjustments section, which indicated when a rating had been adjusted and allowed users to inspect the adjustment history by hovering over or clicking the indicator. It also displayed the HoO Adjusted and TWG Calibrated status indicators to show the adjustment and calibration status of the ratings.

The main performance table was divided into Core Functions, Strategic Functions, and Support Functions. Each category contained fields for Programs, Projects, and Activities; Success Indicators; Allotted Budget; Individuals or Offices Accountable; Actual Accomplishments; and Ratings. The rating section consisted of Quality, Efficiency, Timeliness, and Average. Subtotal values were displayed for each function category to organize the department's performance results. Below the table, the Formula Weights section presented the weight allocations for the three function categories and indicated that all category weights were required to total 100%. The adjacent rating summary displayed the calculated results for Core Functions, Strategic Functions, and Support Functions, together with the Final Average Rating and its current evaluation status. A Remarks/Suggestions/Recommendations section was also provided for recording feedback regarding the ratee's performance. At the bottom of the form, the Name and Signature of Ratee and Final Rating By sections displayed the department head's and evaluator's names, positions, and dates, including fields for additional signatories when applicable.



Republic of the Philippines
Benguet State University
La Trinidad, 2601 Benguet
OVPAA

OFFICE PERFORMANCE COMMITMENT AND REVIEW (OPCR)

I, Ana Santos, Vice President of the OVPAA, commit to deliver and agree to be rated on the attainment of the following targets in accordance with the indicated measure for the period July - December 2026.

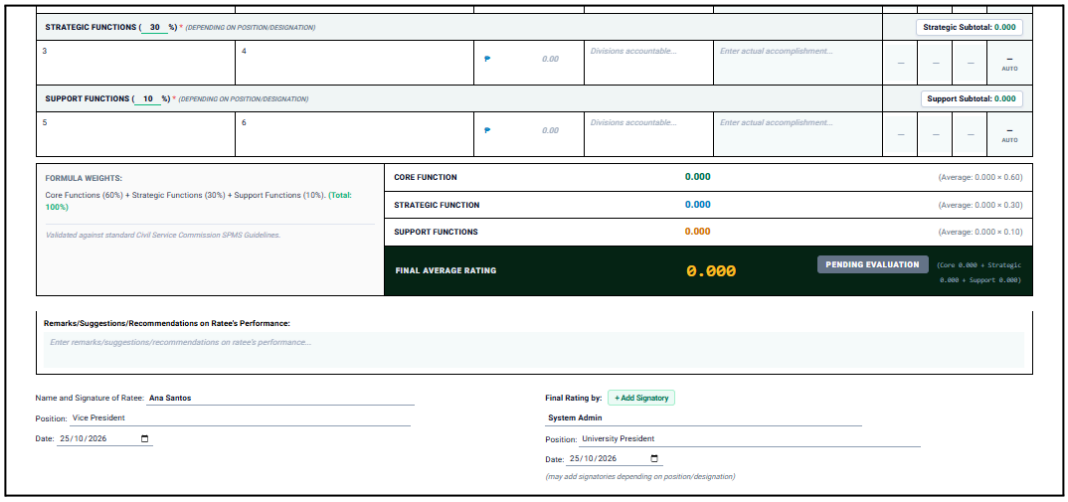
APPROVED BY: + Add Signatory
Name: System Admin (may add signatories depending on position)
Position: University President
Date: 25/10/2026

Ana Santos
Name of Employee
Date: 25/10/2026

Rating Scale:
5 – Outstanding
4 – Very Satisfactory
3 – Satisfactory
2 – Unsatisfactory
1 – Poor

PROJECT/ PROGRAM/ ACTIVITIES	SUCCESS INDICATORS (TARGETS + MEASURES) PERFORMANCE	ALLOTTED BUDGET (₱)	DIVISIONS ACCOUNTABLE	ACTUAL ACCOMPLISHMENT	RATINGS			
					Q	T	E	AVE
CORE FUNCTIONS (60 %) * (DEPENDING ON POSITION/DESIGNATION)								Core Subtotal: 0.000
Enter project, program, activities...	21fgvgfv	₱ 0.00	Divisions accountable...	Enter actual accomplishment...	N/A	–	–	– AUTO
Enter project, program, activities...	Enter success indicators (targets + measures)...	₱ 0.00	Divisions accountable...	Enter actual accomplishment...	–	–	–	– AUTO

Figure 19.1. Office Performance Commitment and Review Form



STRATEGIC FUNCTIONS (30 %) * (DEPENDING ON POSITION/DESIGNATION)								Strategic Subtotal: 0.000
3	4	₱ 0.00	Divisions accountable...	Enter actual accomplishment...	–	–	–	– AUTO
SUPPORT FUNCTIONS (10 %) * (DEPENDING ON POSITION/DESIGNATION)								Support Subtotal: 0.000
5	6	₱ 0.00	Divisions accountable...	Enter actual accomplishment...	–	–	–	– AUTO

FORMULA WEIGHTS:
Core Functions (60%) + Strategic Functions (30%) + Support Functions (10%). (Total: 100%)
Validated against standard Civil Service Commission SPMS Guidelines.

CORE FUNCTION	0.000	(Average: 0.000 × 0.60)
STRATEGIC FUNCTION	0.000	(Average: 0.000 × 0.30)
SUPPORT FUNCTIONS	0.000	(Average: 0.000 × 0.10)
FINAL AVERAGE RATING	0.000	PENDING EVALUATION (Core 0.000 + Strategic 0.000 + Support 0.000)

Remarks/Suggestions/Recommendations on Ratee's Performance:
Enter remarks/suggestions/recommendations on ratee's performance...

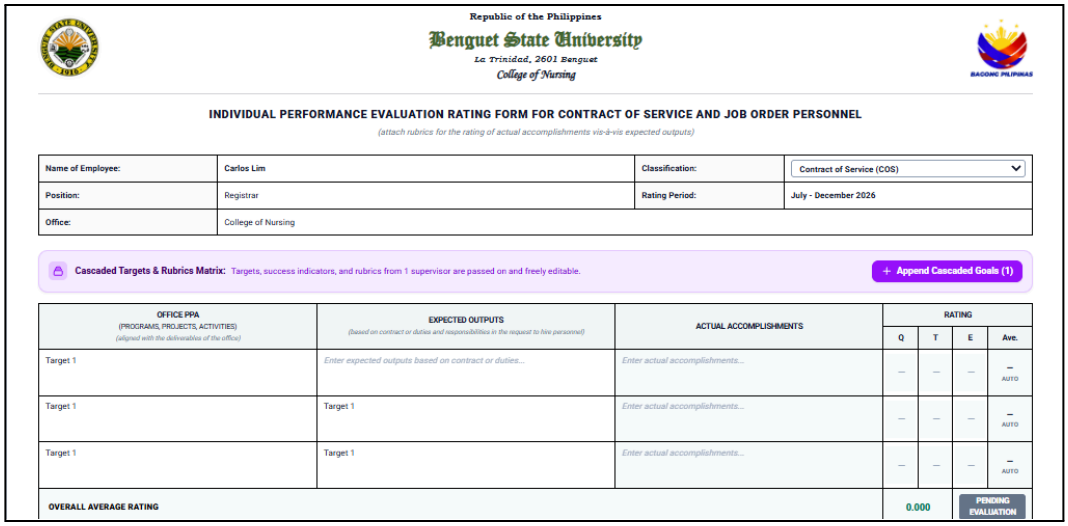
Name and Signature of Ratee: Ana Santos
Position: Vice President
Date: 25/10/2026

Final Rating by: + Add Signatory
System Admin
Position: University President
Date: 25/10/2026
(may add signatories depending on position/designation)

Figure 19.2. Office Performance Commitment and Review Form

The Office Performance Commitment and Review (OPCR) Form Interface displays the department's performance targets, success indicators, budget allocations, accountable individuals or offices, actual accomplishments, and ratings. The upper section presented the university header, department head's details, performance period, approving authority, and rating scale from 1 to 5. It

also included Score Adjustments indicators for viewing rating adjustment history and showing HoO adjustment and TWG calibration statuses. The performance table organized targets into Core Functions, Strategic Functions, and Support Functions, with fields for Programs, Projects, and Activities; Success Indicators; Allotted Budget; Individuals or Offices Accountable; Actual Accomplishments; and ratings for Quality, Efficiency, and Timeliness. Subtotals were displayed for each function category. The Formula Weights section showed the weight allocations, which had to total 100%, alongside the calculated category ratings and Final Average Rating. The interface also provided a remarks section for performance feedback and signatory fields for the ratee and final evaluator, including their names, positions, dates, and additional signatories when applicable.



Republic of the Philippines
Benguet State University
La Trinidad, 2601 Benguet
College of Nursing

INDIVIDUAL PERFORMANCE EVALUATION RATING FORM FOR CONTRACT OF SERVICE AND JOB ORDER PERSONNEL

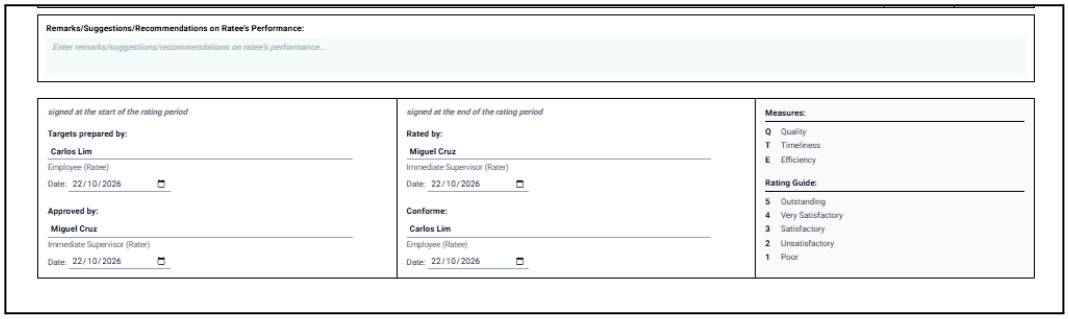
(attach rubrics for the rating of actual accomplishments vis-à-vis expected outputs)

Name of Employee:	Carlos Lim	Classification:	Contract of Service (COS)
Position:	Registrar	Rating Period:	July - December 2026
Office:	College of Nursing		

Cascaded Targets & Rubrics Matrix: Targets, success indicators, and rubrics from 1 supervisor are passed on and freely editable. + Append Cascaded Goals (1)

OFFICE PPA (PROGRAMS, PROJECTS, ACTIVITIES) (aligned with the deliverables of the office)	EXPECTED OUTPUTS (based on contract or duties and responsibilities in the request to hire personnel)	ACTUAL ACCOMPLISHMENTS	RATING Q	T	E	Ave.
Target 1	Enter expected outputs based on contract or duties...	Enter actual accomplishments...	–	–	–	– AUTO
Target 1	Target 1	Enter actual accomplishments...	–	–	–	– AUTO
Target 1	Target 1	Enter actual accomplishments...	–	–	–	– AUTO
OVERALL AVERAGE RATING				0.000		PENDING EVALUATION

Figure 20.1. IPERF for COS/JO Personnel



Remarks/Suggestions/Recommendations on Ratee's Performance:

Enter remarks/suggestions/recommendations on ratee's performance...

signed at the start of the rating period

Targets prepared by:
Carlos Lim
Employee (Ratee)
Date: 22/10/2026

Approved by:
Miguel Cruz
Immediate Supervisor (Rater)
Date: 22/10/2026

signed at the end of the rating period

Rated by:
Miguel Cruz
Immediate Supervisor (Rater)
Date: 22/10/2026

Conforme:
Carlos Lim
Employee (Ratee)
Date: 22/10/2026

Measures:
Q Quality
T Timeliness
E Efficiency

Rating Guide:
5 Outstanding
4 Very Satisfactory
3 Satisfactory
2 Unsatisfactory
1 Poor

Figure 20.2. IPERF for COS/JO Personnel

The Individual Performance Evaluation Rating Form for Contract of Service (COS) and Job Order (JO) Personnel interface displayed the employee's information, expected outputs, actual accomplishments, and evaluation ratings. The upper section presented the official Benguet State University header, followed by the employee's name, position, office, personnel classification, and

rating period. The classification field allowed the appropriate personnel category, such as Contract of Service (COS), to be selected.

The interface also included the Cascaded Targets & Rubrics Matrix, which allowed targets, success indicators, and rubrics from a supervisor to be passed on to the employee and edited as needed. The Append Cascaded Goals button provided an option to add cascaded goals to the form. Below this section, the performance evaluation table was organized into Office Performance (Programs, Projects, and Activities), Expected Outputs, Actual Accomplishments, and Rating columns. The rating section consisted of Quality, Timeliness, Efficiency, and Average, which were used to assess the employee's performance.

The second screenshot displayed the Overall Average Rating and its current evaluation status, which was marked as Pending Evaluation. A Remarks/Suggestions/Recommendations section was provided for recording feedback regarding the employee's performance. The bottom section contained fields for documenting the individuals responsible for preparing, approving, rating, and confirming the evaluation. These included the names, positions, and dates of the employee and immediate supervisor at the beginning and end of the rating period. The Measures and Rating Guide sections also presented the evaluation criteria and their corresponding descriptions, ranging from 1 (Poor) to 5 (Outstanding).

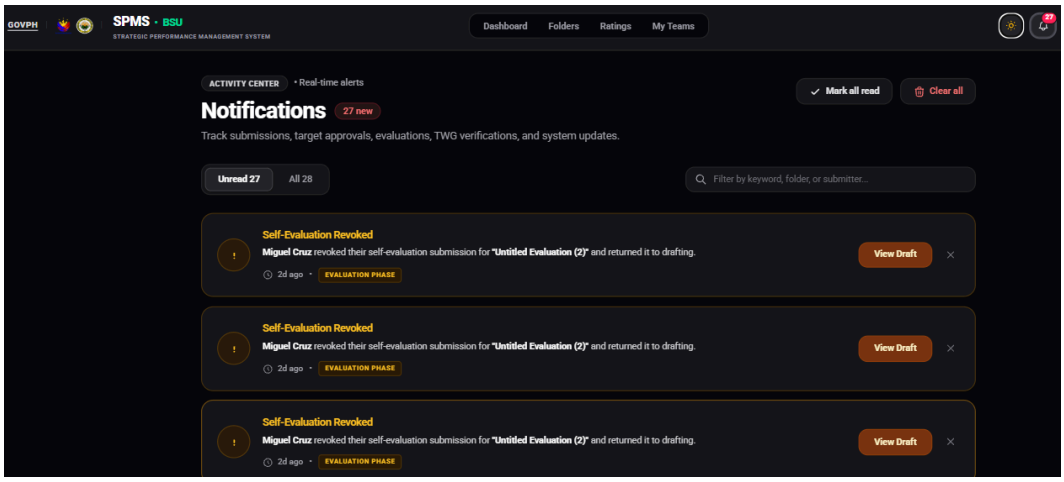


Figure 21. Notifications Interface of the SPMS

The Notifications Interface displayed alerts about performance submissions, target approvals, evaluations, TWG verifications, and system updates. It included Unread and All tabs, allowing users to view unread or all notifications. A search field was also provided to find notifications by keyword, folder, or submitter.

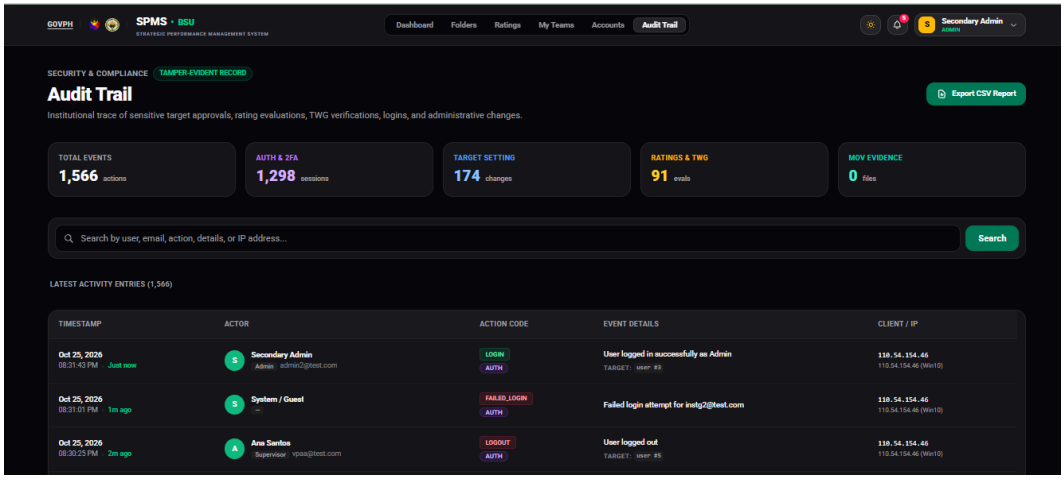


Figure 22. Audit Trail Interface of the SPMS for Admin Only

The Audit Trail Interface displayed records of system activities, including logins, target approvals, rating evaluations, TWG verifications, and administrative changes. It presented summary cards showing the total events, authentication sessions, target-setting changes, rating evaluations, and MOV evidence files. A search bar was provided to find specific records using details such as the user, email, action, or IP address. The activity table displayed the timestamp, actor, action code, event details, and client IP address. An Export CSV Report button was also available for exporting audit records.

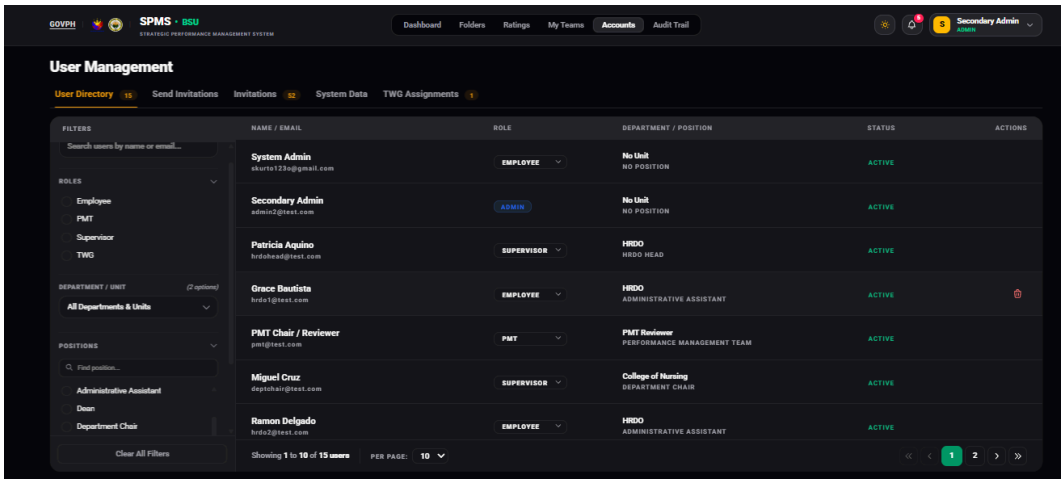


Figure 23. User Management Interface of the SPMS for Admin Only

The User Management Interface displayed the list of users, including their names, email addresses, roles, departments, positions, and account statuses. It provided tabs for managing the user directory, sending invitations, viewing invitations, accessing system data, and managing TWG assignments. The filter panel allowed the administrator to search for users and filter them by role, department, or position. The interface also provided options for managing user roles and displayed pagination controls for navigating through the user list.

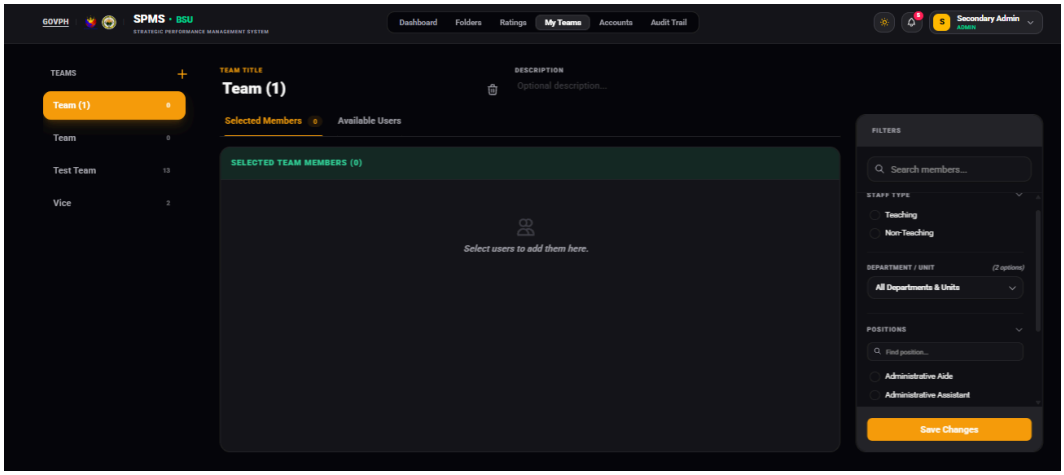


Figure 24.1. My Teams Interface of the SPMS for Admin

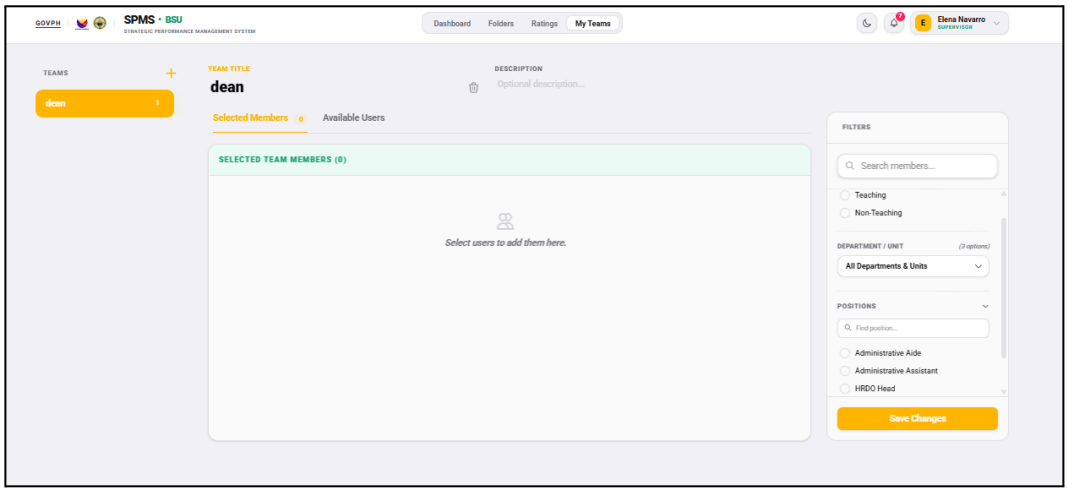


Figure 24.2. My Teams Interface for Supervisors

The My Teams Interface allowed administrators and supervisors to create and manage teams by selecting members from the available users. It displayed team names, selected members, and filters for searching users by staff type,

department, and position. The Save Changes button allowed them to save their team configurations.

Implementation Plan

This section describes how the developed SPMS for Benguet State University, La Trinidad Campus, will be prepared for turnover to the Human Resource Development Office and introduced into the university's performance management process. It covers the deployment environment, implementation phases, maintenance and support, and expected outcomes. The plan serves as a guide for preparing the system for actual use, subject to the university's approval and technical requirements.

Deployment Environment

The developed SPMS uses Vue.js 3, Tailwind CSS, and Axios for the frontend, CodeIgniter 4 for backend processing, and SQLite 3 for database storage. It provides functions for managing performance targets and accomplishments through the OPCR, DPCR, IPCR, and IPERF forms. It also supports performance review, rating computation, evaluation status monitoring, notifications, and report generation. Before the system is used for actual operations, the deployment environment will be reviewed in coordination with the HRDO and the BSU Information and Communications Technology (ICT) Office. The review will cover the server or hosting environment, database configuration,

network access, security settings, email-based one-time password authentication, and data backup procedures. The capacity of the selected environment to support the expected number of users will also be assessed.

The system may be deployed on a university-managed server or an approved hosting service, depending on the university's decision. If a different deployment environment is selected, the application files, database, uploaded files, and configuration settings will be transferred and tested to ensure that the system functions properly. The final deployment arrangement will depend on the university's technical requirements and approval.

Implementation Phases

The implementation will proceed in four phases. The first phase, testing and evaluation, was completed to assess the functionality and usability of the developed SPMS. Unit testing, integration testing, system testing, and user acceptance testing were conducted to check the system's functions and the interaction of its features. The system's usability was also evaluated using the Post-Study System Usability Questionnaire. The results of these activities served as the basis for identifying and addressing issues before proceeding with the next implementation phases.

The second phase is pre-implementation preparation. The researchers will coordinate with the HRDO and BSU ICT Office to review the system

configuration and prepare the information needed for actual use. Test records will be removed or reset as appropriate. User accounts, assigned roles, organizational information, evaluation cycles, deadlines, and other required records will be verified. The security settings, database backup, and recovery procedures will also be checked before the system is used to manage official performance records.

The third phase is user orientation and training. The researchers will conduct a walkthrough for HRDO personnel, supervisors, and employees to familiarize them with the system's functions and procedures. HRDO personnel will be guided in managing evaluation periods, monitoring submissions, maintaining performance records, and generating reports. Supervisors will be oriented on reviewing submissions, evaluating performance, and monitoring assigned employees. Employees will be guided on accessing their performance folders, preparing and submitting performance forms, recording accomplishments, and checking their evaluation status. A user guide will also be provided to serve as a reference during system use.

The fourth phase is system turnover and implementation. After the necessary reviews and revisions have been completed, the researchers will coordinate with the HRDO and BSU ICT Office for the turnover of the system, subject to the university's approval. The turnover will include the application files, database, technical documentation, user guide, and necessary access credentials for authorized personnel. The responsibilities for system administration, technical

support, data backup, and security monitoring will also be established. Once the required preparations and approvals are completed, the system may be introduced for actual use in accordance with the university's procedures.

Maintenance and Support

After turnover, the BSU ICT Office will be responsible for the technical maintenance of the system in coordination with HRDO. Maintenance activities will include application updates, database management, regular backups, security monitoring, and technical support. The ICT Office will also check the backup and recovery procedures periodically to help protect stored performance records.

The HRDO will coordinate with the ICT Office when reporting system errors, requesting technical changes, or updating performance management procedures that may affect the system. Proposed modifications will be reviewed and tested before being applied to the deployed version. Since the SPMS stores employee performance information, the university will need to address the appropriate privacy and security requirements before actual implementation. The responsibilities and resources for continued maintenance will be agreed upon by the concerned university offices.

Expected Outcomes

Once implemented, the SPMS is expected to provide HRDO personnel, supervisors, and employees with a centralized platform for managing performance evaluation activities. Employees will be able to prepare and submit performance forms and monitor their evaluation status. Supervisors will be able to review submissions, evaluate performance, process ratings, and monitor the progress of assigned evaluations. HRDO personnel will be able to manage evaluation periods, monitor submissions, access performance records, and generate reports using the information stored in the system.

The system is also expected to reduce dependence on manual rating computation, repeated follow-ups, and the physical handling of draft performance forms. Through automated rating computation, centralized records, and evaluation status monitoring, the SPMS may help improve the accuracy and accessibility of performance information and support more efficient evaluation activities.

However, the existing process of printing forms for handwritten signatures will remain because electronic signatures are not included in the developed system. The expected benefits will depend on proper deployment, user participation, compliance with university procedures, and continued technical support from the BSU ICT Office.

RECOMMENDATION

Based on the development and evaluation of the SPMS for Benguet State University, La Trinidad Campus, the following recommendations are:

1. **Deployment and Adoption.** The BSU Human Resource Development Office, in coordination with the Information and Communications Technology Office, is encouraged to review the system's deployment requirements and prepare it for actual use. Before implementation, the university should verify the hosting environment, database configuration, security settings, user accounts, and backup procedures.

2. **User Training and Orientation.** The HRDO should conduct orientation and training for personnel, supervisors, and employees who will use the system. A user guide should also be provided to help users understand the functions relevant to their roles and follow the proper procedures for submitting, reviewing, and monitoring performance forms.

3. **Regular Maintenance and Data Security.** The BSU ICT Office should perform regular system maintenance, security updates, database backups, and performance monitoring in coordination with the HRDO. Appropriate privacy and security measures should also be reviewed before the system is used to store and process official employee performance records.

4. **Further Usability and Performance Evaluation.** Future evaluators may conduct additional assessments after the system has been used in actual working conditions. These assessments may include feedback from a wider range of users and observations of the system's performance, reliability, and usability during actual performance evaluation cycles.

5. **Enhancement of Monitoring and Coaching Functions.** Future developers may improve the system's monitoring and coaching capabilities by providing additional features for documenting coaching activities, tracking progress throughout the evaluation period, and recording approved changes to performance targets when permitted by university policies.

6. **Integration with Other University Systems.** Future development may explore integration with existing BSU information systems to reduce duplicate data entry and improve information management. Such integration should be coordinated with the BSU ICT Office and evaluated in accordance with the university's technical, security, and data privacy requirements.

7. **Further System Enhancement.** Future researchers may consider additional improvements based on user feedback and the university's requirements. These may include enhancements to reporting, accessibility,

and compatibility with different devices, subject to technical feasibility and approval by the university.

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APPENDICES

APPENDIX A
System Usability Questionnaire

Republic of the Philippines
BENGUET STATE UNIVERSITY
2601, La Trinidad, Benguet

PSSUQ SOFTWARE EVALUATION QUESTIONNAIRE

Strategic Performance Management System for HRDO

Dear Respondent,
Greetings!

As part of the evaluation of the Strategic Performance Management System developed for the Human Resource Development Office of Benguet State University, we would like to ask a few minutes of your time to evaluate your experience in using the system.

This questionnaire uses the **Post-Study System Usability Questionnaire (PSSUQ)** to measure your satisfaction and experience with the system in terms of its usefulness, information quality and interface quality. Your honest responses will help us determine how easy and satisfactory the system is to use.

All responses will be treated with confidentiality and will be used for academic purposes only.

Thank you for your time and participation.

Respectfully yours,
The Researchers

CONSENT TO PARTICIPATE

Direction: Please put a check mark (✓) in the box that represents your choice.

I understand that this questionnaire is intended to evaluate the EVALHUB – Web-Based Strategic Performance Management System (SPMS). I understand that my responses will be used for academic purposes only and will be treated with confidentiality.

Do you agree to participate in this system evaluation?

☐ **YES,** I agree to participate.
☐ **NO,** I do not agree to participate.

If NO, please indicate your reason:

☐ I do not have enough time.
☐ I am not comfortable participating.
☐ Other reason: _____

Respondent's Signature: _____

Republic of the Philippines
BENGUET STATE UNIVERSITY
2601, La Trinidad, Benguet

RESPONDENT PROFILE

Direction: Please fill in or check (✓) the information below.

Full Name (optional): _____ Date: _____

Role: ☐ HRDO Personnel ☐ Supervisor (pls specify: ☐ Dean ☐ Director ☐ Head of Office)
☐ Permanent Employee ☐ Contractual Employee

Directions:

Below are statements about your experience using the Strategic Performance Management System. Please check (✓) **ONE** box for each statement using the scale in Table 1, based on your own use of the system.

Table 1. Seven-point Likert Scale

Rating	Description
1	Strongly Agree
2	Agree
3	Somewhat Agree
4	Neutral
5	Somewhat Disagree
6	Disagree
7	Strongly Disagree

Questionnaire — SPMS User Evaluation

System Usefulness

Statement	1	2	3	4	5	6	7	N.A.
1. Overall, I am satisfied with how easy it is to use this system.	☐	☐	☐	☐	☐	☐	☐	☐
2. It was simple to use this system.	☐	☐	☐	☐	☐	☐	☐	☐
3. I was able to complete the tasks and scenarios quickly using the system.	☐	☐	☐	☐	☐	☐	☐	☐
4. I felt comfortable using the system.	☐	☐	☐	☐	☐	☐	☐	☐
5. It was easy to learn to use this system.	☐	☐	☐	☐	☐	☐	☐	☐
6. I believe I could become productive quickly using this system.	☐	☐	☐	☐	☐	☐	☐	☐



Republic of the Philippines
BENGUET STATE UNIVERSITY
2601, La Trinidad, Benguet

Information Quality

Statement	1	2	3	4	5	6	7	N.A.
7. The system gave error messages that clearly told me how to fix problems.	☐	☐	☐	☐	☐	☐	☐	☐
8. Whenever I made mistake using the system, I could recover easily and quickly.	☐	☐	☐	☐	☐	☐	☐	☐
9. The information (such as online help, on-screen messages, and other documentation) provided with this system was clear.	☐	☐	☐	☐	☐	☐	☐	☐
10. It was easy to find the information I needed.	☐	☐	☐	☐	☐	☐	☐	☐
11. The information was effective in helping me complete the tasks and scenarios.	☐	☐	☐	☐	☐	☐	☐	☐
12. The organization of information on the system screen was clear.	☐	☐	☐	☐	☐	☐	☐	☐

Interface Quality

Statement	1	2	3	4	5	6	7	N.A.
13. The interface of this system was pleasant.	☐	☐	☐	☐	☐	☐	☐	☐
14. I liked using the interface of this system.	☐	☐	☐	☐	☐	☐	☐	☐
15. This system has all the functions and capabilities I expect it to have.	☐	☐	☐	☐	☐	☐	☐	☐
16. Overall, I am satisfied with this system.	☐	☐	☐	☐	☐	☐	☐	☐

****End of Questionnaire****

APPENDIX B
User Feedback Form



Republic of the Philippines
BENGUET STATE UNIVERSITY
2601, La Trinidad, Benguet

FEEDBACK FORM

Strategic Performance Management System for HRDO

Direction: Please answer the questions based on your experience using the system.

1. What did you like about the system?

2. What part of the system needs improvement?

3. Do you have any other comments or suggestions?

4. Based on your experience, would you recommend the use of web-based SPMS? Why?

Thank you for your valuable feedback and suggestions!

APPENDIX C
User Guide

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1. Institutional SPMS Framework and Overview

1.1 Purpose of this User Guide

The BSU-SPMS User Guide is a task-oriented operational manual written specifically for the employees and academic supervisors of Benguet State University. Unlike technical installation documents, this guide focuses entirely on the practical execution of your semester performance evaluations, explaining step-by-step how to formulate targets, attach documentary evidence, evaluate subordinate deliverables, and export certified Civil Service Commission (CSC) reports.

1.2 System Access and Supported Browsers

BSU-SPMS is an institutional web platform accessible via any modern desktop, laptop, tablet, or smartphone:

Production Cloud Portal: https://perfsys.duckdns.org/

On-Campus Local Intranet: http://localhost:8080/

Supported Browsers: Google Chrome (v100+), Mozilla Firefox (v100+), Microsoft Edge (v100+), and Apple Safari (v15+). JavaScript and LocalStorage must be enabled.

1.3 The Two-Phase Evaluation Lifecycle

Under Civil Service Commission Memorandum Circular No. 6, s. 2012, every academic semester operates under two distinct evaluation phases governed by institutional cutoff dates:

Phase 1: Target Commitment and Planning (Semester Opening):

- Ratees access standard institutional templates.
- Ratees formulate specific Major Final Outputs (MFOs) aligned with their supervisor's cascaded goals.
- Ratees balance their performance weights across Core, Strategic, and Support functions to total exactly 100.0%.
- Immediate supervisors review, suggest revisions, or officially approve the targets. Approved targets are frozen as the baseline for the semester.

Phase 2: Accomplishment Evaluation and Scoring (Semester Closing):

- Ratees input actual outputs delivered against their approved commitments.
- Ratees upload Means of Verification (MOVs) or link cloud portfolios to substantiate claims.
- Ratees input self-ratings across Quality, Efficiency, and Timeliness (Q-E-T).
- Supervisors review evidence, score deliverables on the official 1.00 to 5.00 rubric, and sign off.

- The PMT and TWG calibrate scores to maintain institutional consistency and certify final ratings.

1.4 Forms Taxonomy by Employee Role

Form Code	Form Title	Target Employee Classification	Approving Superior
IPCR	Individual Performance Commitment and Review	Regular, Temporary, and Contractual Faculty Members	Department Chair / Section Head
IPERF	Individual Performance Evaluation and Review Form	Job Order (JO) and Contract of Service (COS) Personnel	Unit Head / Administrative Chief
DPCR	Division Performance Commitment and Review	Department Chairs (Academic Departments)	College Dean

CDPCR	College Division Performance Commitment and Review	College Deans (Collegiate Divisions)	Vice President for Academic Affairs (VPAA)
OPCR	Office Performance Commitment and Review	Apex Executive Leadership (VPAA / University President)	University President / Board of Regents

---

2. Account Access, Security, and Navigation

2.1 First-Time Onboarding via Institutional Email Invitation

To protect institutional data integrity, BSU-SPMS does not offer open public self-registration. New accounts are created via controlled institutional invitations issued by the HRDO:

- Check your official @bsu.edu.ph institutional inbox for an email titled "BSU-SPMS Institutional Account Invitation".
- Click the secure one-time registration link. The link contains a cryptographic onboarding token valid for 48 hours.

- On the registration screen, verify that your pre-assigned collegiate unit and civil service position are correct.
- Set a strong password (minimum 8 characters, containing uppercase, lowercase, numbers, and symbols).
- Click Activate Account. You are redirected to the Login Portal.

2.2 Secure Login and Two-Factor Authentication (2FA OTP)

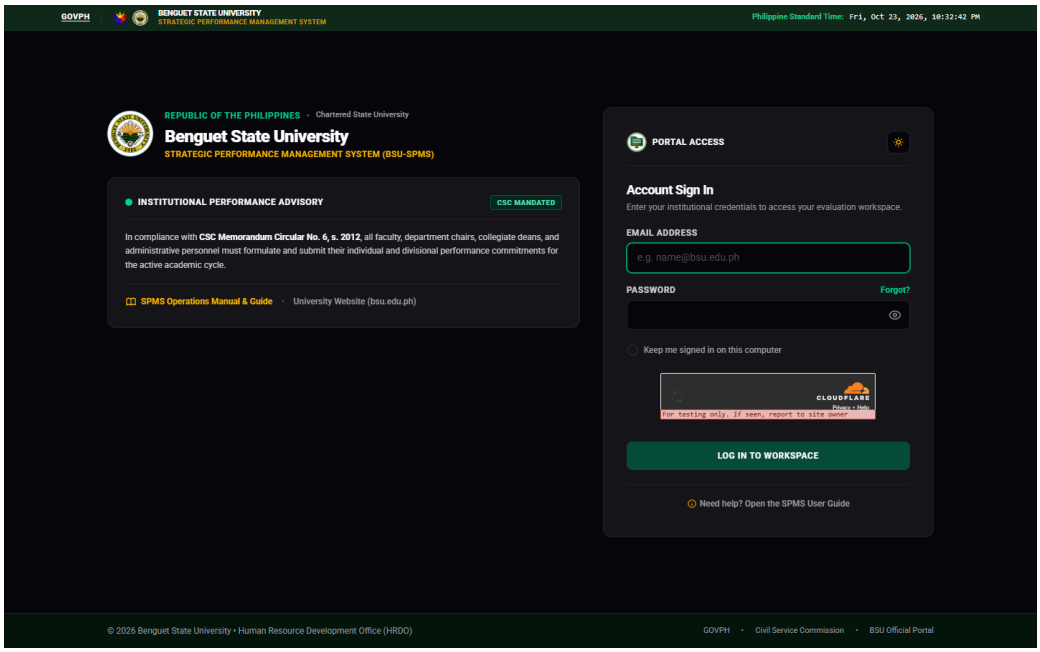


Figure 1: BSU-SPMS Secure Login Portal with 2FA Authentication

- Navigate to the login portal at https://perfsys.duckdns.org/login.
- Enter your official institutional email (@bsu.edu.ph) and your password.
- Click Sign In.
- The system automatically dispatches a 6-digit One-Time Password (OTP) to your email.

- In the 2FA prompt dialog, enter the 6-digit OTP code before the 5-minute countdown timer expires.
- Check Remember this device for 30 days if using a personal computer.
- Click Verify and Continue. Upon successful verification, your user dashboard opens.

> *[!NOTE]*
>
> *If you enter an incorrect password 5 consecutive times, the automated brute-force defense locks your IP address and account for 15 minutes. Wait 15 minutes or contact HRDO to reset your security counter.*

2.3 Navigating the Main Dashboard and Semester Cycles

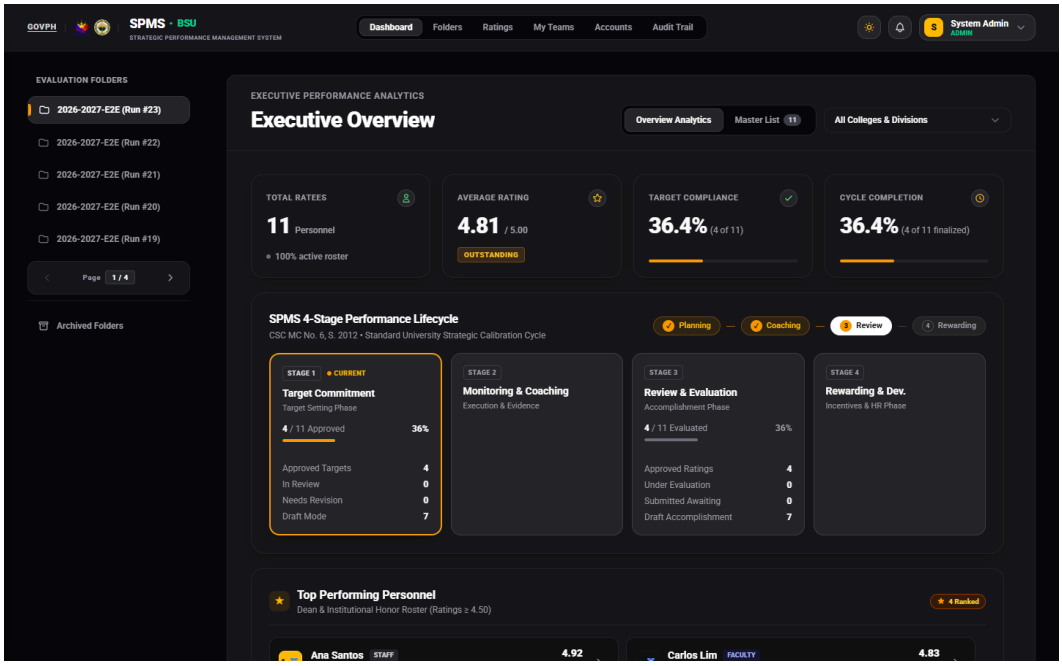


Figure 2: BSU-SPMS User Dashboard and Active Evaluation Milestones

Upon logging in, your dashboard presents the institutional control center:

Active Cycle Header: Displays the current academic term (e.g., "1st Semester AY 2026-2027"), active phase (Target Formulation vs. Accomplishment Evaluation), and real-time Philippine Standard Time (PST) countdown clock.

Progress Meter: Displays the percentage completion of your current folder (Drafted, Submitted, Evaluated, Certified).

Supervisory Review Queue: For Chairs and Deans, this pane displays pending subordinate submissions requiring your review.

Notification Bell: Located in the top navigation bar, clicking the bell opens real-time alerts regarding approved targets, supervisor revision notes, and approaching deadlines.

2.4 Profile Management, Password Updates, and Avatar Customization

- Click your user avatar in the top-right corner of the navigation bar and select My Profile.

Avatar Customization: You may upload an official institutional photo (JPEG or PNG, max 2 MB) or choose a generated letter avatar with customized university color schemes.

Password Modification: Enter your current password followed by your new password. An email confirmation is dispatched upon modification.

Plantilla Appointment Details: Review your registered Civil Service position and college assignment. If your appointment has changed, submit an update request to HRDO.

2.5 Offline Capability (Progressive Web App and Automatic Sync)

BSU-SPMS operates as a certified Progressive Web App (PWA). If campus Wi-Fi drops while you are formulating targets:

- The system displays a yellow Offline Mode Active indicator banner.
- You can continue typing deliverables, revising measures, and entering self-ratings without interruption.
- Changes are automatically saved to your local browser storage every 800 milliseconds via debounce tracking.
- When internet connectivity is restored, the system displays a green Connection Restored badge and synchronizes your changes with the institutional server in the background.

---

3. Faculty and Staff User Guide (IPCR and IPERF)

3.1 Accessing Your Semester Evaluation Folder

- On your dashboard, locate the active semester card and click Open Evaluation Folder.

- The system loads your performance document. If this is the start of a new academic term, the system automatically initializes a fresh document derived from the approved institutional template.

3.2 The Superior Basis Split-Pane (Cascaded Goals)

Under CSC guidelines, individual targets must directly align with division goals:

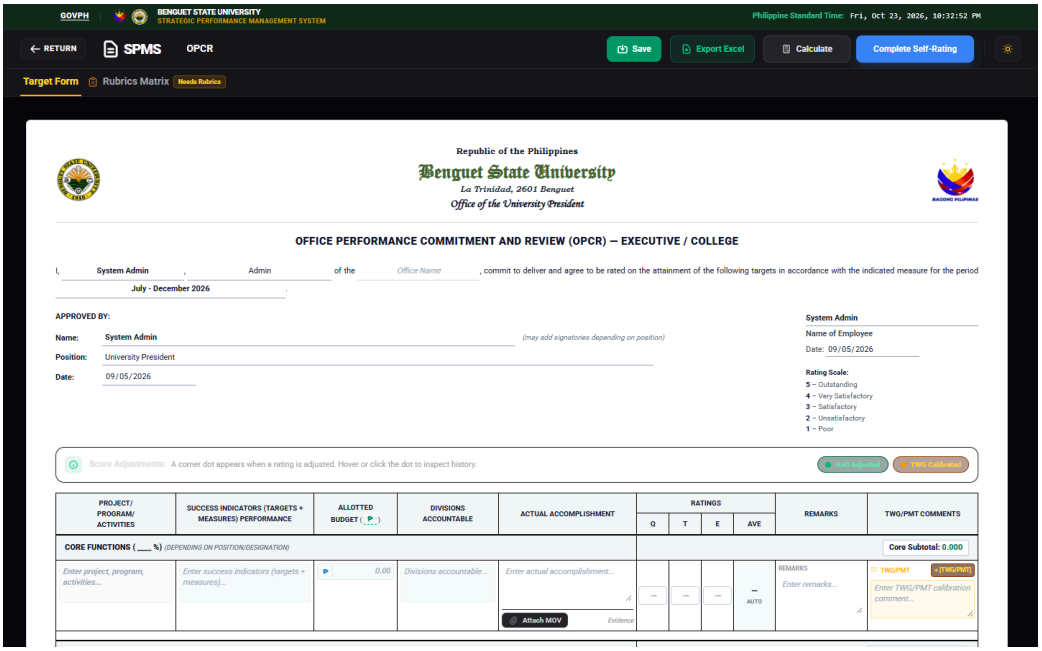


Figure 3: Performance Commitment Matrix Editor with Superior Basis Split-Pane

- In the matrix editor, click the Superior Basis tab in the side drawer.
- The drawer expands into a split-pane displaying your immediate supervisor's approved commitments (e.g., your Department Chair's DPCR commitments).
- Review your department's target deliverables to ensure your individual MFOs directly support the collegiate targets.

3.3 Formulating Targets and the 100.0% Weight Validation Engine

CSC SPMS mandates that every performance matrix balance across three statutory categories:

Core Functions (60% to 70%): Primary instructional, clinical, and laboratory duties.

Strategic Functions (20% to 25%): Research projects, extension programs, and institutional committees.

Support Functions (10% to 15%): Administrative duties, student advising, and campus services.

The 100.0% Validation Engine Rule:

- The sum of all individual row weights must equal exactly 100.0%.
- If your weights total 99.0% or 101.5%, the system displays a warning banner: Weights Total: 99.0% (Deficit: 1.0%).
- The Submit Targets button remains strictly disabled until the 100.0% balance condition is achieved.

Adding and Editing Deliverable Rows:

- Select the appropriate category tab (Core Functions, Strategic Functions, or Support Functions).
- Click + Add Deliverable Row.
- Enter the Major Final Output (MFO) (e.g., "Undergraduate Instruction in Computer Science").

- Enter the Performance Measure / Success Indicator (e.g., "100% of assigned syllabus modules delivered with at least 85% student passing rate").
- Enter the Assigned Weight (%) (e.g., 25.0).
- The system autosaves your inputs within 800 milliseconds.

3.4 Specifying Success Indicators (Quality, Efficiency, and Timeliness)

Under CSC MC No. 6, s. 2012, all accomplishment ratings are evaluated across three dimensions:

Quality (Q): Accuracy, thoroughness, adherence to institutional standards, and stakeholder acceptability.

Efficiency (E): Resource optimization, cost-effectiveness, and output volume per standard unit of time.

Timeliness (T): Delivery of deliverables in accordance with calendar deadlines and statutory schedules.

- When formulating your indicators, clearly state the quantitative benchmark for each dimension (e.g., "Quality: 4.5+ average on student evaluation; Efficiency: 1:1 syllabus compliance; Timeliness: Final grades submitted within 72 hours of exam").

3.5 Attaching Means of Verification (Local Binary Files and Cloud Links)

Every deliverable claim must be accompanied by supporting proof:

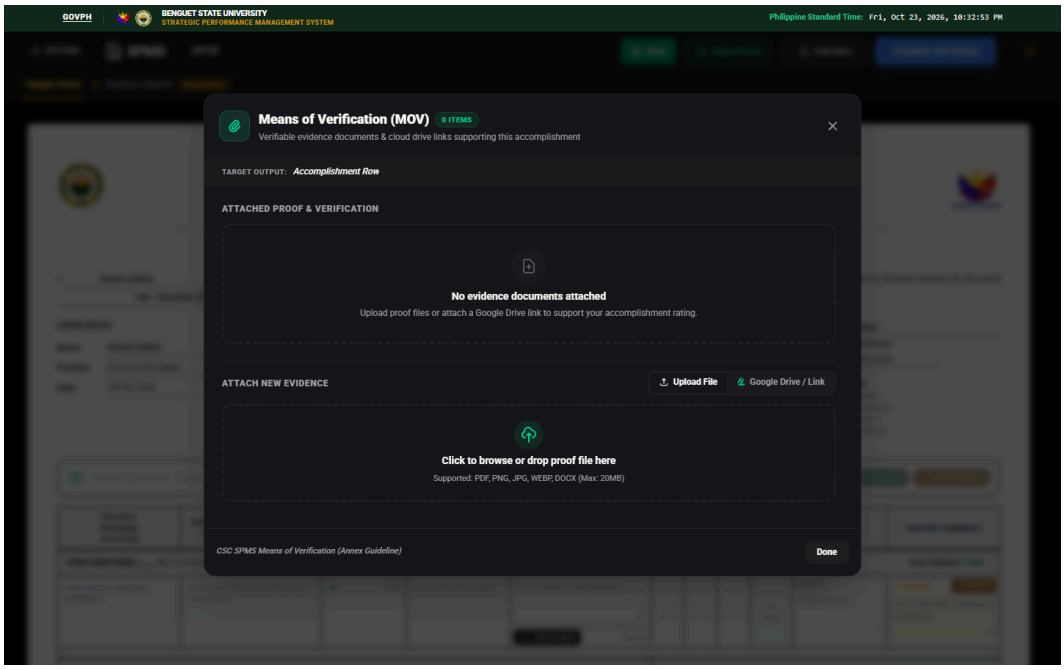


Figure 4: Slide-Out Means of Verification (MOV) Evidence Drawer

- In any deliverable row, click the Paperclip / Evidence icon. The slide-out Evidence Drawer opens.

Uploading Local Files:

- Drag and drop files into the upload area or click Browse Files.
- Supported formats: PDF, Microsoft Word (.docx), Microsoft Excel (.xlsx), PNG, and JPEG (up to 20 MB per file).
- Files are stored under AES-256 encrypted directories on the institutional server.

Attaching Cloud Storage Links (Google Drive / OneDrive / Dropbox):

- If your evidence portfolio is stored on Google Drive, select the Cloud Link tab.
- Paste the public or institutional share link into the URL field.
- Enter a descriptive label (e.g., "Google Drive Portfolio — AY 2026-2027 Research Evidence").
- Click Attach Cloud Link.

Verifying Uploads: Click any attached file to open the in-app previewer to verify that documents display legibly.

> *[!IMPORTANT]*
>
> *When attaching Google Drive links, verify that the file sharing permission is set to "Anyone at Benguet State University with the link can view". If set to Restricted, supervisors cannot inspect your proof.*

3.6 Submitting Targets for Review and Pre-Approval Revocation

- Once all deliverables are formulated and your weights total exactly 100.0%, click Submit Targets for Approval.
- A confirmation modal displays your category breakdown and the designated reviewing supervisor (e.g., "Department Chair — IT Department").
- Click Confirm Submission. Your folder status changes to pending_target_approval.

Pre-Approval Revocation: If you realize you forgot to include a deliverable before your supervisor reviews it, click Revoke Submission. This returns the folder to draft_target for editing. Once your supervisor opens and reviews the document, revocation is locked.

3.7 Accomplishment Reporting and Self-Rating

At the end of the academic term, HRDO opens the Accomplishment Evaluation phase:

- Open your semester folder. The matrix now unlocks the Accomplishment and Self-Rating columns.
- In each row, enter your actual physical outputs delivered under the Actual Accomplishments column.
- Open the Evidence Drawer and attach final MOV proof (e.g., certified grade sheets, published papers).
- Enter your self-scores for Quality (Q), Efficiency (E), and Timeliness (T) from 1.00 to 5.00:

  5 (Outstanding): Performance exceeding expectations by 30% or more.

  4 (Very Satisfactory): Performance exceeding expectations by 15% to 29%.

  3 (Satisfactory): 100% compliance with established targets.

  2 (Unsatisfactory): Performance met 51% to 99% of targets with noticeable deficiencies.

1 (Poor): Performance met 50% or below of targets.

- The system automatically computes the row average score and multiplies it by the category weight.
- Click Submit Accomplishment Report to send your evaluation to your supervisor.

3.8 Dimension Gating (Excluding Non-Applicable Criteria)

Certain deliverables do not possess all three scoring dimensions (for example, a routine administrative report may have Quality and Timeliness measures, but Efficiency is not applicable):

- In the rating inputs for that row, locate the non-applicable dimension.
- Toggle the N/A switch on that dimension.
- The dimension is greyed out.
- The calculation engine automatically recalibrates the row average across the remaining active dimensions (e.g., dividing by 2 instead of 3), preventing an unfair score penalty.

3.9 Exporting Official CSC Excel Workbooks and Printing

Once evaluations are certified by your supervisor, you can export official, formatted spreadsheets:

- In the upper-right corner of the document matrix, click Export CSC Excel.
- The system generates a styled Microsoft Excel workbook (.xlsx) conforming strictly to CSC MC No. 6, s. 2012 Annexes (IPCR Annex A).

- All statutory header banners, collegiate department headers, signature blocks, and live spreadsheet formulas (SUM, AVERAGE) are preserved.
- Open the workbook in Microsoft Excel to print on standard Legal (8.5 x 14 in) or Letter (8.5 x 11 in) paper.

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4. Supervisors and Evaluators Guide (Chairs and Deans)

4.1 4-Tier Target Cascading Workflow

Department Chairs and College Deans hold a dual role in BSU-SPMS: they formulate division commitments while evaluating subordinate staff:

Formulating Division Goals (DPCR): The Dean formulates College DPCR commitments aligned with the VPAA's OPCR. Once approved, Department Chairs formulate Departmental DPCR commitments supporting the Dean's goals.

Cascading Goals Downward: When constituent faculty open their IPCRs, the approved division commitments appear in their Superior Basis drawer, providing unambiguous alignment.

4.2 Reviewing Subordinate Target Submissions

- On your main dashboard, navigate to the Supervisory Review Queue.
- Locate submissions marked pending_target_approval.
- Click Review Targets.
- The subordinate's matrix loads in review mode. Inspect the deliverable rows, measures, and verify that category weights equal 100.0%.

4.3 Returning Targets with Inline Revision Remarks

If a subordinate's targets require adjustments:

- Locate the specific deliverable row requiring revision.
- Click in the Supervisor Remarks field of that row.
- Enter your specific feedback (e.g., [Chair]: Please clarify the target passing rate for Data Structures).
- In the top action bar, click Return for Revision.
- The folder status updates to target_revision. The ratee receives an immediate in-app alert and email notification, and your row remarks are preserved in their editor.

4.4 Approving Targets and Freezing Baseline Commitments

- If all deliverables and measures are satisfactory, click Approve Targets.
- A confirmation dialog prompts for your official supervisory sign-off.
- Click Confirm Approval.
- The folder status updates to target_approved. Commitments are permanently locked, and an immutable snapshot is recorded in activity_logs.

4.5 Inspecting Digital Proof via the In-App Evidence Viewer

During the accomplishment rating phase, supervisors must inspect supporting evidence before scoring:

- In the subordinate's matrix, deliverable rows with uploaded proof display an active paperclip icon with a badge indicating the file count.
- Click the icon to expand the Evidence Drawer.
- Click any document thumbnail to launch the In-App Modal Viewer.
- The document opens directly inside your browser (supporting multi-page PDFs, high-resolution images, and Word previews) without requiring you to download files to your personal computer.

4.6 Conducting Accomplishment Evaluations and 5-Point Scoring

- Review the subordinate's reported physical accomplishments against their approved targets.
- In the Supervisor Rating columns, input your numerical ratings (1.00 to 5.00) for Quality, Efficiency, and Timeliness.
- The system computes the composite row score and overall numerical rating in real time.
- The system automatically derives the qualitative adjectival rating based on official CSC bands:

  4.500 to 5.000: Outstanding (O)

  3.500 to 4.499: Very Satisfactory (VS)

  2.500 to 3.499: Satisfactory (S)

  1.500 to 2.499: Unsatisfactory (US)

  Below 1.500: Poor (P)

4.7 Supervisor Role Stamping and Evaluative Commentary

When entering feedback into the Summary Remarks field, use the single-click role tags:

- Click + [Chair] to insert [Department Chair]:
- Click + [Dean] to insert [College Dean]:
- Enter your evaluative summary, highlighting commendable accomplishments and growth areas.

4.8 Due-Process Concurrence Safeguard

CSC MC No. 6, s. 2012 strictly guarantees due process: an evaluation score cannot be unilaterally imposed on an employee without opportunity for discussion.

- If a supervisor reduces an employee's self-rating, the system flags the row with a Score Adjustment Notice.
- The supervisor must provide written commentary in the row remarks explaining the variance.
- Upon submitting ratings, the ratee receives a notification prompting them to review and sign their concurrence.

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5. Technical Working Group (TWG) and PMT Calibration Guide

5.1 Accessing the College Calibration Dashboard

The Performance Management Team (PMT) and Technical Working Group (TWG) ensure consistent grading standards across Benguet State University's 15 constituent colleges:

- From the navigation menu, select Calibration Console.
- The dashboard displays rating curves, submission statistics, and score distributions for all 15 colleges.

5.2 Analyzing Grade Distributions and Bell Curves

- Select a college from the selector (e.g., College of Nursing or College of Teacher Education).
- The system plots the college's grade distribution against the institutional benchmark:
- Displays the percentage of faculty scoring in the Outstanding and Very Satisfactory bands.
- Highlights statistically anomalous grade distributions requiring committee inspection.

5.3 Score Overrides with Mandatory Written Rationale

If the PMT or TWG identifies an unjustified rating during evidence audit:

- Click the faculty member's row to open their evaluation matrix.
- Click Calibrate Score.
- Enter the adjusted score.

- **Mandatory Justification Rule:** The Save Calibration button remains strictly disabled until the committee inputs a formal written explanation in the Committee
- Calibration Note field.
- Click Save Calibration.

5.4 Understanding Multi-Tier Audit Dots

To provide full transparency, calibrated scores display visual audit indicators:

Blue Dot: Ratee's original self-rating.

Green Dot: Immediate supervisor's initial rating.

Amber Dot: TWG technical calibration.

Purple Dot: Final PMT certified score.

Hovering your cursor over any audit dot displays the original score, the adjusted score, the timestamp, and the committee rationale.

5.5 Certifying Final Calibrated Ratings

- Once collegiate calibration is complete, the PMT Chair clicks Certify College Ratings.
- The semester folders for that college are locked into status calibrated.
- Certified ratings are recorded in the institutional ledger for faculty promotion and Performance-Based Bonus (PBB) eligibility.

6. Executive Monitoring and Compliance Tracking

6.1 Real-Time College Compliance Tracking

University leadership (University President, VPAA, HRDO Director) can track compliance in real time across the university's 15 colleges:

- Open the Executive Compliance Monitor from the navigation bar.
- View visual status meters for each college indicating target submission rates, evaluation completion rates, and pending reviews.

6.2 Plantilla Roster Compliance vs. Campus Census

The monitor cross-references active evaluation folders against the official institutional census in plantillas.

- Identifies any permanent, temporary, or contractual personnel who have not submitted their required forms before the institutional cutoff deadline.

6.3 One-Click Semestral Masterlist Excel Export

- On the Compliance Dashboard, click Export Masterlist (.xlsx).
- The system generates an institutional compliance spreadsheet listing all employees, collegiate assignments, submitted form types, numerical ratings, adjectival ratings, and approval dates.
- This report serves as official documentation for the Civil Service Commission and CHED.

7. Comprehensive User FAQ and Troubleshooting

Q1: Why is my "Submit Targets" button disabled and greyed out?

Answer:

- The statutory validation engine requires that your performance weights total exactly 100.0%.
- Check your weight summary at the top of the matrix. If your weights total 99.0% or 100.5%, adjust the percentage in one of your deliverable rows until the total reads 100.0% (Balanced). The button will unlock automatically.

Q2: My supervisor returned my target for revision. How do I fix it?

Answer:

- Open your evaluation folder. Rows with feedback will display a yellow banner containing your supervisor's remarks.
- Modify the MFO, success indicators, or weights as requested.
- Ensure your total weight remains exactly 100.0%.
- Click Resubmit Targets for Review.

Q3: I did not receive my 6-digit Two-Factor Authentication (2FA) OTP code.

Answer:

- Check your spam or junk folder in your @bsu.edu.ph institutional email account.

- Verify that your inbox is not over its storage quota.
- Wait 60 seconds and click Resend OTP Code.
- If the code still does not arrive, contact the HRDO Administrator to verify that your email address is spelled correctly in the system directory.

Q4: My Google Drive attachment link shows "Access Denied" to my supervisor.

Answer:

- Open the file or folder in Google Drive.
- Click Share.
- Under General Access, change the setting from Restricted to Benguet State University (Anyone in this group with the link can view).
- Save the setting. Your supervisor will immediately be able to view the file in the Evidence Drawer without re-uploading.

Q5: Can I edit my targets or accomplishments after the institutional deadline?

Answer:

- In compliance with Civil Service Commission guidelines and Republic Act 10535 (Philippine Standard Time), the system automatically locks folders when the cutoff deadline passes. Once locked, submissions cannot be altered without formal administrative permission. Contact your College

Dean or the HRDO Administrator to request an official deadline extension.

Q6: How do I print my approved IPCR form?

Answer:

- Open your approved evaluation folder.
- Click Export CSC Excel in the upper-right corner.
- Open the downloaded .xlsx file in Microsoft Excel.
- Select File → Print. The spreadsheet is pre-formatted for standard Legal or Letter paper with appropriate page margins, headers, and signature blocks.

APPENDIX D
Definition of Terms

Architecture – The overall structure of the system and how its components interact.

Authentication – The process of verifying a user's identity before granting access.

Authorization – The process of controlling user access and permissions.

Backend – The server-side part of the system that handles data processing and logic.

CodeIgniter 4 – A PHP framework used for developing the system backend.

Dashboard – A user interface that displays summarized data and system information.

Database – A structured collection of data stored electronically.

Frontend – The client-side part of the system that users interact with.

SQLite 3– A relational database management system used to store system data.

State Universities and Colleges (SUC) – Public higher education institutions in the Philippines established by law and funded by the national government.

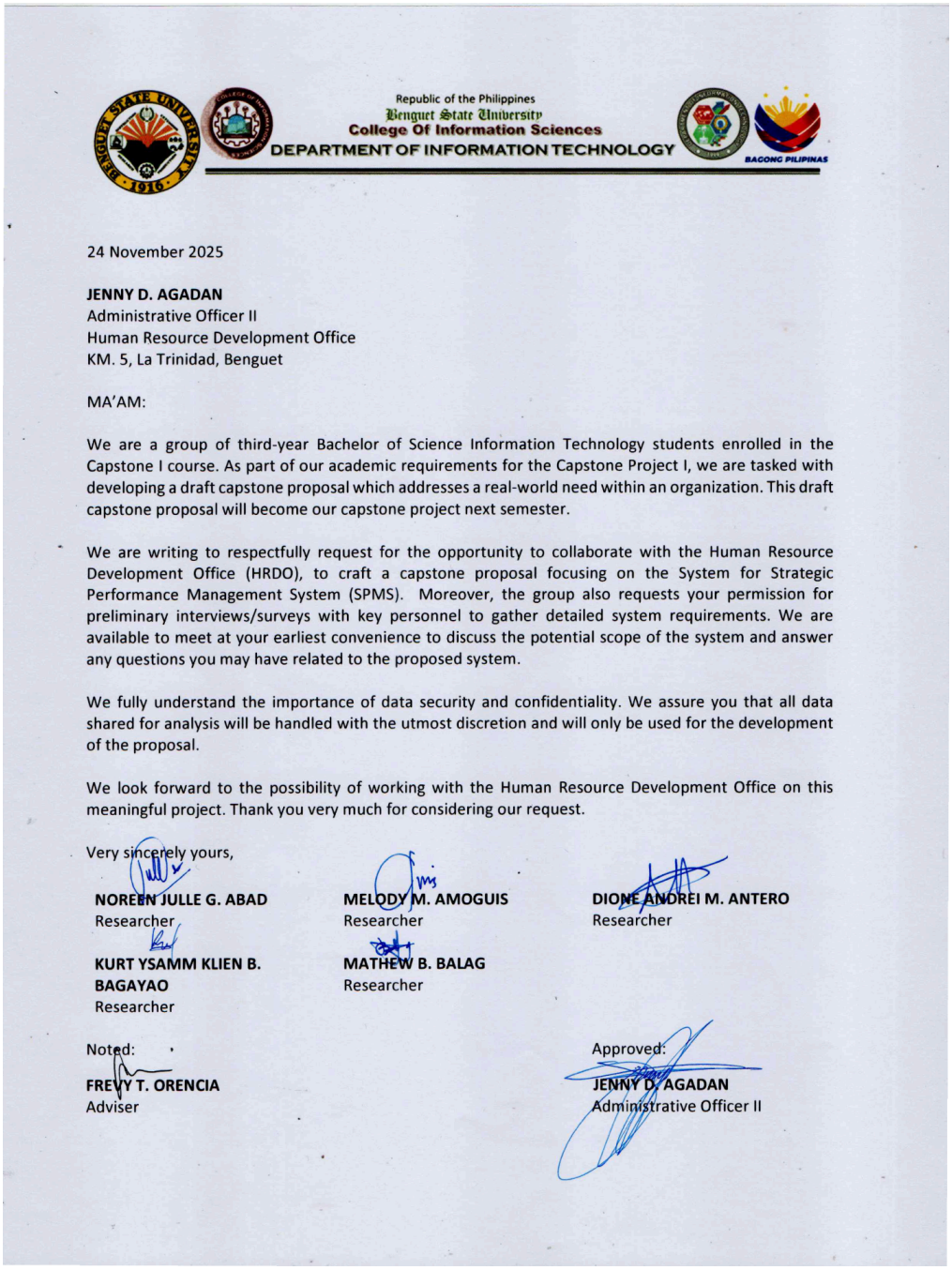
System Design – The process of planning the system's structure, features, and functions.

Tailwind CSS – A CSS framework used for designing user interfaces.

User Interface (UI) – This refers to the visual layout and interactive elements of the system through which users engage with the software.

Web-based System – A system that runs on a web server and is accessed through a browser.

APPENDIX E
Letter of Request (Signed)



Republic of the Philippines
Benguet State University
College Of Information Sciences
DEPARTMENT OF INFORMATION TECHNOLOGY

24 November 2025

JENNY D. AGADAN
Administrative Officer II
Human Resource Development Office
KM. 5, La Trinidad, Benguet

MA'AM:

We are a group of third-year Bachelor of Science Information Technology students enrolled in the Capstone I course. As part of our academic requirements for the Capstone Project I, we are tasked with developing a draft capstone proposal which addresses a real-world need within an organization. This draft capstone proposal will become our capstone project next semester.

We are writing to respectfully request for the opportunity to collaborate with the Human Resource Development Office (HRDO), to craft a capstone proposal focusing on the System for Strategic Performance Management System (SPMS). Moreover, the group also requests your permission for preliminary interviews/surveys with key personnel to gather detailed system requirements. We are available to meet at your earliest convenience to discuss the potential scope of the system and answer any questions you may have related to the proposed system.

We fully understand the importance of data security and confidentiality. We assure you that all data shared for analysis will be handled with the utmost discretion and will only be used for the development of the proposal.

We look forward to the possibility of working with the Human Resource Development Office on this meaningful project. Thank you very much for considering our request.

Very sincerely yours,

NOREEN JULLE G. ABAD
Researcher

MELODY M. AMOGUIS
Researcher

DIONE ANDREI M. ANTERO
Researcher

KURT YSAMM KLIEN B. BAGAYAO
Researcher

MATHEW B. BALAG
Researcher

Noted:

FREVY T. ORENCIA
Adviser

Approved:

JENNY D. AGADAN
Administrative Officer II

APPENDIX F
Interview Guide Questions

1. Current SPMS Processes and Workflow
    - Can you walk us through the complete SPMS cycle as currently implemented in your office from performance planning to rewarding and development planning?
    - What are the roles of HRDO personnel, supervisors, and employees in the current SPMS process?
    - What forms are currently used in the SPMS cycle, and how are these documents prepared, submitted, and stored?
    - How are performance records currently organized, retrieved, and updated when needed?
    - How is the approval and consolidation of performance ratings currently carried out?
    - What specific forms are used at each stage of the SPMS cycle, and how are they currently prepared, submitted, and stored?
2. Problems and Challenges with the Current System
    - Which stage of the SPMS process do you find most prone to errors or delays, and what causes these problems?
    - At what stage of the SPMS process do delays or bottlenecks usually occur?

- Have you experienced instances where performance records were lost, damaged, or submitted late due to the manual process? Can you describe the impact of such incidents?
- How much time does it typically take to complete one full SPMS cycle from distribution of forms to final submission of ratings?
- Are there any compliance issues with CSC Memorandum Circular No. 06, s. 2012 that arises because of the current manual workflow?

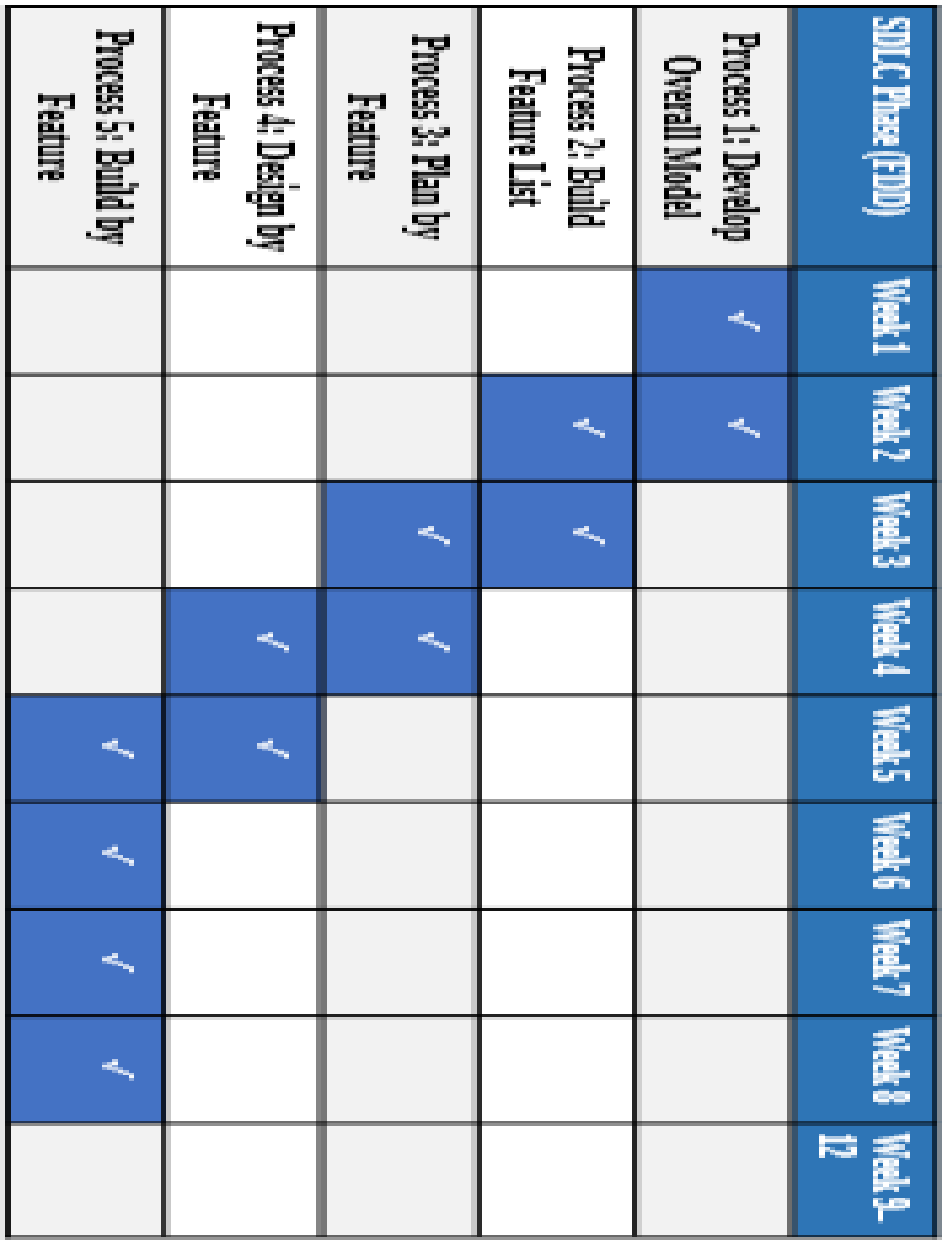
3. System Needs and Functional Requirements
    - What are the most critical features you would want in a digital SPMS, for example, automated computation, real-time tracking, or centralized records?
    - Should the system generate OPCR, IPCR, and DPCR reports automatically? What format would be most useful for institutional submission?
    - How important is the ability to track the status of submitted evaluations in real time?
    - What level of reporting and analytics would be useful — for example, per-employee trend tracking or office-level performance summaries?

- How important is it for the system to support real-time tracking of SPMS status and progress?
- Would automated notifications (e.g., deadlines, pending approvals, submissions) be helpful in your workflow? Why or why not?

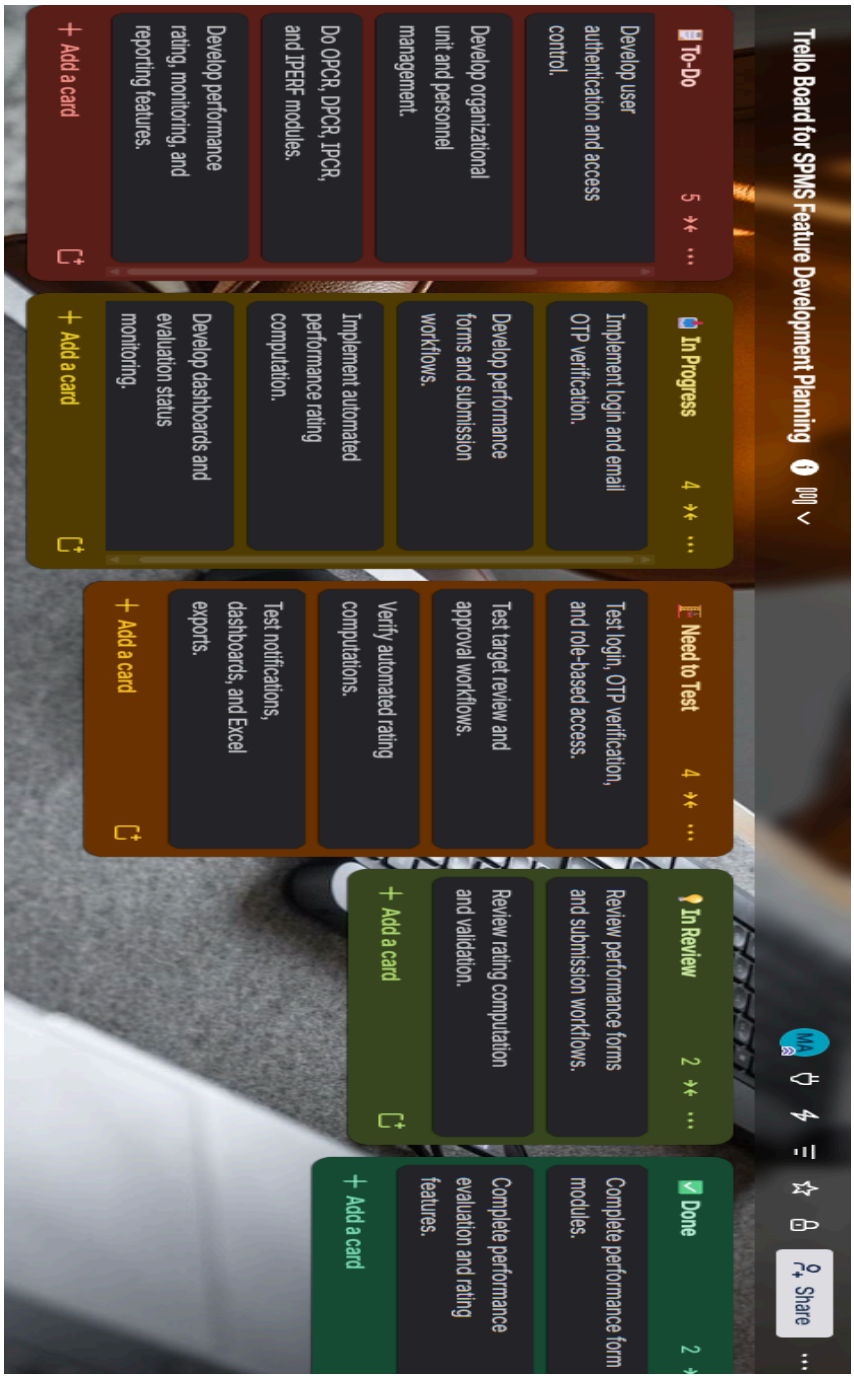
4. System Expectations and Constraints
    - How important is data security and confidentiality in handling SPMS records in your office?
    - What level of access control should be implemented for different users (HRDO, supervisors, employees)?
    - Would you prefer a login system using institutional accounts (e.g., BSU email)? Why?
    - Should the system include audit logs or tracking of changes made to performance records?
5. System Acceptance and Expectations
    - Are there existing BSU systems or databases that the SPMS should connect with?
    - What devices are commonly used by staff and employees when accessing work-related systems (e.g., laptop, desktop, mobile)?
    - What challenges should be considered in terms of internet access or system availability?

- Would offline access or printable outputs still be necessary in some cases? Please explain.

APPENDIX G
Gantt Chart



APPENDIX H
Trello Board



APPENDIX I
Database Schema

Table: Users

Column	Data Type	Constraint	Default	Description
`id`	INTEGER	PK AUTO	-	Unique internal user identifier.
`email`	VARCHAR(191)	NOT NULL, UNIQUE	-	Official institutional email (`@bsu.edu.ph`).
`password`	VARCHAR(255)	NOT NULL	-	Bcrypt cryptographic password hash (`$2y$10$`).
`first_name`	VARCHAR(100)	NOT NULL	-	Employee given first name.
`last_name`	VARCHAR(100)	NOT NULL	-	Employee family surname.
`reset_code`	VARCHAR(64)	NULL	NULL	Cryptographic password reset token.
`reset_code_expires_at`	DATETIME	NULL	NULL	Expiration timestamp of reset token.
`reset_attempts`	INT	NOT NULL	0	Failed reset attempts counter.
`reset_last_attempt_at`	DATETIME	NULL	NULL	Timestamp of last password reset attempt.
`remember_token`	VARCHAR(64)	NULL	NULL	SHA-256 hash of 30-day persistent cookie token.
`remember_token_expiry`	DATETIME	NULL	NULL	Expiration timestamp of remember cookie.
`doc_type`	TEXT	NULL	NULL	Default statutory form classification (`ipcr` , `iperf` , `dpcr` , `opcr`).
`two_factor_code`	VARCHAR(6)	NULL	NULL	6-digit One-Time Password (OTP) for 2FA.
`two_factor_expires_at`	DATETIME	NULL	NULL	Expiration timestamp of active 2FA OTP (5 min).
`email_change_code`	VARCHAR(64)	NULL	NULL	Verification code for email modification workflow.
`email_change_new_email`	VARCHAR(191)	NULL	NULL	Pending new institutional email address.
`email_change_expires_at`	DATETIME	NULL	NULL	Expiration timestamp for email change token.
`email_change_attempts`	INT	NOT NULL	0	Failed email change verification attempts.
`email_change_last_attempt_at`	DATETIME	NULL	NULL	Timestamp of last email change verification attempt.
`password_change_attempts`	INT	NOT NULL	0	Failed password modification attempts.

Column	Data Type	Constraint	Default	Description
password_change_last_attempt_at	DATETIME	NULL	NULL	Timestamp of last password change attempt.
is_active	TINYINT(1)	NOT NULL	1	Account status flag (1=Active, 0=Deactivated).
avatar_image	VARCHAR(255)	NULL	NULL	Stored profile image path.
avatar_color	VARCHAR(20)	NULL	NULL	Hex color code for fallback profile avatar.
avatar_letter	VARCHAR(5)	NULL	NULL	Single letter initial for fallback profile avatar.
created_at	DATETIME	NULL	NULL	Account creation timestamp.
updated_at	DATETIME	NULL	NULL	Last profile modification timestamp.
deleted_at	DATETIME	NULL	NULL	Soft deletion timestamp.

Table: roles

Column	Data Type	Constraint	Default	Description
id	INTEGER	PK AUTO	-	Unique role identifier.
name	VARCHAR(50)	NOT NULL, UNIQUE	-	Role name (Admin, Supervisor, Employee, TWG, PMT).
created_at	DATETIME	NULL	NULL	Role record creation timestamp.
updated_at	DATETIME	NULL	NULL	Last update timestamp.
deleted_at	DATETIME	NULL	NULL	Soft deletion timestamp.

Table: user_roles

Column	Data Type	Constraint	Default	Description
id	INTEGER	PK AUTO	-	Unique mapping identifier.
user_id	INT	FK users(id)	-	Foreign key referencing the assigned user.
role_id	INT	FK roles(id)	-	Foreign key referencing the granted role.
created_at	DATETIME	NULL	NULL	Role assignment timestamp.
updated_at	DATETIME	NULL	NULL	Last update timestamp.
deleted_at	DATETIME	NULL	NULL	Soft deletion timestamp.

Table: invitations

Column	Data Type	Constraint	Default	Description
id	INTEGER	PK AUTO	-	Unique invitation record identifier.
email	VARCHAR(191)	NOT NULL	-	Target institutional email address (@bsu.edu.ph).
token	VARCHAR(64)	NOT NULL	-	Cryptographically secure one-time onboarding token.
status	VARCHAR(30)	NOT NULL	-	Invitation state (pending, accepted, expired).
role_id	INT	FK roles(id)	-	Pre-assigned role to be granted upon registration.
expires_at	DATETIME	NOT NULL	-	Token expiration timestamp (default 48 hours).
created_at	DATETIME	NULL	NULL	Invitation issuance timestamp.

Table: login_attempts

Column	Data Type	Constraint	Default	Description
id	INTEGER	PK AUTO	-	Unique attempt record identifier.
ip_address	VARCHAR(45)	NOT NULL	-	Client IPv4 or IPv6 address.
email	VARCHAR(191)	NULL	NULL	Institutional email attempted.
device_id	VARCHAR(255)	NULL	NULL	Client browser device fingerprint string.
attempts	INT	NOT NULL	0	Cumulative failed authentication count.
last_attempt_at	DATETIME	NULL	NULL	Timestamp of most recent failed submission.

Table: email_queue

Column	Data Type	Constraint	Default	Description
id	INTEGER	PK AUTO	-	Unique queue job identifier.
to_email	VARCHAR(191)	NOT NULL	-	Recipient email address.
subject	VARCHAR(255)	NOT NULL	-	Message subject line.
body	TEXT	NOT NULL	-	Formatted HTML or plaintext email body.
status	VARCHAR(30)	NOT NULL	-	Queue status (pending, sent, failed).
attempts	TINYINT	NOT NULL	0	Dispatch retry attempt counter (max 3 retries).
created_at	DATETIME	NULL	NULL	Job enqueue timestamp.

Table: units

Column	Data Type	Constraint	Default	Description
id	INTEGER	PK AUTO	-	Unique organizational unit identifier.
name	VARCHAR(255)	NOT NULL	-	Official title of the college, department, or office.
parent_id	INT	FK units(id)	NULL	Parent college ID (NULL for root colleges/executive divisions).
created_at	DATETIME	NULL	NULL	Unit creation timestamp.
updated_at	DATETIME	NULL	NULL	Last update timestamp.
deleted_at	DATETIME	NULL	NULL	Soft deletion timestamp.

Table: positions

Column	Data Type	Constraint	Default	Description
id	INTEGER	PK AUTO	-	Unique plantilla position identifier.
title	VARCHAR(255)	NOT NULL	-	Official civil service job title (e.g., Associate Professor II).
is_teaching	TINYINT	NOT NULL	0	Flag: 1 for Faculty (IPCR); 0 for Administrative Staff (IPERF).
created_at	DATETIME	NULL	NULL	Record creation timestamp.
updated_at	DATETIME	NULL	NULL	Last update timestamp.
deleted_at	DATETIME	NULL	NULL	Soft deletion timestamp.

Table: plantillas

Column	Data Type	Constraint	Default	Description
id	INTEGER	PK AUTO	-	Unique plantilla appointment record identifier.
user_id	INT	FK users(id)	-	Appointed employee user ID.
position_id	INT	FK positions(id)	NULL	Assigned plantilla position ID.
unit_id	INT	FK units(id)	NULL	Assigned college or academic department ID.
started_at	DATE	NULL	NULL	Appointment effective start date.
ended_at	DATE	NULL	NULL	Appointment termination date (NULL indicates active appointment).
employment_status	VARCHAR(50)	NOT NULL	Permanent	Appointment classification (Permanent, Temporary, Contractual, COS).
created_at	DATETIME	NULL	NULL	Record creation timestamp.
updated_at	DATETIME	NULL	NULL	Last update timestamp.
deleted_at	DATETIME	NULL	NULL	Soft deletion timestamp.

Table: twg_unit_assignments

Column	Data Type	Constraint	Default	Description
id	INTEGER	PK AUTO	-	Unique assignment identifier.
user_id	INT	FK users(id)	-	Assigned TWG evaluator user ID.
unit_id	INT	FK units(id)	-	Designated college or organizational unit ID.
created_at	DATETIME	NULL	NULL	Assignment timestamp.

Table: document_folders

Column	Data Type	Constraint	Default	Description
id	VARCHAR(36)	PK	-	Unique UUID identifying the evaluation folder.
user_id	INT	FK users(id)	-	Owner employee user ID.
title	VARCHAR(255)	NOT NULL	-	Evaluation cycle title (e.g., 1st Semester AY 2026-2027).
parent_folder_id	VARCHAR(36)	FK folders(id)	NULL	Parent folder UUID in cascading institutional tree.
final_rating	DOUBLE	NULL	NULL	Calibrated composite numerical score (1.00 to 5.00).
status	VARCHAR(50)	NOT NULL	-	Lifecycle status (draft_target, pending_target_approval, target_approved, target_revision, pending_eval, evaluated, calibrated).
routing_preset_id	VARCHAR(36)	FK routing_presets(id)	NULL	Applied routing preset configuration UUID.
ipcr_target_start	DATETIME	NULL	NULL	Faculty target formulation opening timestamp.
ipcr_target_end	DATETIME	NULL	NULL	Faculty target submission cutoff deadline.
dpcr_target_start	DATETIME	NULL	NULL	Department/Dean target formulation opening timestamp.
dpcr_target_end	DATETIME	NULL	NULL	Department/Dean target submission cutoff deadline.
opcr_target_start	DATETIME	NULL	NULL	Institutional/Office target formulation opening timestamp.

Column	Data Type	Constraint	Default	Description
opcr_target_end	DATETIME	NULL	NULL	Institutional/Office target submission cutoff deadline.
iperf_target_start	DATETIME	NULL	NULL	Staff target formulation opening timestamp.
iperf_target_end	DATETIME	NULL	NULL	Staff target submission cutoff deadline.
ipcr_eval_start	DATETIME	NULL	NULL	Faculty accomplishment reporting opening timestamp.
ipcr_eval_end	DATETIME	NULL	NULL	Faculty accomplishment submission cutoff deadline.
dpcr_eval_start	DATETIME	NULL	NULL	Department accomplishment rating opening timestamp.
dpcr_eval_end	DATETIME	NULL	NULL	Department accomplishment submission cutoff deadline.
opcr_eval_start	DATETIME	NULL	NULL	Institutional accomplishment rating opening timestamp.
opcr_eval_end	DATETIME	NULL	NULL	Institutional accomplishment submission cutoff deadline.
iperf_eval_start	DATETIME	NULL	NULL	Staff accomplishment rating opening timestamp.
iperf_eval_end	DATETIME	NULL	NULL	Staff accomplishment submission cutoff deadline.
submitted_at	DATETIME	NULL	NULL	Accomplishment submission timestamp.
rated_at	DATETIME	NULL	NULL	Final supervisor evaluation timestamp.
deadline_reminder_sent_at	DATETIME	NULL	NULL	Timestamp of automated accomplishment reminder dispatch.
target_submitted_at	DATETIME	NULL	NULL	Target formulation submission timestamp.
target_approved_at	DATETIME	NULL	NULL	Official target approval timestamp.
target_approved_by	INT	FK users(id)	NULL	User ID of supervisor who officially approved targets.
target_approver_role	VARCHAR(50)	NULL	NULL	Designation/role tag of approving supervisor.
target_deadline_reminder_sent_at	DATETIME	NULL	NULL	Timestamp of automated target deadline alert.
target_period_open_sent_at	DATETIME	NULL	NULL	Timestamp of target opening announcement.

Column	Data Type	Constraint	Default	Description
eval_period_open_sent_at	DATETIME	NULL	NULL	Timestamp of evaluation opening announcement.
cdpcr_target_start	DATETIME	NULL	NULL	College DPCR target start milestone.
cdpcr_target_end	DATETIME	NULL	NULL	College DPCR target cutoff deadline.
cdpcr_eval_start	DATETIME	NULL	NULL	College DPCR evaluation start milestone.
cdpcr_eval_end	DATETIME	NULL	NULL	College DPCR evaluation cutoff deadline.
created_at	DATETIME	NULL	NULL	Folder instantiation timestamp.
updated_at	DATETIME	NULL	NULL	Last state update timestamp.
deleted_at	DATETIME	NULL	NULL	Archival / freeze timestamp.

Table: documents

Column	Data Type	Constraint	Default	Description
id	VARCHAR(36)	PK	-	Unique document UUID.
document_folder_id	VARCHAR(36)	FK folders(id)	-	Parent folder container UUID.
title	VARCHAR(255)	NOT NULL	-	Document title (e.g., Individual Performance Commitment and Review).
is_target	TINYINT	NOT NULL	0	Boolean flag indicating statutory baseline target commitment.
tabs	LONGTEXT	NULL	NULL	JSON encoded structure of form tabs, deliverable rows, MFOs, and Q-E-T scores.
created_at	DATETIME	NULL	NULL	Document creation timestamp.
updated_at	DATETIME	NULL	NULL	Last modification timestamp.
deleted_at	DATETIME	NULL	NULL	Soft deletion timestamp.

Table: templates

Column	Data Type	Constraint	Default	Description
id	INTEGER	PK AUTO	-	Unique template identifier.
title	VARCHAR(255)	NOT NULL	-	Template title (e.g., Standard IPCR Faculty Template 2026).
tabs	LONGTEXT	NULL	NULL	Default JSON matrix layout including statutory category weights and rubric columns.
created_at	DATETIME	NULL	NULL	Template creation timestamp.

Column	Data Type	Constraint	Default	Description
updated_at	DATETIME	NULL	NULL	Last update timestamp.
deleted_at	DATETIME	NULL	NULL	Soft deletion timestamp.

Table: document_attachments

Column	Data Type	Constraint	Default	Description
id	INTEGER	PK AUTO	-	Unique attachment identifier.
document_id	INT	FK documents(id)	-	Associated document ID.
row_id	VARCHAR(64)	NOT NULL	-	Specific matrix row ID or 'rubric' for matrix-level rubrics.
file_name	VARCHAR(255)	NOT NULL	-	Original client filename.
file_path	VARCHAR(255)	NOT NULL	-	Server storage path under writable/uploads/ .
file_type	VARCHAR(100)	NOT NULL	-	MIME type or cloud provider tag (link/google_drive).
file_size	INT	NOT NULL	-	File size in bytes (max 20 MB).
uploaded_by	INT	FK users(id)	-	User ID of uploader.
is_link	TINYINT	NULL	0	Boolean flag (1 = External URL, 0 = Uploaded Binary).
link_url	TEXT	NULL	NULL	External cloud storage URL (Google Drive, OneDrive).
created_at	DATETIME	NULL	NULL	Upload timestamp.
updated_at	DATETIME	NULL	NULL	Last update timestamp.
deleted_at	DATETIME	NULL	NULL	Soft deletion timestamp.

Table: routing_presets

Column	Data Type	Constraint	Default	Description
id	VARCHAR(36)	PK	-	Unique routing preset UUID.
owner_id	INT	FK users(id)	-	User ID of supervisor/administrator who configured preset.
name	VARCHAR(100)	NOT NULL	-	Preset title (e.g., CAS Faculty 2-Tier Approval Chain).
description	TEXT	NULL	NULL	Descriptive summary of routing scope and criteria.
created_at	DATETIME	NULL	NULL	Preset creation timestamp.
updated_at	DATETIME	NULL	NULL	Last update timestamp.
deleted_at	DATETIME	NULL	NULL	Soft deletion timestamp.

Table: routing_preset_members

Column	Data Type	Constraint	Default	Description
id	INTEGER	PK AUTO	-	Unique member mapping identifier.
preset_id	VARCHAR(36)	FK routing_presets(id)	-	Associated routing preset UUID.
user_id	INT	FK users(id)	-	Designated supervisor/evaluator user ID.
created_at	DATETIME	NULL	NULL	Member assignment timestamp.
updated_at	DATETIME	NULL	NULL	Last update timestamp.
deleted_at	DATETIME	NULL	NULL	Soft deletion timestamp.

Table: evaluation_routings

Column	Data Type	Constraint	Default	Description
id	INTEGER	PK AUTO	-	Unique routing step identifier.
folder_id	VARCHAR(36)	FK folders(id)	-	Evaluated folder UUID.
evaluator_id	INT	FK users(id)	-	Designated supervisor/evaluator user ID.
status	VARCHAR(50)	NOT NULL	-	Operational review status (pending, approved, returned).
evaluator_folder_id	VARCHAR(36)	NOT NULL	-	Evaluator subordinate folder container UUID.
created_at	DATETIME	NULL	NULL	Routing creation timestamp.
updated_at	DATETIME	NULL	NULL	Last update timestamp.
deleted_at	DATETIME	NULL	NULL	Soft deletion timestamp.

Table: activity_logs

Column	Data Type	Constraint	Default	Description
id	INTEGER	PK AUTO	-	Unique log entry identifier.
user_id	INT	FK users(id)	NULL	User ID who initiated the event (NULL for unauthenticated events).
action	VARCHAR(100)	NOT NULL	-	Action verb tag (LOGIN, TARGET_SUBMIT, SCORE_CALIBRATE, PASSWORD_CHANGE).
category	VARCHAR(50)	NOT NULL	-	Event domain (auth, folder, evaluation, security, admin).
entity_type	VARCHAR(50)	NULL	NULL	Affected domain entity (folder , document , user).
entity_id	INT	NULL	NULL	Identifier of affected entity.

Column	Data Type	Constraint	Default	Description
ip_address	VARCHAR(45)	NULL	NULL	Client IPv4/IPv6 address.
user_agent	VARCHAR(255)	NULL	NULL	Client browser user-agent header.
details	TEXT	NULL	NULL	JSON or structured textual payload documenting state modifications.
created_at	DATETIME	NULL	NULL	Immutable event timestamp (PST).

Table: notifications

Column	Data Type	Constraint	Default	Description
id	INTEGER	PK AUTO	-	Unique notification record identifier.
user_id	INT	FK users(id)	-	Recipient user ID.
sender_id	INT	FK users(id)	NULL	Dispatching user ID (NULL for automated system triggers).
type	VARCHAR(50)	NOT NULL	system	Notification category tag (system , deadline , routing , approval).
title	VARCHAR(150)	NOT NULL	-	Short notification summary.
message	TEXT	NOT NULL	-	Full explanatory notification message.
link	VARCHAR(255)	NULL	NULL	Direct application relative route.
icon	VARCHAR(50)	NULL	bell	UI icon descriptor.
read_at	DATETIME	NULL	NULL	Timestamp when notification was marked read.
created_at	DATETIME	NULL	NULL	Dispatch timestamp.
updated_at	DATETIME	NULL	NULL	Last update timestamp.